

TECHNICAL REPORT

IEC TR 60909-1

Second edition
2002-07

Short-circuit currents in three-phase a.c. systems –

Part 1: Factors for the calculation of short-circuit currents according to IEC 60909-0

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INTERNATIONAL ELECTROTECHNICAL COMMISSION

SHORT-CIRCUIT CURRENTS IN THREE-PHASE AC SYSTEMS –**Part 1: Factors for the calculation of short-circuit currents
according to IEC 60909-0**

FOREWORD

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Technical reports do not necessarily have to be reviewed until the data they provide are considered to be no longer valid or useful by the maintenance team.

IEC 60909-1, which is a technical report, has been prepared by IEC technical committee 73: Short-circuit currents.

This technical report shall be read in conjunction with IEC 60909-0.

The text of this technical report is based on the following documents:

Enquiry draft	Report on voting
73/120/DTR	73/125/RVC

Full information on the voting for the approval of this technical report can be found in the report on voting indicated in the above table.

This publication has been drafted in accordance with the ISO/IEC Directives, Part 3.

This document, which is purely informative, is not to be regarded as an International Standard.

The committee has decided that the contents of this publication will remain unchanged until 2010. At this date, the publication will be

- reconfirmed;
- withdrawn;
- replaced by a revised edition, or
- amended.

SHORT-CIRCUIT CURRENTS IN THREE-PHASE AC SYSTEMS –

Part 1: Factors for the calculation of short-circuit currents according to IEC 60909-0

1 General

1.1 Scope and object

This part of IEC 60909 is a technical report applicable to short-circuit currents in three-phase a.c. systems. This technical report aims at showing the origin and the application, as far as necessary, of the factors used to meet the demands of technical precision and simplicity when calculating short-circuit currents according to IEC 60909-0.

Thus this technical report is an addition to IEC 60909-0. It does not, however, change the basis for the standardized calculation procedure given in IEC 60909-0.

NOTE References are given in some cases to offer additional help, not to change the procedure laid down in the standard.

1.2 Reference documents

IEC 60038:1983, *IEC standard voltages*

IEC 60909-0:2001, *Short-circuit currents in three-phase a.c. systems – Part 0: Calculation of currents*

IEC/TR 60909-2:1992, *Electrical equipment – Data for short-circuit current calculations in accordance to IEC 60909 (1988)*

IEC/TR 60909-4:2000, *Short-circuit currents in three-phase a.c. systems – Part 4: Examples for the calculation of short-circuit currents*

1.3 Application of the factors

1.3.1 Factor c

The voltage factors $c_{\max}$ and $c_{\min}$ are used together with the equivalent voltage source at the short-circuit location in order to calculate maximum and minimum initial short-circuit currents (see 2.1).

1.3.2 Factors K_G and K_S or K_{SO}

The impedance correction factors K_G and K_S or K_{SO} are introduced when calculating the short-circuit impedances of generators and power station units (with or without on-load tap changer) (see 2.2).

1.3.3 Factors $K_{G,S}$, $K_{T,S}$ or $K_{G,SO}$, $K_{T,SO}$

The impedance correction factors $K_{G,S}$, $K_{T,S}$ or $K_{G,SO}$, $K_{T,SO}$ are introduced when calculating the partial short-circuit currents in case of a short circuit between generator and unit transformer (with or without on-load tap changer) of a power station unit (see 2.2.3.2 or 2.2.4.2).

1.3.4 Factor K_T

The impedance correction factor K_T is used when calculating the short-circuit impedances of network transformers (see 2.3).

1.3.5 Factor κ

The peak short-circuit current is calculated by using this factor (see 2.4).

1.3.6 Factors μ , λ and q

Factors used when calculating the decay of the a.c. component of the short-circuit current of a near-to-generator or a near-to-asynchronous-motor short circuit (see 2.5, 2.6 and 2.7).

1.3.7 Factors m and n

The factors m and n are used for the calculation of the Joule integral or the thermal equivalent short-circuit current (see 2.8).

1.3.8 Contribution of asynchronous motors to the initial symmetrical short-circuit current

Derivation and validity of relevant equations for checking the contribution of asynchronous motors or groups of asynchronous motors to the initial symmetrical short-circuit current (see 2.9).

1.4 Symbols, subscripts and superscripts

The following symbols, subscripts and superscripts are used in addition to those already defined in IEC 60909-0.

1.4.1 Symbols

E	Voltage behind the synchronous direct axis reactance X_d of a synchronous machine
E'	Voltage behind the transient direct axis reactance X_d' of a synchronous machine
E''	Voltage behind the subtransient direct axis reactance X_d'' of a synchronous machine
E_Q''	Subtransient voltage behind the impedance of a network feeder connected at Q
$E_0(I_f)$	Terminal voltage of a saturated synchronous machine at no-load ($I_G = 0$)
I^b	Branch current (load current) before the short circuit
I_f	Field current of a synchronous machine
$i_k(t)$	Time-dependent short-circuit current
I_{kUb}''	Initial symmetrical short-circuit current caused by the voltage $-U^b$, when calculating short-circuit currents using the superposition method
p_G or p_T	Relative values to define the region for the variation of terminal voltages, for instance $U_G = U_{rG} (1 \pm p_G)$ or $U_{THV} = U_{rTHV} (1 \pm p_T)$
T_{AC}	AC time constant of an asynchronous motor
T_N''	Mean value of the time constants T_{dN}'' and T_{qN}''
$T_{\mu q}$	Time constant calculated with the product μq (see 2.7.2)
t_p	Time duration from the beginning of a short circuit until the peak short-circuit current
U^b	Voltage at the short-circuit location before the short circuit
X_p	Potier reactance
Y	Admittance
γ	Impedance angle
Δ	Deviation
φ_U	Voltage angle

1.4.2 Subscripts

0	No load (T_{d0}'')
a	Approximation
ad	Admissible
d	Direct axis
e	Exact
f	Field of the synchronous machine
IEC	according to IEC 60909-0, for example $K_{S(IEC)}$
i	Internal
L	Load
MAX	Maximum (short-circuit current at worst-case load flow)
N	Network
q	Quadrature axis
S	Superposition method
*	Per unit quantity

1.4.3 Superscripts

b	Before
'	Transient
"	Subtransient

2 Factors used in IEC 60909-0

2.1 Voltage factor c for the equivalent voltage source at the short-circuit location

2.1.1 General

The magnitude of a short-circuit current in a three-phase a.c. system (maximum or minimum short-circuit current) at any location depends primarily on the network configuration, the generators or power-station units and the motors in operation and secondarily on the operational stage of the network before the short circuit.

The variations during operation in a three-phase a.c. system are very large. Therefore, it is difficult to find the special load flow that leads either to a maximum or to a minimum short-circuit current at the different locations of the network. In a given system, there are as many different short-circuit current magnitudes as there are possible different load-flow conditions for every location. Normally, extreme load-flow cases are not empirically known.

IEC 60909-0 therefore recommends a calculation method with the equivalent voltage source $cU_n / \sqrt{3}$ at the short-circuit location. This method, described in IEC 60909-0, is an approximation method without special conditions of operation. The aim of this standard is to find the maximum short-circuit currents with sufficient accuracy, mainly taking into account safety aspects and as far as possible economical aspects.

During the planning stage of a network, the different future load-flow conditions are unknown. Therefore, the equivalent voltage source $cU_n / \sqrt{3}$ is based on the nominal system voltage U_n and the voltage factor $c = c_{\max}$ or $c = c_{\min}$ for the calculation of the maximum or the minimum short-circuit currents. These factors c are given in table 1 of IEC 60909-0. The introduction of a voltage factor c is necessary for various reasons (IEC 60909-0, 1.3.15). These are:

- voltage variation depending on time and place;
- changing of transformer taps;

- neglecting loads and capacitances by calculating according to IEC 60909-0 (see 2.3.1);
- the subtransient behaviour of generators, power-station units and motors.

The meaning of the voltage factor c is illustrated for a simple model of a radial network in 2.1.4. Furthermore, results of extended calculations given in 2.2.5 and 2.3.4 demonstrate the possible deviations of calculations with the equivalent voltage source at the short-circuit location against the worst-case values found with a special procedure using the superposition method.

2.1.2 Calculation methods

In principle, there are two methods for the calculation of the initial symmetrical short-circuit current at the short-circuit location (IEC 60909-0, figures 1 and 2):

- the superposition method, derived from Helmholtz's or Thevenin's principle;
- the method using the equivalent voltage source at the short-circuit location (see 2.1.3).

Examples for the superposition method are given in 2.2 and 2.3. There the results of the superposition method are compared with the results found with the method using the equivalent voltage source at the short-circuit location.

If a certain load flow in an existing network is known, then it is possible to calculate the initial symmetrical short-circuit current with the superposition method, but this method gives the short-circuit current only related to the load flow presupposed. Therefore, it does not necessarily lead to the maximum short-circuit current. The reason is that for one short-circuit location there are as many short-circuit currents as load-flow conditions without exceeding the boundary conditions of voltages and currents during normal operation, even if the same operational voltage at the short-circuit location is given.

To overcome this problem and to find the worst-case load flow that leads to the maximum short-circuit current at one short-circuit location, a special method was developed by varying the operation conditions [9], [13], [26]¹⁾. Further information is given in 2.2.5 and 2.3.4.

2.1.3 Equivalent voltage source at the short-circuit location and voltage factor c

The procedure for the calculation of the initial symmetrical short-circuit current using the equivalent voltage source at the short-circuit location is described in IEC 60909-0. This method, which is normally based only on the rated data of electrical equipment, is an essential simplification compared to the superposition method or a transient calculation, because also in this case it is necessary to know the load-flow conditions before the short circuit.

Using this simplified procedure, the equivalent voltage source $cU_n / \sqrt{3}$ at the short-circuit location is the only active voltage in the positive-sequence system. All network feeders, synchronous machines and asynchronous motors are short-circuited behind their internal (subtransient) reactances (IEC 60909-0, 3.6.1). All the shunt capacitances and the shunt admittances (loads), except those of the motors, are to be omitted in the positive- and the negative-sequence system (IEC 60909-0, 2.3). Capacitances of the zero-sequence system are to be considered in general. The zero-sequence capacitances are to be omitted in low-voltage systems and in high-voltage effectively grounded systems (i.e. earth fault factor $\delta \leq 1,4$). Special considerations are necessary in high-voltage networks with long-distance lines and, of course, in the case of isolated neutral or resonant earthed networks (IEC 60909-0, 1.1). An example for the application of the calculation using the equivalent voltage source at the short-circuit location F is given in IEC 60909-0, figure 4.

¹⁾ The figures in square brackets refer to the bibliography.

It shall be shown that the short-circuit current deviation $\Delta i_k''$ depends on the voltage deviation Δu . The following definitions are used:

$$\Delta u = \left| \frac{U_Q}{U_B} \right| - 1 \quad (1)$$

$$\Delta i_k'' = \left| \frac{I_{ka}''}{I_{ke}''} \right| - 1 \quad (2)$$

where

a is the subscript for the approximation of the short-circuit current when Z_A is neglected;

e is the subscript for the exact value of the short-circuit current when Z_A is taken into account.

The following equation, giving the voltage conditions before the short circuit, can be derived from figure 1b if U_Q is taken as a constant value:

$$U_Q = U_A (1 + \alpha Z_L Y_A) = U_B (1 + \alpha Z_L Y_A) \quad (3)$$

with $U_A = U_B$ (see figure 1b) and $Y_A = 1/Z_A$.

Using equation (3) the voltage deviation can be expressed as follows:

$$\Delta u = \left| \frac{U_Q}{U_B} \right| - 1 = |1 + \alpha Z_L Y_A| - 1 \quad (4)$$

The exact value of the initial symmetrical short-circuit current can be calculated with the help of figure 1b:

$$I_{ke}'' = I_Q \cdot \frac{1}{1 + (1 - \alpha) Z_L Y_A} \quad (5a)$$

with

$$I_Q = \frac{U_Q}{\sqrt{3} Z_L} \cdot \frac{1}{\alpha + \frac{(1 - \alpha)}{1 + (1 - \alpha) Z_L Y_A}} \quad (5b)$$

Introducing I_Q from equation (5b) to equation (5a) leads to

$$I_{ke}'' = \frac{U_Q}{\sqrt{3} Z_L} \cdot \frac{1}{1 + \alpha(1 - \alpha) Z_L Y_A} \quad (6)$$

The approximation I_{ka}'' is found from figure 1b with $Z_A = \infty$ (or $Y_A = 0$) and the voltage U_B at the prospective short-circuit location, using equation (3):

$$I_{ka}'' = \frac{U_B}{\sqrt{3} Z_L} = \frac{U_Q}{\sqrt{3} Z_L} \cdot \frac{1}{1 + \alpha Z_L Y_A} \quad (7)$$

The current deviation $\Delta i_k''$ follows from equation (6) and (7):

$$\Delta i_k'' = \left| \frac{I_{ka}''}{I_{ke}''} \right| - 1 = \left| \frac{1 + \alpha(1 - \alpha)Z_L Y_A}{1 + \alpha Z_L Y_A} \right| - 1 \quad (8)$$

The product $Z_L Y_A$ can be written as:

$$Z_L Y_A = Z_L e^{j\gamma} Y_A e^{-j\varphi} = Z_L Y_A e^{j(\gamma - \varphi)} = Z_L Y_A e^{j\eta} \quad (9)$$

with $Z_L = R_L + jX_L = Z_L e^{j\gamma}$ and $Y_A = G_A - jB_A = Y_A e^{-j\varphi}$

The following results can be found from figure 2.

- The largest negative deviation $\Delta i_k''$ occurs in the case $\alpha = 1$ and $\eta = 0$. That means that the load Z_A is at the end of the line ($\alpha = 1$) and that the load is inductive ($\eta = 0$).
- The absolute values of the negative deviations $\Delta i_k''$ are smaller than the voltage deviations Δu :

$$|\Delta i_k''| < \Delta u \quad (10)$$

- In the case of $\alpha = 1$ and $\eta = 0$ equation (4) leads to (see figure 2):

$$\Delta u = Z_L Y_A = \Delta u_{\max} \quad (11)$$

- The following relation can be found from equation (8) established for the highest negative deviations $\Delta i_{k \max}''$ at $\alpha = 1$ (broken line in figure 2a), depending on the maximum positive voltage deviations $\Delta u_{\max}$:

$$\Delta i_{k \max}'' = \frac{1}{1 + \Delta u_{\max}} - 1 \quad (12)$$

- The following conditions are found from equation (12):

$$\Delta u_{\max} = +5 \%: -0,048 \leq \Delta i_{k \max}''$$

$$\Delta u_{\max} = +10 \%: -0,091 \leq \Delta i_{k \max}''$$

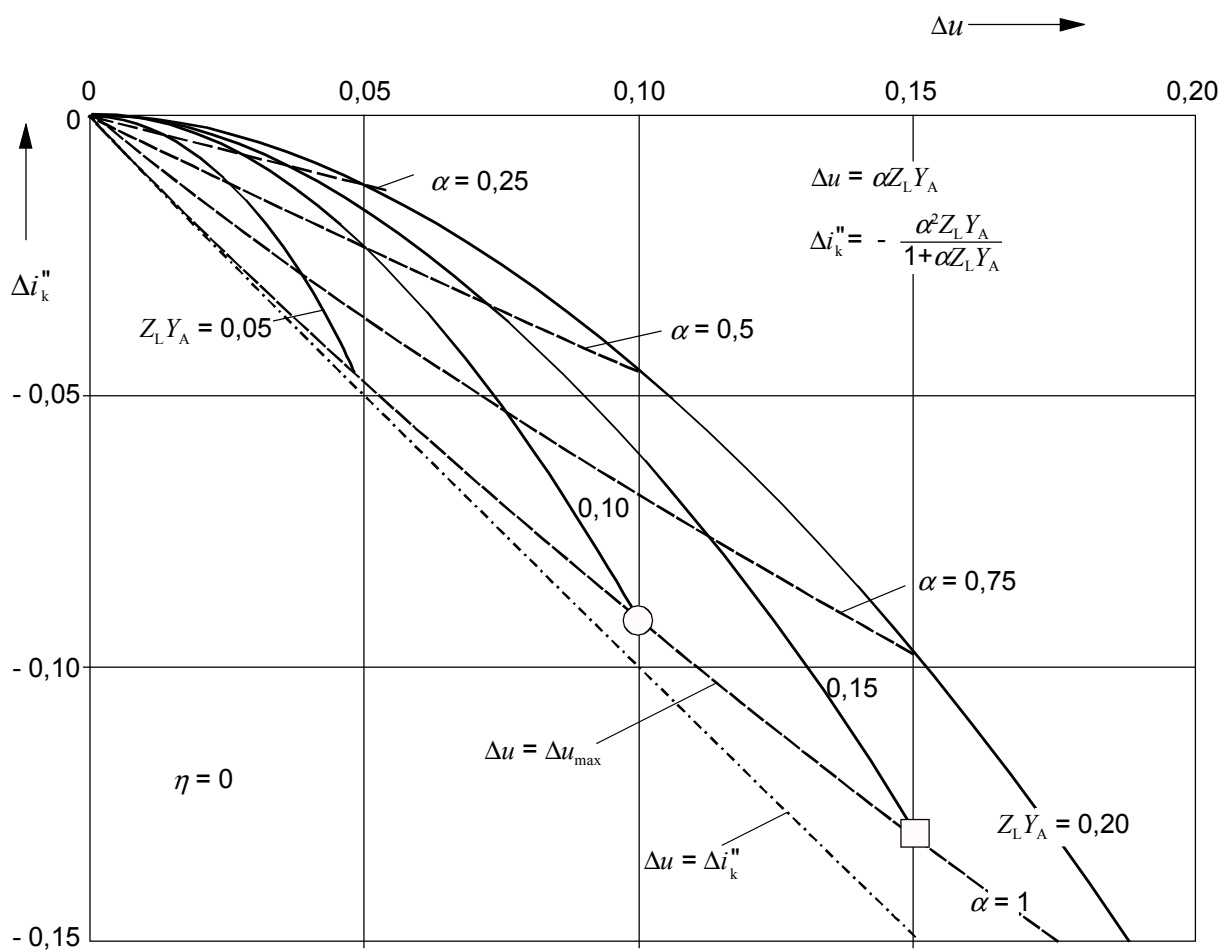
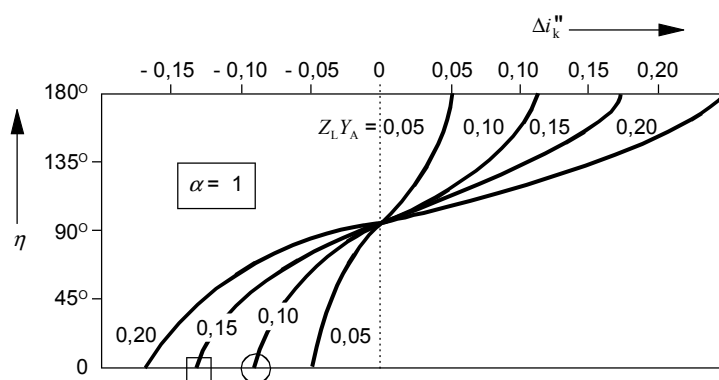
$$\Delta u_{\max} = +15 \%: -0,130 \leq \Delta i_{k \max}''$$

Together with the approximation I_{ka}'' from equation (7) and $\Delta u_{\max} = Z_L Y_A$ according to equation (11), equation (12) leads to

$$I_{ke \max}'' = I_{ka}'' (1 + \Delta u_{\max}) = \frac{U_Q}{\sqrt{3} Z_L} \quad (13)$$

If the method with the equivalent voltage source $cU_n / \sqrt{3}$ at the short-circuit location is used (see 2.1.3) to calculate the initial symmetrical short-circuit current $I_{k(IEC)}''$ and if the result shall be approximately the same as $I_{ke \max}''$ then the following relation is found with $I_{ke \max}''$ from equation (13):

$$I_{k(IEC)}'' = \frac{cU_n}{\sqrt{3} Z_L} \approx I_{ke \max}'' \quad (14)$$

Figure 2a – $\Delta i_k'' = f(\Delta u, \alpha)$ with $\eta = 0$ Figure 2b – $\Delta i_k'' = f(\eta, Z_L Y_A)$ with $\alpha = 1$ Figure 2 – Calculation of $\Delta i_k''$ according to equation (8) for different parameters

This leads, if the maximum short-circuit current $\underline{I}_{k(IEC)max}''$ shall be calculated, to the maximum of factor c [17]:

$$c_{max} \approx \frac{U_{Qmax}}{U_n} \quad (15)$$

Due to the relation given in equation (15), even if it is only found for the simple model with $Z_Q \Rightarrow 0$ in figure 1, the maximum voltage $U_{Q\max}$ at the feeding point is the relevant voltage for the calculation of the maximum short-circuit current.

However caution is necessary if voltages higher than $c_{\max}U_n$ (or $c_{\max}U_n/\sqrt{3}$ in the positive-sequence system) exist in a network. One example is the internal voltage E'' behind the subtransient reactance X_d'' of a synchronous machine (figures 4 and 6).

It can be shown [10] that this is also the most unfavourable case, even in comparison with results of meshed networks and long-distance transmissions up to about 400 km.

The factor $c = c_{\max}$ (IEC 60909-0, 2.3.1 and table 1) takes into account these results ($c_{\max} = 1,1$ for medium- and high-voltage networks and $c_{\max} = 1,05$ or $1,10$ for low-voltage networks depending on the tolerance for the voltage during operation), considering that the highest voltage in a normal system does not differ, on average, by more than +5 % (LV) or by +10 % (HV) approximately from the nominal system voltage U_n (IEC 60038, table III, note 2). In a normal system of Series I (50 Hz or 60 Hz systems), the highest and the lowest voltage do not differ by more than approximately ± 10 % from the nominal voltage of the system. In a normal system of Series II (60 Hz systems – North American practice), the highest voltage does not differ by more than +5 % and the lowest voltage by more than –10 % from the nominal voltage of the system. If in a network this assumption for a normal system is not fulfilled, and the highest voltage during operation does differ more than above, then a higher factor $c_{\max}$ may be necessary, see equation (15).

Special conditions occur in the case of generators and power-station units with high values of x_d'' and u_k which lead to voltage deviations higher than 10 %. Therefore, impedance correction factors are introduced for the calculation of the impedances of this electrical equipment (IEC 60909-0, 3.6 and 3.7).

The short-circuit voltage of network transformers may in special cases reach values up to 35 %; the rated voltages of these transformers (two-winding or three-winding transformers) may be considerably different from the nominal voltages of the connected networks and in many cases the transformers are equipped with on-load tap changers. Therefore, impedance correction factors are introduced for the calculation of the impedances of network transformers (IEC 60909-0, 3.3.3).

NOTE The note given in 8.3.2.2 of IEC 60909:1988 [28], did not give any help for the user, when carrying out practical calculations with network transformers, if one of the conditions given there is valid. Therefore, it became necessary to develop an impedance correction factor for network transformers (see 2.3).

2.2 Impedance-correction factors when calculating the short-circuit impedances of generators, unit transformers and power-station units

2.2.1 General

One of the main criteria for the rating of electrical equipment is the maximum short-circuit current and, in many cases, also the maximum partial short-circuit current. It is necessary to introduce impedance-correction factors K_G (see 2.2.2) for generators and impedance-correction factors K_S for power-station units with (see 2.2.3) and K_{SO} without (see 2.2.4) on-load tap changer of the unit transformer in addition to the voltage factor $c_{\max}$, especially if the subtransient reactances x_d'' of the generators are high and the transformation ratio of the unit transformers (with or without on-load tap changer) is different from the ratio of the system voltages during operation on both sides of the transformer. The correction factors for generators and power-station units are given in IEC 60909-0, 3.6 and 3.7.

Special considerations and impedance-correction factors are necessary for the calculation of the partial short-circuit currents in the case of a short circuit between generator and unit transformer or auxiliary transformer of a power-station unit (see IEC 60909-0, 4.2.1.3 and 4.2.1.4). These calculations are normally carried out only once during the construction of the power-station unit.

The impedance-correction factors given in IEC 60909-0, 3.6 and 3.7 are derived for the maximum partial short-circuit currents ($c = c_{\max}$). The factor $c_{\max}$ shall be taken from IEC 60909-0, table 1 in accordance with the voltage U_{rG} when using IEC 60909-0, equation (18) and in accordance with the voltage U_{nQ} at the high-voltage side of the unit transformer when using IEC 60909-0, equations (22) and (24). Additional investigations in meshed networks have shown that these correction factors are also adequate when calculating short-circuit currents at different locations of the network (see 2.2.5).

Special considerations are necessary for the correction factors when calculating minimum short-circuit currents, because the special boundary conditions for the different power-station units must be known. These conditions, for instance, are given by the maximum extent of underexcited operation, the minimum active power of thermal power stations during long-term operation or the maximum reactive power (overexcited or underexcited) of units in hydro-pumping stations as well as by special devices for the limitation of the torque angle. Furthermore, attention shall be given to the fact that even during low load conditions in a network, the number of power-station units operating with partial load within the underexcited region is usually minimized. Therefore, a rough estimation for minimum short-circuit currents may be found using the instructions given in IEC 60909-0, 2.5 and given impedance correction factors, even though these are based on operation at rated conditions before the short circuit (IEC 60909-0, 3.7.1).

2.2.2 Correction factor K_G

The impedance correction factor K_G is used for the impedance $\underline{Z}_G = R_G + j X_d''$ of a generator directly connected to a low- or a medium-voltage network (without unit transformer) as shown in figure 3 (IEC 60909-0, figure 11b).

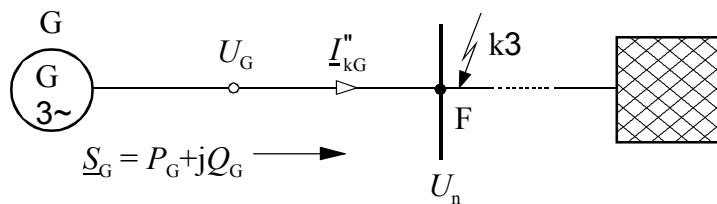


Figure 3 – Partial short-circuit current $I''_{kG(S)}$ of a generator directly connected to a network

The impedance correction factor K_G is derived from a phasor diagram for normal overexcited operation of the generator taking into account the subtransient reactance X_d'' and the subtransient internal voltage E'' .

Figure 4a gives the equivalent circuit diagram (positive-sequence system) of a generator. The terminal voltage U_G of the generator is controlled and therefore constant during operation before the short circuit. Normally $U_G = U_{rG} = 1,05 U_n$ is valid. In special cases the setpoint may vary: $U_G = U_{rG} (1 \pm p_G)$.

Figure 4 describes the procedure for the calculation of the partial short-circuit current $I_{-kG(S)}''$ with the superposition method.

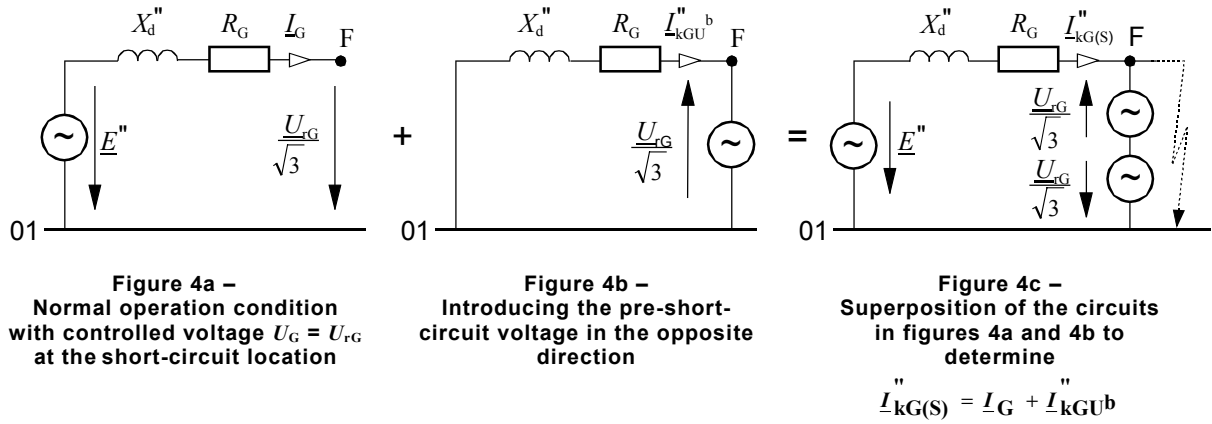


Figure 4 – Calculation of $I_{-kG(S)}''$ with the superposition method

The currents I_G and I_{-kGU^b}'' are derived from figures 4a and 4b at $U_G = U_{rG}$ as follows:

$$I_G = \frac{E'' - U_{rG} / \sqrt{3}}{R_G + jX_d''} \quad (16)$$

$$I_{-kGU^b}'' = \frac{U_{rG} / \sqrt{3}}{R_G + jX_d''} \quad (17)$$

The index U^b indicates that the current I_{-kGU^b}'' is found when the voltage $U_{rG} / \sqrt{3}$ before the short circuit is introduced in the opposite direction (figure 4b).

The superposition according to figure 4c leads to the partial short-circuit current $I_{-kG(S)}''$:

$$I_{-kG(S)}'' = I_G + I_{-kGU^b}'' = \frac{E'' - U_{rG} / \sqrt{3}}{R_G + jX_d''} + \frac{U_{rG} / \sqrt{3}}{R_G + jX_d''} = \frac{E''}{R_G + jX_d''} \quad (18)$$

On the other hand, when applying the method with the equivalent voltage source $cU_n / \sqrt{3}$ at the short-circuit location, the following equation for the calculation of the short-circuit current I_{-kG}'' is found:

$$I_{-kG}'' = \frac{E''}{R_G + jX_d''} = \frac{cU_n}{\sqrt{3}(R_G + jX_d'')K_G} \quad (19)$$

Taking $E'' = U_{rG} / \sqrt{3} + I_G Z_G = U_{rG} / \sqrt{3} + I_G (\cos \varphi_G - j \sin \varphi_G)(R_G + jX_d'')$, the impedance correction factor K_G is found:

$$K_G = \frac{U_n}{U_{rG}} \cdot \frac{c}{1 + \frac{\sqrt{3}I_G}{U_{rG}}(R_G \cos \varphi_G + X_d'' \sin \varphi_G) + j \frac{\sqrt{3}I_G}{U_{rG}}(X_d'' \cos \varphi_G - R_G \sin \varphi_G)} \quad (20)$$

If R_G is very small compared to X_d'' ($R_G \ll X_d''$) the following approximation can be applied:

$$K_G \approx \frac{U_n}{U_{rG}} \cdot \frac{c}{1 + (I_G / I_{rG}) x_d'' \sin \varphi_G} \quad (21)$$

The per unit (p.u.) value of X_d'' is introduced with $x_d'' = X_d'' / Z_{rG}$ and $Z_{rG} = U_{rG}^2 / S_{rG}$ (IEC 60909-0, 3.6.1).

The maximum short-circuit current I_{kG}'' is found if the denominator of the second part of the right side of equation (21) reaches the maximum. Normally this will be the case at the point of rated operation (S_{rG} , U_{rG} , $\cos \varphi_{rG}$) or in its vicinity if the extreme overexcited operation with $Q > Q_r$ is excluded. Thus, it seems acceptable to choose $I_G = I_{rG}$ and $\sin \varphi_G = \sin \varphi_{rG}$.

By introducing $c = c_{\max}$ the approximation of equation (21) then assumes the following form:

$$K_G = \frac{U_n}{U_{rG}} \cdot \frac{c_{\max}}{1 + x_d'' \sin \varphi_{rG}} \quad (22)$$

Example:

$$S_{rG} = 10 \text{ MVA}$$

$$U_{rG} = 10,5 \text{ kV}$$

$$\cos \varphi_{rG} = 0,8$$

$$x_d'' = 0,12$$

The generator feeds a network with the nominal voltage $U_n = 10 \text{ kV}$ (see figure 3).

A calculation of the maximum partial short-circuit current according to the right side of equation (19) with $K_G = 0,977$, found from equation (22) with $c_{\max} = 1,1$ (IEC 60909-0, table 1), leads to $I_{kG}'' = 4,900 \text{ kA}$.

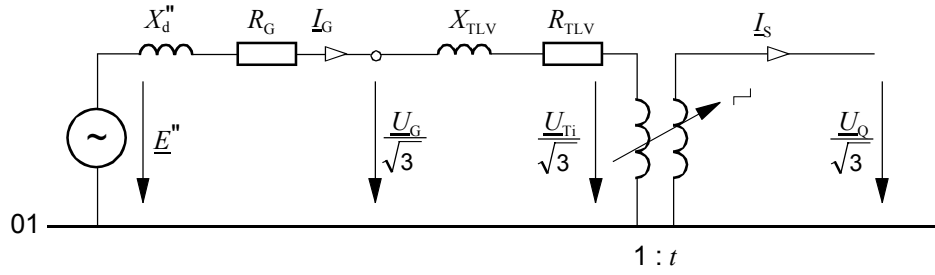
From a calculation with the superposition method (index (S)) according to equation (18) the short-circuit current $I_{-kG(S)}'' = (0,759 - j4,889) \text{ kA}$; $I_{kG(S)}'' = 4,948 \text{ kA}$ is found. The approximation value I_{kG}'' is about 1 % smaller than the value $I_{kG(S)}''$.

If the terminal voltage of the generator is different from U_{rG} , then it may be necessary to introduce $U_G = U_{rG}(1 \pm p_G)$ instead of U_{rG} to the equations (20), (21) and (22). Then the maximum short-circuit current is found with the voltage U_G at the upper limit $U_G = U_{rG}(1 + p_G)$ [17].

2.2.3 Correction factors for power station units with on-load tap changer

2.2.3.1 Impedance correction factor K_S

The determination of the impedance correction factor K_S (IEC 60909-0, equation (22)) for power-station units with on-load tap-changer according to figure 5 is based on the representation given in figure 6a with the transformation ratio t of the unit transformer $t \neq t_r = U_{rTHV} / U_{rTLV}$.



$$t \neq t_r = U_{rTHV} / U_{rTLV}$$

$$\underline{U}_Q = t \underline{U}_{Ti}$$

$$\underline{I}_S = (1/t) \underline{I}_G$$

Figure 6a – Equivalent circuit diagram in the positive-sequence system with $\underline{Z}_G = R_G + jX_d''$

and $\underline{Z}_{TLV} = R_{TLV} + jX_{TLV}$

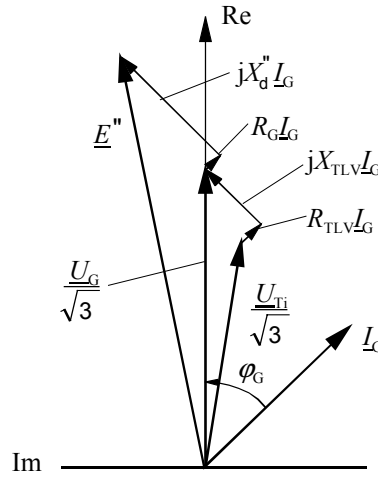


Figure 6b – Phasor diagram for overexcited operation with U_G , I_G , φ_G

Figure 6 – Simulation of a power-station unit with on-load tap-changer

Without knowing the load flow in the network, neither the voltage U_Q nor the transformation ratio $t = U_Q / U_{Ti}$ of the unit transformer ($t \neq t_r = U_{rTHV} / U_{rTLV}$) can be known.

The internal voltage $\underline{E}''$ of the generator is found from figure 6b, depending on $\underline{U}_G / \sqrt{3} = U_G / \sqrt{3}$ and $\underline{I}_G = I_G (\cos \varphi_G - j \sin \varphi_G)$ before the short circuit at the high-voltage side of the unit transformer (figure 5):

$$\underline{E}'' = \frac{\underline{U}_G}{\sqrt{3}} + \underline{I}_G \underline{Z}_G = \frac{\underline{U}_G}{\sqrt{3}} \left[1 + \frac{\sqrt{3} I_G}{U_G} (R_G \cos \varphi_G + X_d'' \sin \varphi_G) + j \frac{\sqrt{3} I_G}{U_G} (X_d'' \cos \varphi_G - R_G \sin \varphi_G) \right] \quad (26)$$

The transformation ratio can be substituted by $t = \underline{U}_Q / \underline{U}_{Ti}$ with $\underline{U}_{Ti} / \sqrt{3}$ found from the phasor diagram in figure 6b.

$$\begin{aligned} \frac{\underline{U}_{Ti}}{\sqrt{3}} &= \frac{\underline{U}_G}{\sqrt{3}} - \underline{I}_G \underline{Z}_{TLV} \\ &= \frac{\underline{U}_G}{\sqrt{3}} \left[1 - \frac{\sqrt{3} I_G}{U_G} (R_{TLV} \cos \varphi_G + X_{TLV} \sin \varphi_G) - j \frac{\sqrt{3} I_G}{U_G} (X_{TLV} \cos \varphi_G - R_{TLV} \sin \varphi_G) \right] \quad (27) \end{aligned}$$

Introducing $\underline{E}''$ from equation (26), $\underline{U}_{Ti} / \sqrt{3}$ from equation (27) and the equivalent voltage source $cU_{nQ} / \sqrt{3}$ at the short-circuit location F (equal to the connection point Q in figure 5), the following equation can be found:

$$\underline{I}_{kS(S)}'' = \frac{cU_{nQ}}{\sqrt{3}(t_r^2 \underline{Z}_G + \underline{Z}_{THV}) \underline{K}_S} \quad (28)$$

with $t_r = U_{rTHV} / U_{rTLV}$

$$\underline{Z}_G = R_G + jX_d''$$

$$\underline{Z}_{THV} = t_r^2 \underline{Z}_{TLV}$$

(without impedance correction factor K_T) and

$$\underline{K}_S = \frac{U_{nQ} U_Q}{U_G^2} \cdot \frac{U_{rTLV}^2}{U_{rTHV}^2} \cdot \frac{c}{1 + \frac{\sqrt{3} I_G}{U_G} (R_G \cos \varphi_G + X_d'' \sin \varphi_G) + j \frac{\sqrt{3} I_G}{U_G} (X_d'' \cos \varphi_G - R_G \sin \varphi_G)} \cdot \frac{1}{1 - \frac{\sqrt{3} I_G}{U_G} (R_{TLV} \cos \varphi_G + X_{TLV} \sin \varphi_G) - j \frac{\sqrt{3} I_G}{U_G} (X_{TLV} \cos \varphi_G - R_{TLV} \sin \varphi_G)} \quad (29)$$

It can be assumed that $R_G \ll X_d''$ and $R_{TLV} \ll X_{TLV}$. Equation (29) then leads to the following approximation if $U_G = U_{rG}$ is valid:

$$K_S \approx \frac{U_{nQ} U_Q}{U_{rG}^2} \cdot \frac{U_{rTLV}^2}{U_{rTHV}^2} \cdot \frac{c_{\max}}{\sqrt{1 + 2x_d'' \left(\frac{I_G}{I_{rG}} \right) \sin \varphi_G + x_d''^2 \left(\frac{I_G}{I_{rG}} \right)^2}} \cdot \frac{1}{\sqrt{1 - 2x_T \left(\frac{I_G}{I_{rG}} \right) \sin \varphi_G + x_T^2 \left(\frac{I_G}{I_{rG}} \right)^2}} \quad (30)$$

with $x_d'' = X_d'' / Z_{rG}$

$x_T = X_T / Z_{rT}$ and

$Z_{rG} = Z_{rT} = U_{rG}^2 / S_{rG}$ at $U_{rG} = U_{rTLV}$ and $S_{rG} = S_{rT}$.

The investigations for a large number of power-station units with on-load tap-changer have shown that, similar to the case of generators directly connected to the network, the short-circuit current $\underline{I}_{kS(S)}''$ reaches its maximum in many cases if the power-station unit is operated at its rated point before the short circuit [7], [11], [16], [17]. Details are given in [22]. 24 power-station units out of the 47 under consideration with $S_{rG} = 32 \text{ MVA} \dots 1\,640 \text{ MVA}$ have their maximum partial short-circuit currents if they are operated at their rated point $\underline{S}_{rG}$ before the short circuit. Further 16 power-station units if they are operated at $Q > Q_{rG}$ in the over-excited region. Only in the remaining seven cases the maximum short-circuit current is to be expected after underexcited operation. In six of these seven cases, $x_T > x_d''$ is valid ($S_{rG} = 32 \text{ MVA} \dots 147 \text{ MVA}$).

Figure 7 gives one example from these investigations with maximum short-circuit current from the generator after operation at rated conditions before the short circuit. The maximum short-circuit current $\underline{I}_{kS(S)}''$ is reached for the lowest value U_Q at the high-voltage side of the unit transformer with on-load tap-changer. It was assumed that it is sufficient to take U_{nQ} as the lowest value during normal operation.

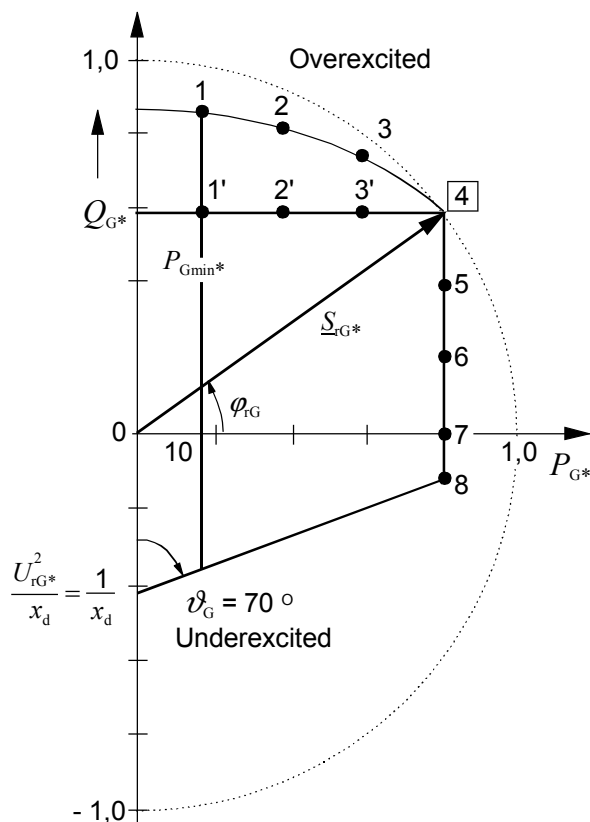


Figure 7a – Power chart of a turbine generator with limits for over- and underexcited operation

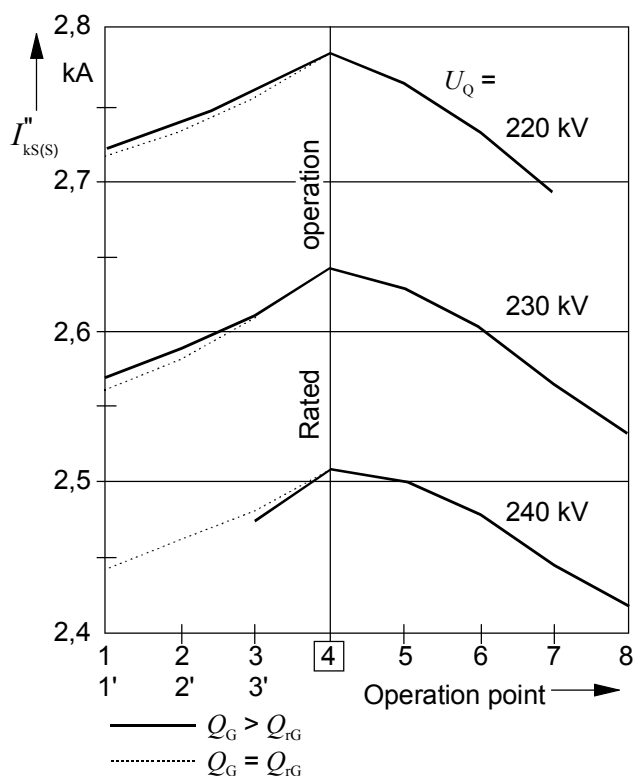


Figure 7b – Partial short-circuit currents found with the superposition method (S) from equation (28) together with K_s from equation (30)

Example: $S_{rG} = S_{rT} = 395 \text{ MVA}$
 $U_{rG} = 21 \text{ kV}$
 $\cos \varphi_{rG} = 0,8$
 $x_d'' = 0,25$
 $x_d = 2,38$

$P_{G \min} = 0,2 P_{rG}$
 $U_{rTHV} = 245 \text{ kV}$
 $U_{rTLV} = 21 \text{ kV}$
 $t = U_{rTHV} (1 \pm 0,112) / U_{rTLV}$
 $u_{kr} = 15 \% ; u_{k+} = 15,8 \% ; u_{k-} = 14,5 \% ; U_{nQ} = 220 \text{ kV}$

Figure 7 – Partial short-circuit current of a power station unit found with the superposition method

A further simplification of equation (30) leads to

$$K_S = \frac{U_{nQ}^2}{U_{rG}^2} \cdot \frac{U_{rTLV}^2}{U_{rTHV}^2} \cdot \frac{c_{\max}}{1 + |x_d'' - x_T| \sin \varphi_{rG}} \quad (31)$$

because $U_Q = U_{nQ}$ can be presumed to give the highest partial short-circuit current $I''_{ks(S)}$ (see figure 7 and other examples given in [7], [11], [16], [17] and [22]).

Because the pre-short-circuit voltage U_Q and the actual position of the on-load tap-changer are not known, IEC 60909-0 proposes the use of the following equation for the calculation of the partial short-circuit current:

$$I''_{ks} = \frac{c U_{nQ}}{\sqrt{3} (t_r^2 Z_{\underline{G}} + Z_{\underline{THV}}) K_S} \quad (32)$$

with K_S from equation (31) and $\underline{Z}_{\text{THV}}$ in equation (32) as the impedance of the unit transformer, calculated from u_{kr} , see IEC 60909-0, 3.3.1. Only if the minimum operating voltage $U_{Q \min} > U_{nQ}$ at the high-voltage side of the unit transformer is well known from long-term operating experience, then the use of the product $U_{nQ} U_{Q \min}$ instead of U_{nQ}^2 in equation (31) is allowed, see IEC 60909-0, 3.7.1.

It is assumed that the operating voltage at the terminals of the generator is equal to U_{rG} . If the voltage U_G is permanently higher than U_{rG} , then $U_{G \max} = U_{rG}(1 + p_G)$ should be introduced instead of U_{rG} , with, for instance, $p_G = 0,05$ [17].

In order to demonstrate the efficiency of the derived impedance correction factor K_S in equation (31), figure 8 gives the cumulative frequency H of the deviations Δ_S , found with the simplified calculations in accordance to IEC 60909-0 in comparison to the values found with the superposition method, for 47 power-station units with on-load tap-changer of the unit transformer [22] and [23].

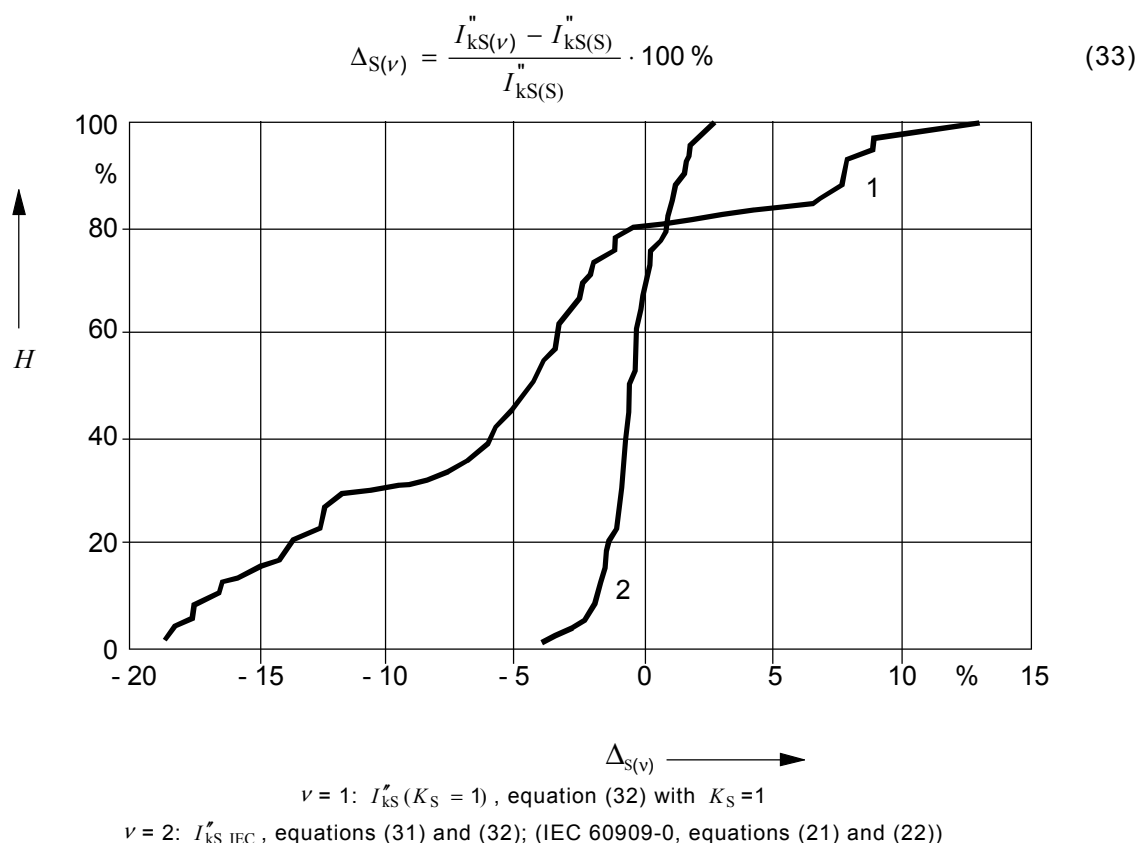


Figure 8 – Cumulative frequency H of the deviations calculated with equation (33) [22] and [23]

Curve 1 of figure 8 demonstrates the necessity for the introduction of an impedance correction factor for the impedance of power-station units together with the equivalent voltage source, because otherwise approximately 50 % of the cases would lead to results on the non-conservative (unsafe) side (deviations more than -5 %).

The simplified impedance correction factor in equation (31) leads to a sufficient approximation for the maximum partial short-circuit current $I''_{kS, IEC}$ (curve 2 in figure 8). All the results are within the region $\Delta_S = \pm 5$ %. Safety and economical aspects are sufficiently met, if K_S is used.

The partial short-circuit currents I_{kG}'' and I_{kT}'' as well as the partial short-circuit current I_{kF2}'' can be calculated with the superposition method (S) as follows:

$$I_{kG(S)}'' = I_{-G} + I_{kGU^b}'' \quad (34a)$$

$$I_{kT(S)}'' = -I_{-G} + I_{kTU^b}'' \quad (35a)$$

$$I_{kF2(S)}'' = I_{kG(S)}'' + I_{kT(S)}'' = I_{kGU^b}'' + I_{kTU^b}'' \quad (36a)$$

If $U_G / \sqrt{3}$ and $I_{-G} = I_G (\cos \varphi_G - j \sin \varphi_G)$ are introduced, the following partial short-circuit currents are found when taking reactances only instead of impedances ($R_G \ll X_d''$; $R_T \ll X_T$) [23].

$$I_{kG(S)}'' = I_G \cos \varphi_G - j \left(I_G \sin \varphi_G + \frac{U_G}{\sqrt{3} X_d''} \right) \quad (34b)$$

$$I_{kT(S)}'' = -I_G \cos \varphi_G + j \left(I_G \sin \varphi_G - \frac{U_G}{\sqrt{3} \left(X_{TLV} + \frac{1}{t^2} X_Q \right)} \right) \quad (35b)$$

$$I_{kF2(S)}'' = I_{kG(S)}'' + I_{kT(S)}'' = -j \frac{U_G}{\sqrt{3}} \left(\frac{1}{X_d''} + \frac{1}{X_{TLV} + \frac{1}{t^2} X_Q} \right) \quad (36b)$$

In the case of on-load tap-changer of the unit transformer (figure 9) the quantities X_{TLV} , t , I_G and $\cos \varphi_G$ may change. It is presupposed that U_G is equal to U_{rG} . From equation (34b) it follows under these conditions that:

$$I_{kG(S)}'' = \frac{E''}{X_d''} = \frac{U_{rG}}{\sqrt{3} X_d''} \sqrt{1 + 2x_d'' \frac{I_G}{I_{rG}} \sin \varphi_G + \left(x_d'' \frac{I_G}{I_{rG}} \right)^2} \quad (34c)$$

The calculation according to IEC 60909-0, 4.2.1.3, introduces the impedance correction factor $K_{G,S}$:

$$I_{kG}'' = \frac{c U_{rG}}{\sqrt{3} K_{G,S} Z_G} \quad [\text{IEC 60909-0, equation (35)}] \quad (37)$$

with
$$K_{G,S} = \frac{c_{\max}}{1 + x_d'' \sin \varphi_{rG}} \quad [\text{IEC 60909-0, equation (36)}] \quad (38)$$

The cumulative frequency H of the deviations $\Delta_{G(v)}$ is given in figure 10.

$$\Delta_{G(v)} = \frac{I_{kG(v)}'' - I_{kG(S)}''}{I_{kG(S)}''} \cdot 100 \% \quad (39)$$

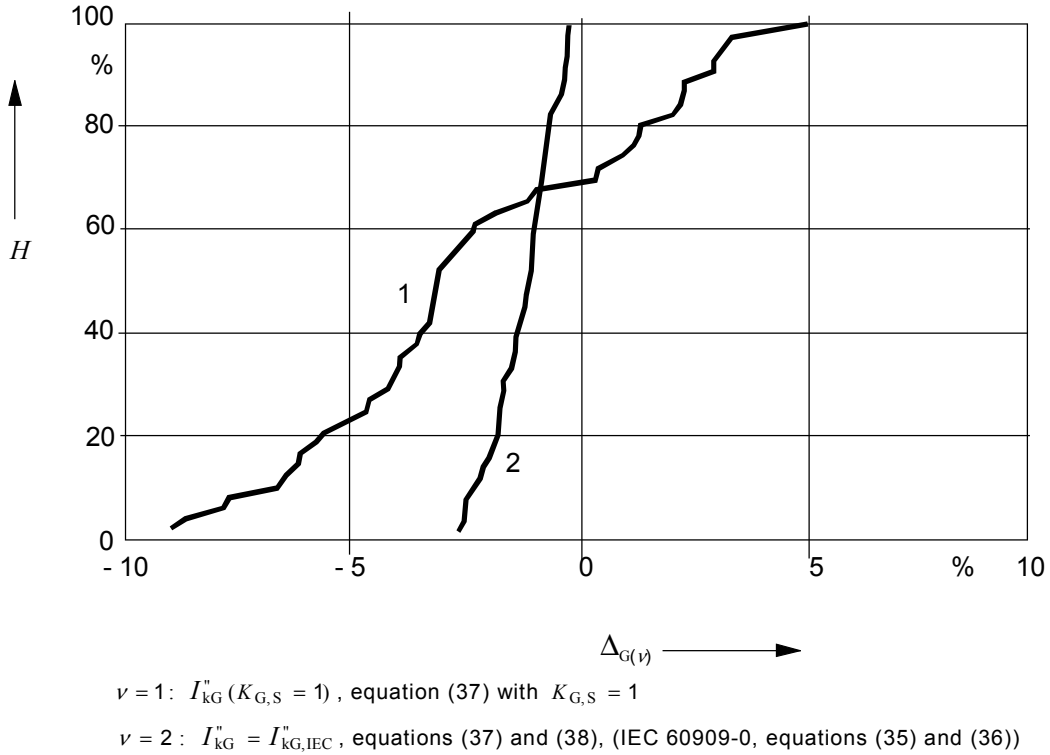


Figure 10 – Cumulative frequency H of the deviations $\Delta_{G(\nu)}$ according to equation (39) for partial short-circuit currents of generators in 47 power-station units with on-load tap changer [23]. Short-circuit location F1 in figure 9

From equation (35b), with reactances only, the following expression is found for the partial short-circuit current $I''_{kT(S)}$ [24]:

$$I''_{kT(S)} = \frac{U_{rG}}{\sqrt{3} \left(X_{TLV} + \frac{1}{t^2} X_Q \right)} \sqrt{1 - 2 \left(x_{TLV} + \frac{1}{t^2} x_Q \right) \frac{I_G}{I_{rG}} \sin \varphi_{rG} + \left[\left(x_{TLV} + \frac{1}{t^2} x_Q \right) \frac{I_G}{I_{rG}} \right]^2} \quad (40)$$

with $X_T = x_T U_{rG} / (\sqrt{3} I_{rG})$ and $X_Q = x_Q U_{rG} / (\sqrt{3} I_{rG})$.

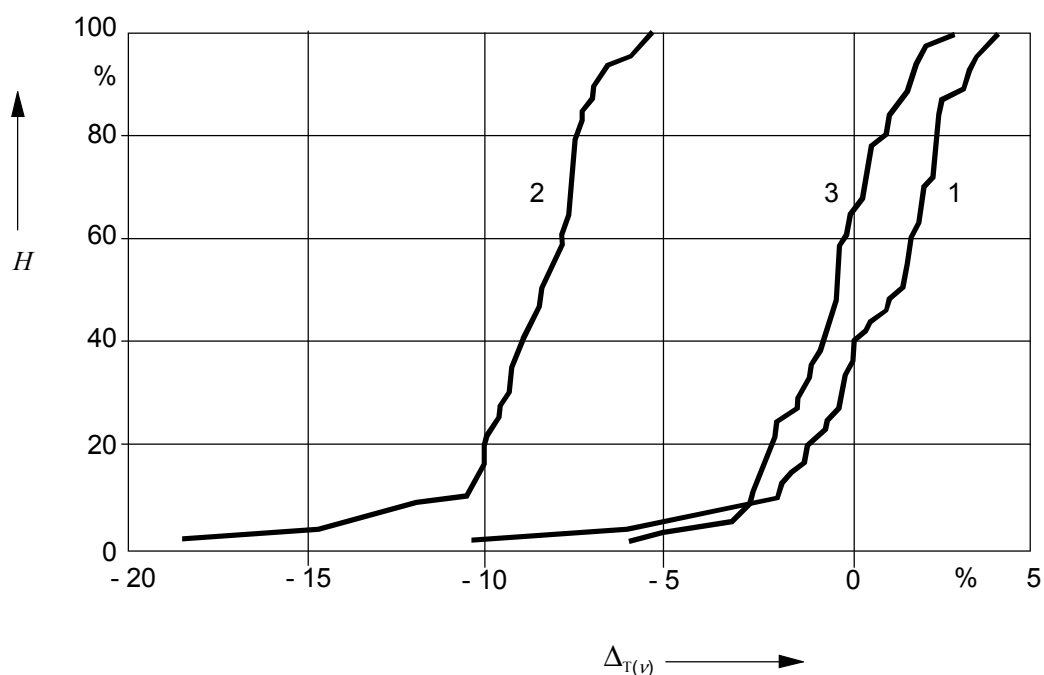
The calculation according to IEC 60909-0, 4.2.1.3, equation (37), does not introduce an impedance correction factor in this case (i.e. $K_{T,S} = 1$):

$$I''_{kT(\nu)} = \frac{c U_{rG}}{\sqrt{3} \left| Z_{TLV} K_{T,S} + \frac{1}{t^2} Z_Q \right|} \quad (41)$$

NOTE In IEC 60909:1988 [28] an impedance correction factor $K_{T,S} = 1,1$ and a fictive transformation ratio $t_f = U_{nQ}/U_{rG} \neq t_r$ had been introduced, see figure 11, curve 2.

The cumulative frequency H of the deviations $\Delta_{T(\nu)}$ is given in figure 11 for three cases.

$$\Delta_{T(\nu)} = \frac{I''_{kT(\nu)} - I''_{kT(S)}}{I''_{kT(S)}} \quad 100 \% \quad (42)$$



$\nu = 1$: $K_{T,S} = 1$; $t = t_r$ (IEC 60909-0, equation (37))

$\nu = 2$: $K_{T,S} = 1,1$; $t = t_t$ (see Note below table I of IEC 60909:1988 [28]. This is given for historic reasons)

$\nu = 3$: $*K_{T,S}$ as given in equation (43), $t = t_r$

Figure 11 – Cumulative frequency H of the deviations $\Delta_{T(\nu)}$ according to equation (42) for the partial short-circuit currents of unit transformers in 47 power station units with on-load tap-changer [23] – Short-circuit location F1 in figure 9

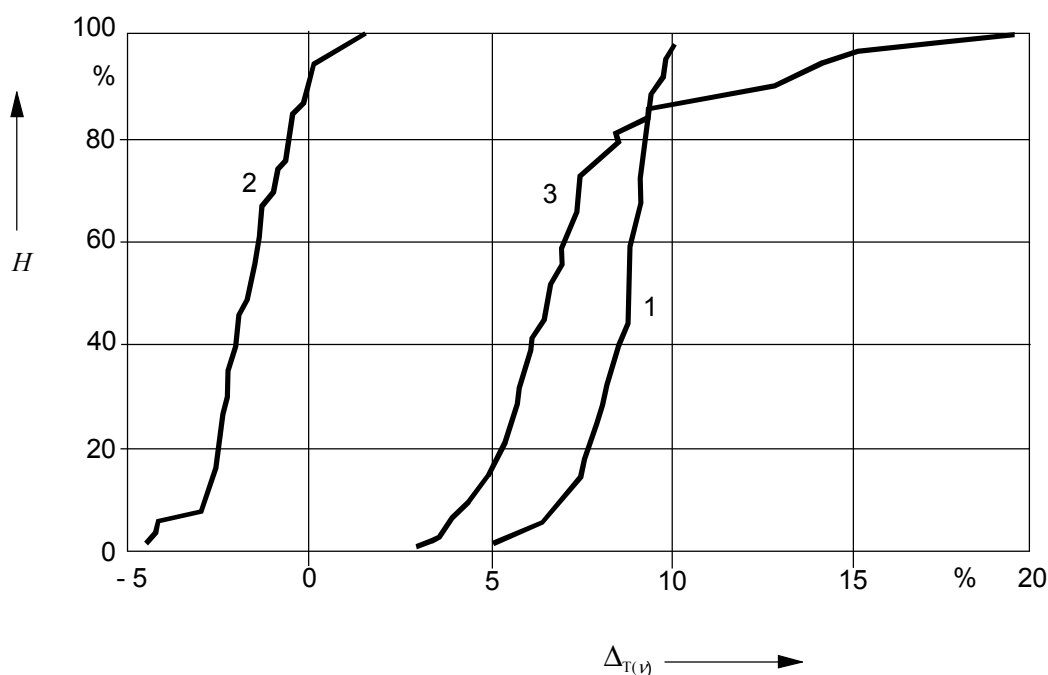
Curve 3 in figure 11 is based on calculations for 47 transformers of power-station units [23] with a representative range of short-circuit currents I_{kQmax}'' using the following impedance correction factor [23]:

$$*K_{T,S} = \frac{c_{max}}{1 + x_T / x_d} \quad (43)$$

found from equation (40) under special conditions [23] with x_T as the rated reactance of the unit transformer with on-load tap-changer related to $Z_{iT} = U_{iTLV} / (\sqrt{3} I_{iTLV})$.

The results of figure 11 are valid whether operation before the short circuit is in the overexcited or underexcited region.

In special cases, it may be possible that the generator of a power-station unit only works in the overexcited region during its lifetime. Then the results of figure 12 are found using the same boundary conditions as in figure 11.



$\nu = 1$: $K_{T,S} = 1$; $t = t_r$ (IEC 60909-0, equation (37))

$\nu = 2$: $K_{T,S} = 1,1$; $t = t_r$ (see Note below table I of IEC 60909:1988. This is given for historic reasons)

$\nu = 3$: $*K_{T,S}$ as given in equation (43), $t = t_r$

Figure 12 – Cumulative frequency H of the deviations $\Delta_{T(\nu)}$ according to equation (42), see figure 11, for the calculation of $I''_{kT(S)}$ if only overexcited operation is anticipated [23]

The partial short-circuit current I''_{kf2} flowing in the busbar to the auxiliary transformer in figure 9, if the short circuit occurs in F2, shall be calculated from IEC 60909-0, equation (38) [17]:

$$I''_{kf2} = \frac{cU_{rG}}{\sqrt{3}} \left(\frac{1}{K_{G,S} Z_G} + \frac{1}{K_{T,S} Z_{TLV} + \frac{1}{t_f^2} Z_{Q \min}} \right) = \frac{cU_{rG}}{\sqrt{3} Z_{rsI}} \quad (44)$$

with

$$K_{G,S} = \frac{c_{\max}}{1 + x_d'' \sin \varphi_{rG}} \quad (45a)$$

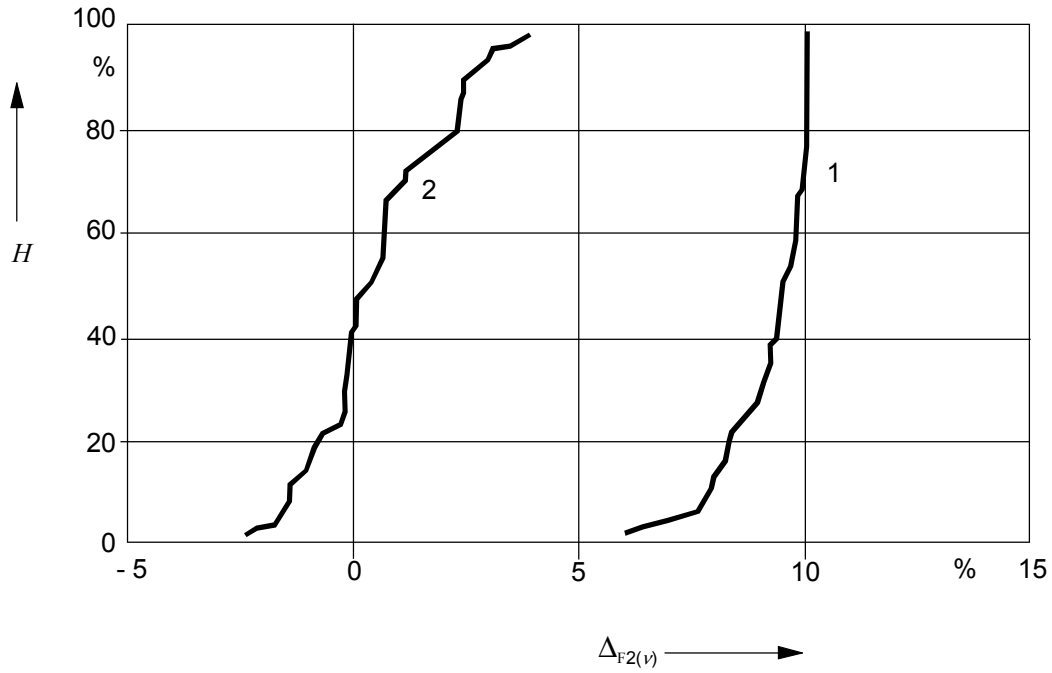
and

$$K_{T,S} = \frac{c_{\max}}{1 - x_T \sin \varphi_{rG}} \quad (45b)$$

Figure 13 gives the cumulative frequency H of the deviations $\Delta_{F2(\nu)}$ of the results found from equation (44) against the results found from the superposition method according to equation (36b):

$$\Delta_{F2(\nu)} = \frac{I''_{kf2(\nu)} - I''_{kf2(S)}}{I''_{kf2(S)}} \cdot 100 \% \quad (46)$$

The curves $\nu = 1$ and $\nu = 2$ in figure 13 are with good approximation also valid if the generator is always overexcited before the short circuit [23].



$\nu = 1$: $K_{G,S} = K_{T,\Sigma} = 1$

$\nu = 2$: $K_{G,S}$ and $K_{T,S}$ according to equation (45a) and (45b), (IEC 60909-0, equations (36) and (39))

Figure 13 – Cumulative frequency H of the deviations $\Delta_{F2(\nu)}$ according to equation (46) for the partial short-circuit current I''_{kF2} (figure 9) in the case of over- or under-excited operation before the short-circuit

The calculation of the partial short-circuit current I''_{kF3AT} in figure 9 is carried out with $\underline{Z}_{rs1}$ from equation (44) and the impedance of the auxiliary transformer AT corrected with the impedance correction factor K_T (IEC 60909-0, 3.3.3).

2.2.4 Correction factors for power station units without on-load tap-changer

2.2.4.1 Correction factor K_{SO}

The determination of the impedance correction factor K_{SO} (IEC 60909-0, equation (24)) for power-station units without (additional subscript O) on-load tap-changer similar to figure 5, is different from the determination of K_S because t is constant during operation. The maximum value of the partial short-circuit current is found if the operating voltage U_Q before the short circuit reaches its maximum. In these cases, normally U_{rTHV} is higher than U_{nQ} ; in some cases even higher than U_{mQ} [16], [17], [19], [22] and [23].

$$I''_{kSO(S)} = \frac{E''}{t_r(X_d'' + X_{TLV})} = \frac{U_G}{\sqrt{3}t_r(X_d'' + X_{TLV})} \sqrt{1 + 2x_d'' \left(\frac{U_{rG}}{U_G} \frac{I_G}{I_{rG}} \right) \sin \varphi_{rG} + x_d''^2 \left(\frac{U_{rG}}{U_G} \frac{I_G}{I_{rG}} \right)^2} \quad (47)$$

where $U_G = U_{rG}(1 \pm p_G)$, I_G and φ_G are variable and if the transformer has off-load taps $\pm p_T$, then $t = t_r(1 \pm p_T)$ should be introduced instead of t_r . Fundamental investigations are given in [17].

The calculation with the equivalent voltage source at the short-circuit location with $\underline{Z}_{SO}$ from IEC 60909-0, equation (23) leads to

$$\underline{I}_{kSO}'' = \frac{cU_{nQ}}{\sqrt{3}(t_r^2 \underline{Z}_G + \underline{Z}_{THV})K_{SO}} \quad (48)$$

with

$$K_{SO} = \frac{U_{nQ}}{U_{rG}(1 + p_G)} \frac{U_{rTLV}}{U_{rTHV}} \frac{c_{\max}}{1 + x_d'' \sin \varphi_{rG}} \quad (49a)$$

and if the transformer has off-load taps $\pm p_T$, to the correction factor according to IEC 60909-0, equation (24) [17]:

$$K_{SO} = \frac{U_{nQ}}{U_{rG}(1 + p_G)} \frac{U_{rTLV}}{U_{rTHV}} (1 \pm p_T) \frac{c_{\max}}{1 + x_d'' \sin \varphi_{rG}} \quad (49b)$$

If the highest partial short-circuit current is searched for, then $(1 - p_T)$ shall be used in equation (49b) according to IEC 60909-0, 3.7.2. Investigations are given in [17] and [19].

Figure 14 gives the cumulative frequency H of the deviations Δ_{SO} found by the simplified calculations with equation (48) in comparison with the results found with the superposition method (equation (47)) for 27 power station units without tap-changer or if taps are adjusted [22] and [23].

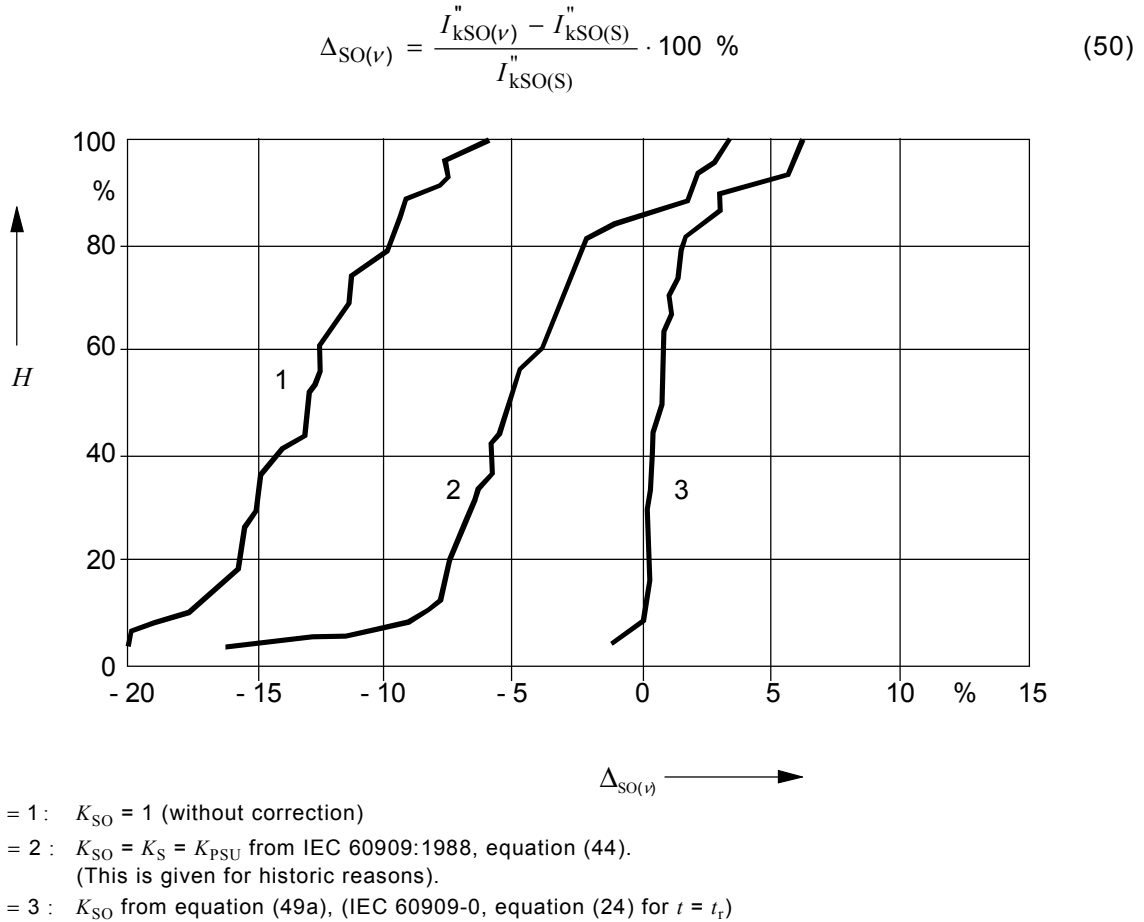


Figure 14 – Cumulative frequency H of the deviations calculated with equation (50), [22] and [23]

2.2.4.2 Correction factors $K_{G,SO}$ and $K_{T,SO}$

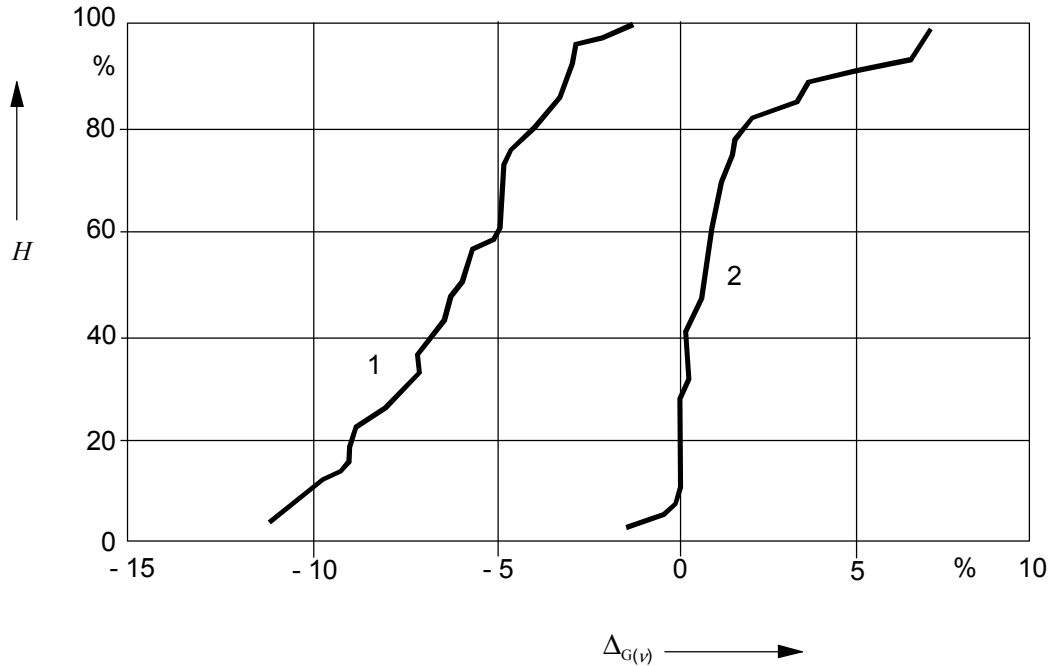
The equation for the calculation of the partial short-circuit current of a generator in a power station unit without on-load tap-changer follows from equation (34c) taking into account $U_G = U_{rG}(1 \pm p_G)$ and $X_d'' = x_d'' U_{rG} / (\sqrt{3} I_{rG})$:

$$I_{kG(S)}'' = \frac{E''}{X_d''} = \frac{U_{rG}(1 \pm p_G)}{\sqrt{3} X_d''} \sqrt{1 + 2 \frac{x_d''}{1 \pm p_G} \frac{I_G}{I_{rG}} \sin \varphi_G + \left(\frac{x_d''}{1 \pm p_G} \frac{I_G}{I_{rG}} \right)^2} \quad (51)$$

The calculation in IEC 60909-0 uses equation (40) with $K_{G,SO}$, taking into account the variable voltage at the terminals of the generator [17]:

$$K_{G,SO} = \frac{1}{1 + p_G} \cdot \frac{c_{\max}}{1 + x_d'' \sin \varphi_{rG}} \quad (52)$$

Figure 15 gives the cumulative frequency H of the deviations $\Delta_{G(v)}$ calculated with equation (39) for 27 generators of power-station units without on-load tap-changer [23].



$v = 1$: $K_{G,SO} = 1$ (without correction)

$v = 2$: $K_{G,SO}$ from equation (52), (IEC 60909-0, equation (41))

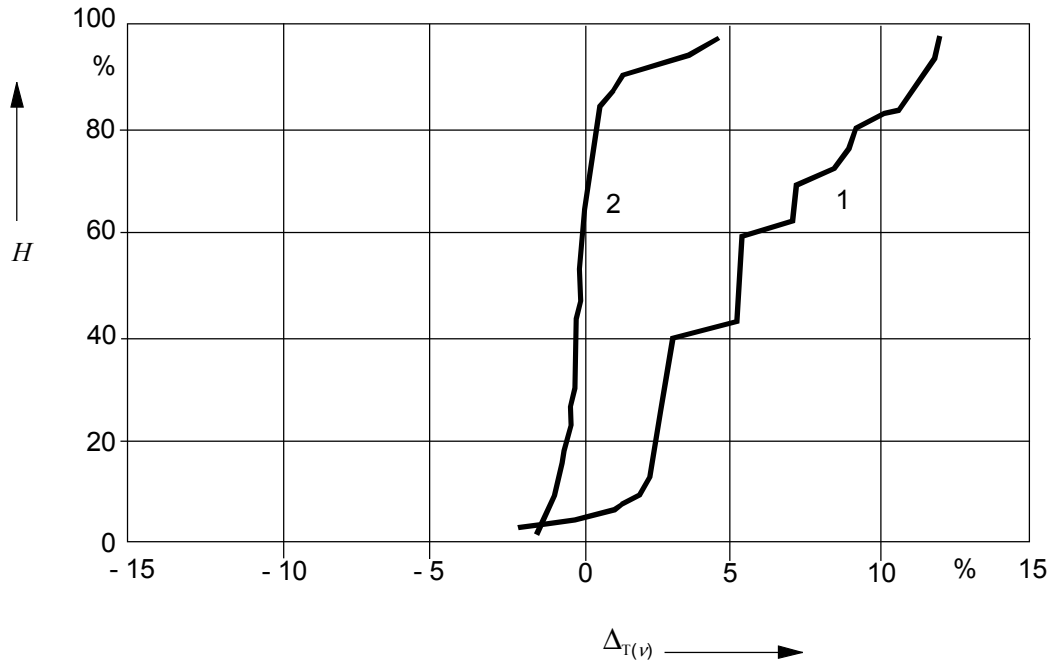
Figure 15 – Cumulative frequency H of the deviations calculated with equation (39) for 27 generators of power-station units without on-load tap-changer

The equation for the calculation of the partial short-circuit current of the unit transformer in a power station without on-load tap-changer follows from equation (35b) taking into account $U_G = U_{rG}(1 \pm p_G)$, $X_T = x_T U_{rG} / (\sqrt{3} I_{rG})$ and $X_Q = x_Q U_{rG} / (\sqrt{3} I_{rG})$, where X_T is the rated reactance of the unit transformer at the low-voltage side.

$$I_{kT(S)}'' = \frac{U_G}{\sqrt{3} \left(X_T + \frac{1}{t_r^2} X_Q \right)} \sqrt{1 - 2 \left(x_T + \frac{1}{t_r^2} x_Q \right) \frac{I_G}{(1 \pm p_G) I_{rG}} \sin \varphi_G + \left[\left(x_T + \frac{1}{t_r^2} x_Q \right) \frac{I_G}{(1 \pm p_G) I_{rG}} \right]^2} \quad (53)$$

If the maximum value is searched for, the minimum value of the network impedance in Q shall be introduced. If the transformer has off-load taps $\pm p_T$, then in equation (53) $t = t_r(1 \pm p_T)$ should be introduced instead of t_r . Fundamental investigations are given in [17].

The calculation in IEC 60909-0 uses equation (41) with $K_{T,SO} = 1$. Figure 16 gives the cumulative frequency H of the deviations $\Delta_{T(\nu)}$ (equation (42)) for 27 unit transformers without tap-changer or if taps are adjusted in the main position [23].



$\nu = 1$: $K_{T,SO} = 1$ (without correction), (IEC 60909-0, 4.2.1.4)

$\nu = 2$: $*K_{T,SO}$ from equation (54), $t = t_r$

Figure 16 – Cumulative frequency H of the deviations calculated with equation (42) for 27 unit transformers of power station units without on-load tap changer

$$*K_{T,SO} = \frac{c_{\max} U_{rTHV}}{U_{mQ}} \quad (54)$$

The introduction of the impedance correction factor from equation (54) [23] will lead to a slightly better approximation for the results found with the superposition method (see figure 16, curve 2, if $t = t_r$ is assumed).

The partial short-circuit current I_{-kF2}'' flowing in the busbar to the auxiliary transformer in figure 9, if the short circuit occurs in F2, shall be calculated with IEC 60909-0, equation (43) [17]:

$$I_{-kF2(\nu)}'' = \frac{c U_{rG}}{\sqrt{3}} \frac{1}{K_{G,SO} Z_G} + \frac{1}{K_{T,SO} Z_{TLV} + \frac{1}{t_r^2} Z_{Qmin}} \quad (55)$$

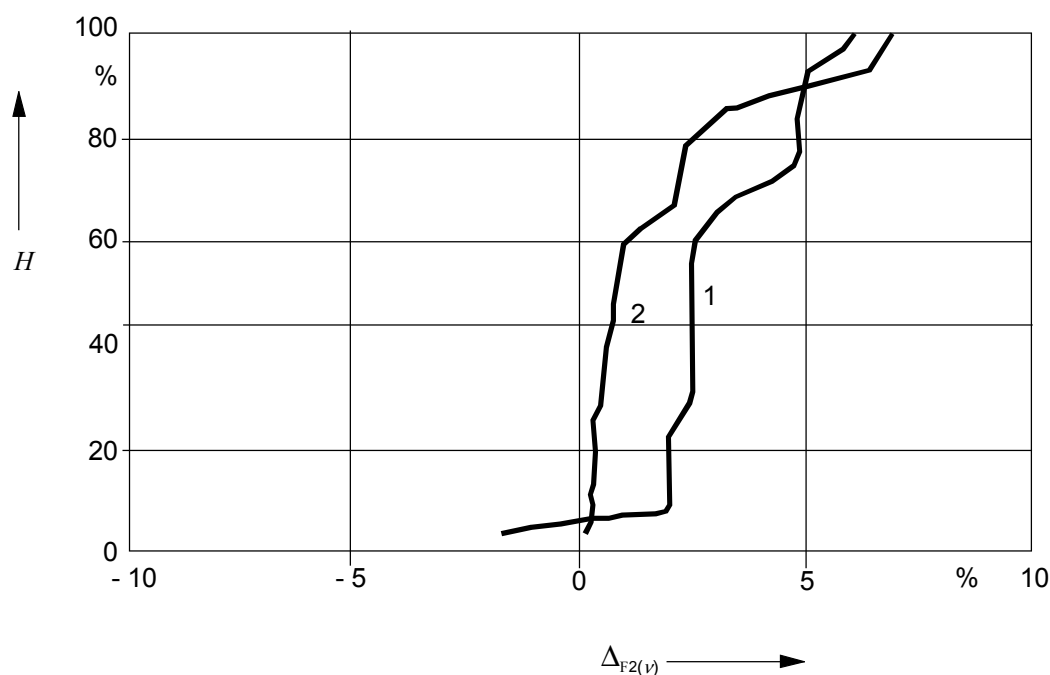
with

$$K_{G,SO} = \frac{1}{1 + p_G} \frac{c_{\max}}{1 + x_d'' \sin \phi_{rG}} \quad (56a)$$

and

$$K_{T,SO} = \frac{1}{1 + p_G} \frac{c_{\max}}{1 - x_T \sin \phi_{rG}} \quad (56b)$$

Figure 17 gives the cumulative frequency H of the deviations $\Delta_{F2(\nu)}$ (equation (46)) of the results found from equation (55) against the results found from the superposition method according to equation (36b) with $t = t_T$.



$\nu = 1$: $K_{G,SO} = 1$; $K_{T,SO} = 1$ (without correction)

$\nu = 2$: $K_{G,SO}$ and $K_{T,SO}$ from equation (56) (IEC 60909-0, 4.2.1.4)

Figure 17 – Cumulative frequency H of the deviations calculated with equation (46) for the partial short-circuit current I''_{kF2} (figure 9) of power-station units without on-load tap-changer

2.2.5 Influence of the impedance correction factor for power-station units when calculating short-circuit currents in meshed networks and maximum short-circuit currents at worst-case load flow

The aim of the IEC 60909-0 is to find a good approximation for the partial short-circuit currents of generators and power station units as well as for the short-circuit currents at different short-circuit locations in meshed networks.

As already shown in figures 8 and 14, the application of the correction factor K_S or K_{SO} leads to a good approximation of the partial short-circuit currents of power-station units with or without on-load tap-changer.

The remaining question was whether the approximation method using the equivalent voltage source $cU_n / \sqrt{3}$ with $c = c_{\max}$ (IEC 60909-0, table 1) and the impedance correction factors according to 2.2.3.1 and 2.2.4.1, also leads to results for short-circuit currents in meshed networks of sufficient accuracy in comparison to the results found with the superposition method. To answer this question, high-voltage networks with $U_n \geq 110$ kV and parts thereof were examined [7], [9], [13] and [26]. Short-circuit currents were calculated at 260 nodes using the superposition method and using the approximation method given in IEC 60909:1988 [28] with an impedance correction factor K_{PSU} according to equation (44) in IEC 60909:1988. This impedance correction factor K_{PSU} is nearly identical to the impedance correction factor K_S given in IEC 60909-0, equation (22) if the assumptions $U_Q = U_{nQ}$ and $U_G = U_{rG}$ are fulfilled.

The only difference is that the expression $x_d'' - x_T$ is now taken as an absolute value to take account of those cases where x_T is significantly higher than x_d'' . Networks with $U_n = 380$ kV ($U_m = 420$ kV) are included in these investigations though this voltage level was not within the scope of IEC 60909:1988, because a nominal system voltage belonging to U_m has not yet been standardized (IEC 60038). The inclusion seems justified because in all cases the same relation $U_m/U_n \approx 1,1$ is valid ($U_m/U_n = 123$ kV/110 kV $\approx 1,12$; $U_m/U_n = 245$ kV/220 kV $\approx 1,11$; $U_m/U_n = 420$ kV/380 kV $\approx 1,105$) (see also IEC 60909-0, footnotes in table 1).

First of all high-voltage networks with $U_n \geq 110$ kV with 260 nodes were treated to find the exact values $I_{k(S)}''$ with the superposition method and thereafter to find the approximate values of I_{kIEC}'' using the equivalent voltage source with $c = c_{max} = 1,1$ (IEC 60909-0, table 1) and the impedance correction factors K_{PSU} according to equation (44) of IEC 60909:1988. The result is given in curve 1 of figure 18 as a cumulative frequency curve of the deviations Δ with a median value of about +4 % at the conservative side. The variation between $H = 10$ % and $H = 90$ % is relatively small (figure 18, left side).

The load flow used to find $I_{k(S)}''$ from the superposition method was a particular load flow for the network, heavy loaded during winter. As already stated in 2.1.2, this one distinct load flow cannot lead to the maximum possible short-circuit currents at each node of the network. To find out to what extent the maximum possible short-circuit currents I_{kMAX}'' , calculated with a worst-case load flow, may exceed the short-circuit currents of curve 1 in figure 18 (left side), a special optimization method has been developed with the help of linear programming [9], [13] and [26]. The results are indicated as I_{kMAX}'' . The variables, or boundary conditions, in this optimization procedure are the permissible voltage band (voltage between U_n and $1,1U_n \approx U_m$), the variation of the transformation ratio of transformers with on-load tap changers, the dependency of the short-circuit voltage or reactance of these transformers from the tap-changer position $x_T = f(s)$, the underexcited region of generators and the possible ranges of power at the connection points of generators or power-station units and loads.

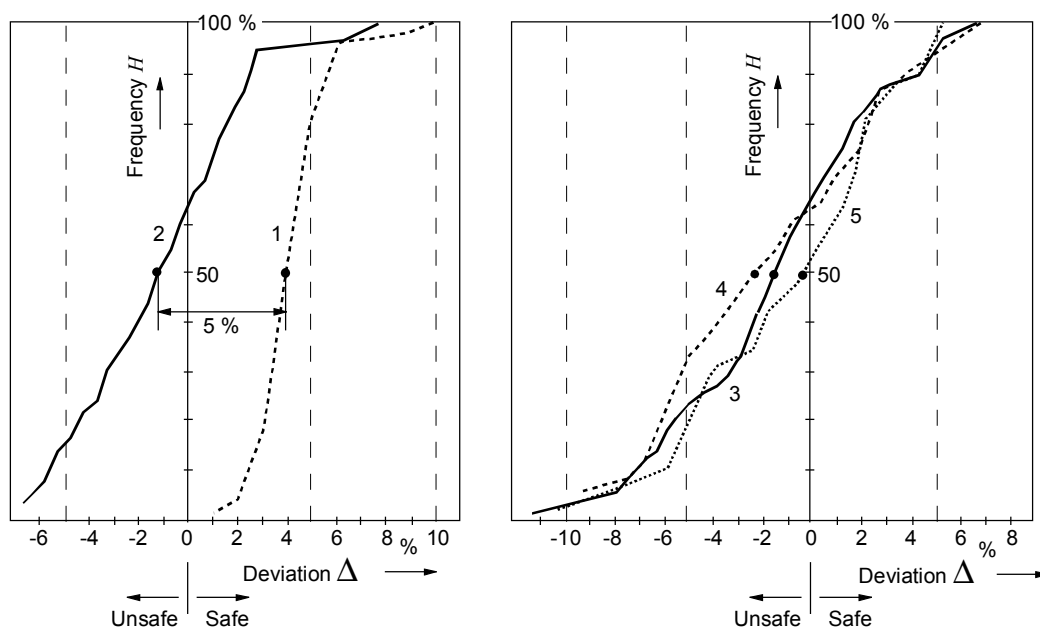
The results are given in figure 18 as a cumulative frequency H of the deviations Δ related to the maximum possible short-circuit current I_{kMAX}'' at worst-case load flow for every node of the network.

The following conclusions may be drawn from figure 18.

- The median value (50 % value) of the cumulative frequency curve for the deviations of the initial symmetrical short-circuit currents I_{kIEC}'' related to I_{kMAX}'' , as shown in curve 2 of figure 18, is about 5 % higher than the results found from the normal superposition method (curve 1 in figure 18) using constant reactances of the transformers: $u_k = u_{kr} = \text{constant}$, independent of the position of the tap-changer.
- There are only small differences between the median values of the curves 2 and 3 in figure 18. Curve 3 considers the changing of the short-circuit voltage u_k of the transformers with tap-changer depending on tap-changer position.
- The restriction of the area of underexcited operation of the generator reduces the highest possible short-circuit currents in general, as can be seen from the shifting of curve 4 to curve 5 in figure 18.

The investigations in 2.2.2 up to 2.2.4 and the results of figure 18 show that the introduction of impedance-correction factors for generators and power-station units when calculating short-circuit currents with the equivalent voltage source leads to sufficient results. Only the negative deviations in figure 18 are not satisfactory.

It will be shown in 2.3 that it was necessary for the revision of IEC 60909 (IEC 60909-0) to introduce impedance correction factors for network transformers. The high negative deviations at the right side of figure 18 are related to results at short-circuit locations on the low-voltage side of network transformers if the short circuit is only fed through the transformer (see figure 19).



$$\text{Curve 1: } \Delta = \frac{I_{k \text{ IEC}}'' - I_{k(S)}''}{I_{k(S)}''} \cdot 100\% ;$$

$$\text{Curve 2 ... 5 } \Delta = \frac{I_{k \text{ IEC}}'' - I_{k \text{ MAX}}''}{I_{k \text{ MAX}}''} \cdot 100\%$$

$I_{k \text{ IEC}}''$ Initial symmetrical short-circuit current with $c_{\text{max}} = 1,1$ and K_{PSU} from equation (44) of IEC 60909:1988

$I_{k(S)}''$ Initial symmetrical short-circuit current found with the superposition method for one heavy load condition before the short circuit (see text)

$I_{k \text{ MAX}}''$ Initial symmetrical short-circuit current for the worst-case load flow before the short circuit for every point of the network [9], [13] and [26]

Curve 2: Calculation of $I_{k \text{ MAX}}''$ with $u_k = u_{kr} = \text{constant}$

Curve 3: Calculation of $I_{k \text{ MAX}}''$ with $u_k = f(s)$

Curve 4: Calculation of $I_{k \text{ MAX}}''$ for a special minor group of nodes with $u_k = f(s)$

Curve 5: Calculation of $I_{k \text{ MAX}}''$ in the same way as for curve 4 but with restricted areas of underexcited operation

Figure 18 – Cumulative frequency H of the deviations Δ [13]

2.3 Impedance correction factor K_T when calculating the short-circuit impedances of network transformers

2.3.1 General

IEC 60909:1988 [28] gave the following general information in 8.3.2.2, but only in a note, about the treatment of network transformers, which are often equipped with on-load tap-changers.

This note stated that it is sufficient for network transformers with tap-changers to determine Z_T in accordance with formula (6) [IEC 60909-0, equation (7)] for the main position and to convert the impedances, currents and voltages according to subclause 8.4 using the rated transformation ratio t_r corresponding to the tap-changer in the main position.

It also specified that special considerations are necessary only if

- a single-fed short-circuit current is calculated and the short-circuit current has the same direction as the operational current before the short circuit occurs (short circuit on the low-voltage side of one transformer or parallel transformers with tap-changers according to figure 3, or figure 6b);
- it is possible to change the transformation ratio of a transformer with the tap-changer in a wide range, $U_{THV} = U_{rTHV}(1 \pm p_T)$ with $p_T > 0,05$;
- the minimum short-circuit voltage $u_{k \min}$ is considerably lower than the rated short-circuit voltage in the main position ($u_{k \min} < u_{kr}$);
- the voltage during operation is considerably higher than the nominal system voltage ($U \geq 1,05 U_n$).

This note was neither relevant for the standardized procedure given in IEC 60909:1988, nor gave any help for the user of the standard because at least one of the conditions given in the note often applies. To improve this unsatisfactory situation, a statistical research was carried out based on data collected from about 150 network transformers [19] and [25] with on-load tap changer.

This investigation had the aim of improving the standard procedure with the equivalent voltage source at the short-circuit location by introducing an impedance-correction factor for network transformers, taking into account the viewpoints of both safety and economy [19].

Figure 19 describes the configuration for the calculation of the initial symmetrical short-circuit current $I''_{kT(S)}$ with the superposition method (S).

To describe the modelling of the network transformer and the dependency of the magnitude of the short-circuit current from the voltage relation, the load-flow condition before the short circuit, the short-circuit voltage $u_k(t)$ and the short-circuit power $S''_{kQ} = \sqrt{3}U_{nQ}I''_{kQ}$ or the short-circuit current I''_{kQ} of the network feeder, an example is given in 2.3.2 for a 300-MVA-transformer with on-load tap-changer at the high-voltage side.

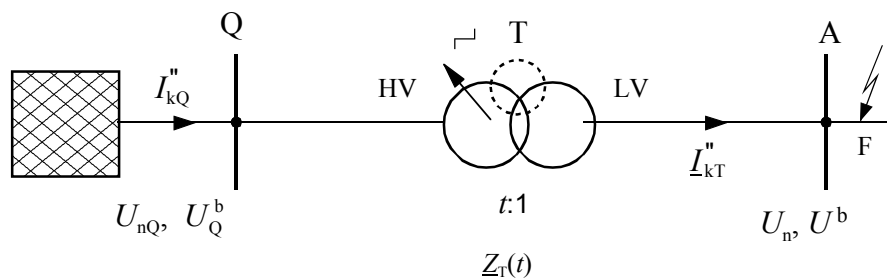
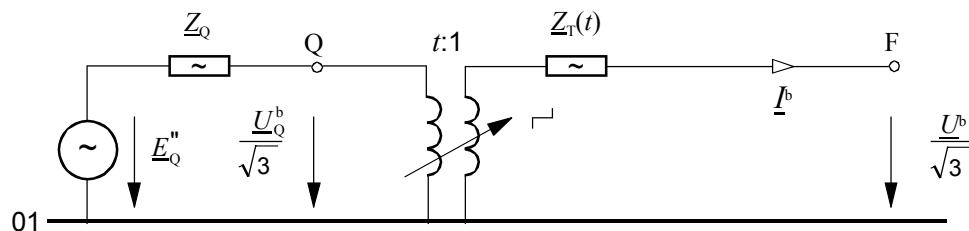
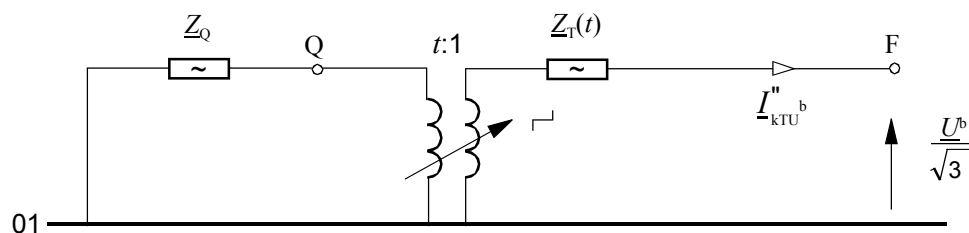


Figure 19a – System diagram, single-fed short circuit

Figure 19b – Current I^b and voltage $U^b/\sqrt{3}$ before the short circuitFigure 19c – Introduction of the pre-short-circuit voltage $U^b/\sqrt{3}$ in the opposite direction at the short-circuit locationFigure 19 – Calculation of $I_{kT(S)}'' = I^b + I_{kTU}''$ with the superposition method [19] and [25]

The statistical evaluation for about 150 network transformers is given in 2.3.3 for the configuration of single-fed short circuits at the low-voltage side (figure 19). This configuration has been used for the development of the impedance correction factor K_T [19] and [25].

2.3.2 Example for a network transformer $S_{rT} = 300$ MVA

For this example the following data are given:

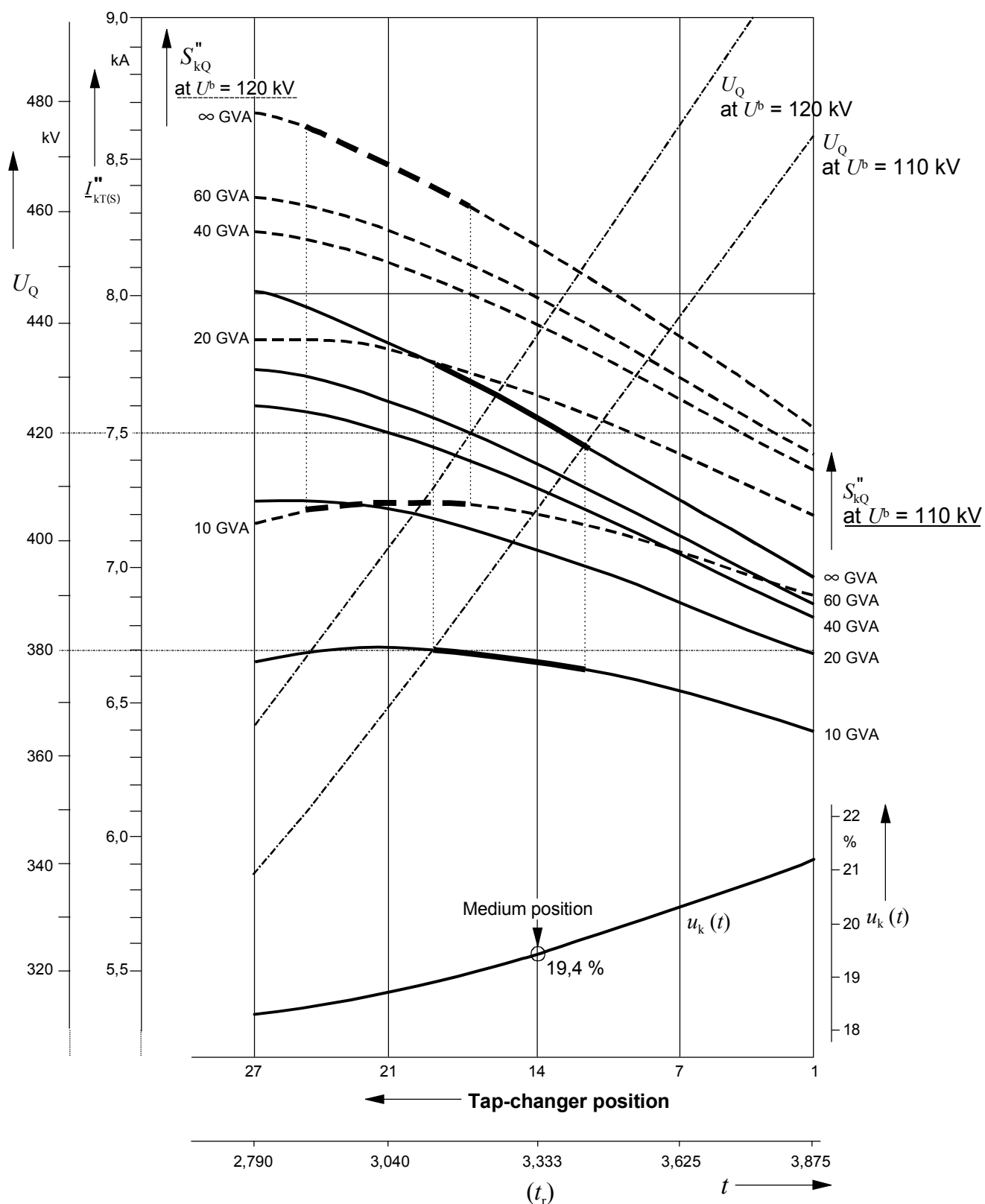
$S_{rT} = 300$ MVA; $U_{rTHV} / U_{rTLV} = 400$ kV/120 kV; $p_T = \pm 16,25$ % with ± 13 taps (27 taps);

$u_{kr} = 19,4$ %; $u_{k+} = 21,1$ % (tap 1); $u_{k-} = 18,2$ % (tap 27).

From figure 19 the following equation is found for the calculation of $I_{kT(S)}''$ with the superposition method:

$$I_{kT(S)}'' = I^b + I_{kTU}'' = I^b \cos \varphi^b - j \left(I^b \sin \varphi^b + \frac{U^b}{\sqrt{3}(X_T(t) + X_{Qt})} \right) \quad (57)$$

The resistances R_T and R_{Qt} are small compared with the reactances and can be neglected.



NOTE (Bold lines for the admissible region $U_Q = 380$ kV ... 420 kV, calculated with equation (58).)

Figure 20 – Short-circuit currents $I''_{kT(S)}$ depending on t , U^b and S''_{kQ} for the network transformer $S_{rT} = 300$ MVA (data see text)

$$U_Q^b = tU^b \sqrt{1 + 2x_T(t) \frac{U_{rTLV}}{U^b} \frac{I^b}{I_{rTLV}} \sin \varphi^b + \left(x_T(t) \frac{U_{rTLV}}{U^b} \frac{I^b}{I_{rTLV}} \right)^2} \quad (58)$$

t is the transformation ratio ($t \neq t_r = U_{rTHV} / U_{rTLV}$) and $x_T(t)$ is the relative value of the short-circuit reactance with $x_T(t) = X_T(t) / Z_{rT}$ and $Z_{rT} = U_{rTLV} / (\sqrt{3} I_{rTLV})$, depending on the transformation ratio t .

Figure 20 gives the short-circuit currents found with equation (57). The short-circuit power at the connection point Q varies between 10 GVA and ∞ GVA for voltages $U^b = 110$ kV and $U^b = 120$ kV ($U_{\max}^b \leq U_m = 123$ kV, IEC 60038). The results of figure 20 are found under the conditions $I^b = I_{rTLV}$ and $\cos \varphi^b = 0,9$. The region of the possible short-circuit currents between 6,7 kA and 8,6 kA in figure 20 depends on the boundary conditions.

Following the method with the equivalent voltage source at the short-circuit location and introducing an impedance-correction factor K_T leads to equation (59):

$$I_{kT}'' = \frac{cU_n}{\sqrt{3} |Z_{rT} K_T + Z_{Qt}|} \approx \frac{cU_n}{\sqrt{3} (X_T K_T + X_{Qt})} \quad (59)$$

with

$$X_T = X_{TLV} = \frac{u_{Xr}}{100\%} \cdot \frac{U_{rTLV}^2}{S_{rT}} \quad (60)$$

and

$$X_{Qt} = \frac{c_Q U_{nQ}}{\sqrt{3} I_{kQ}''} \cdot \frac{U_{rTLV}^2}{U_{rTHV}^2} \quad (61)$$

To find an adequate expression for K_T , it is useful to compare the imaginary part of $I_{kT(S)}''$ with I_{kT}'' from equation (59). With $c = c_{\max}$ for the maximum short-circuit current, the following expression is found on the conservative side [19]:

$$K_T = \frac{U_n}{U^b} \cdot \frac{X_T(t)}{X_T} \cdot \frac{c_{\max}}{1 + \frac{I^b}{U^b / \sqrt{3}} X_T(t) \sin \varphi^b} \quad (62)$$

From the data collected from about 150 network transformers [19] a quotient $X_T(t) / X_T \approx 0,92 \dots 1,12$ and a quotient $U_{rTLV} / U_n = 1,0 \dots 1,125$ with an average value of 1,075 is found. These investigations lead to the following simplification of equation (62), written in the form given in IEC 60909-0, equation (12b):

$$K_T = \frac{U_n}{U^b} \cdot \frac{c_{\max}}{1 + x_T (I_T^b / I_{rT}) \sin \varphi_T^b} \quad (63)$$

with $x_T = X_T / Z_{rT}$ at $Z_{rT} = U_{rTLV} / (\sqrt{3} I_{rTLV})$ and U^b as the highest operating voltage before the short circuit and I_T^b as the highest current of the transformer before the short circuit depending on the network configuration and the relevant reliability philosophy. Equation (63) is identical with IEC 60909-0, equation (12b). The factor $c_{\max}$ is related to the nominal voltage of the network connected to the low-voltage side of the network transformer.

Figure 21a gives the deviations Δ_{NT} found with I_{kT}'' from equation (59) and $K_T = 1$ and with $I_{kT(S)}''$ found from equation (57).

$$\Delta_{NT} = \frac{I_{kT}'' - I_{kT(S)}''}{I_{kT(S)}''} 100 \% \quad (64)$$

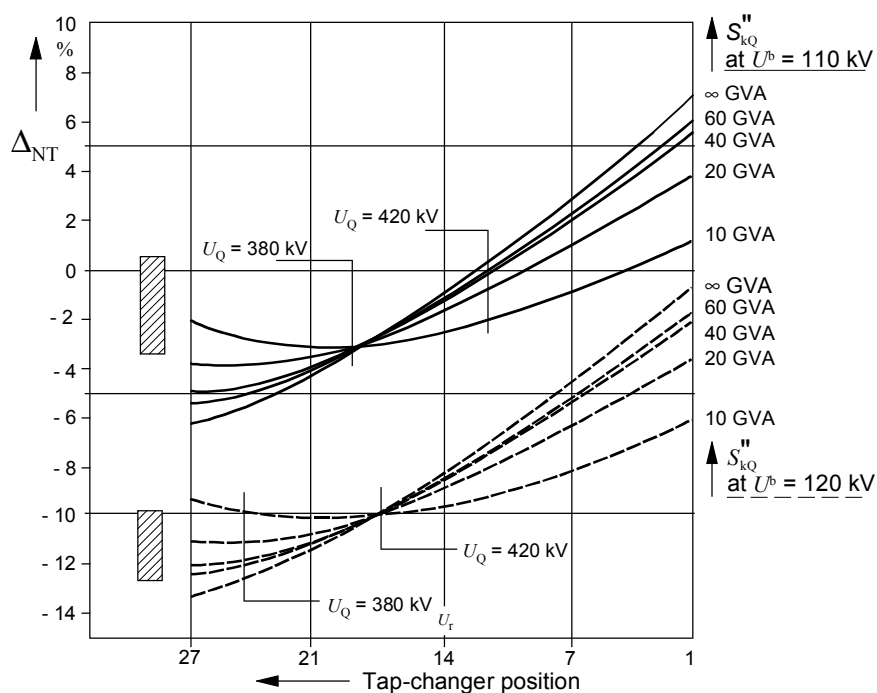


Figure 21a – Correction factor $K_T = 1$

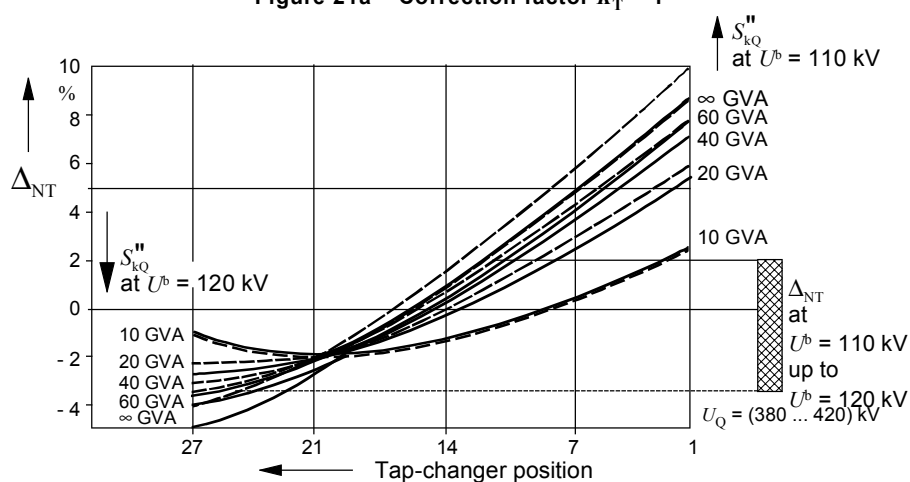


Figure 21b – Correction factor from equation (63) with $I_T^b / I_{rT} = 1$ and $\cos \varphi^b = 0,8$

Figure 21 – Deviations Δ_{NT} calculated with equation (64) for the transformer $S_{rT} = 300$ MVA

From figure 21a with $K_T = 1$ it can be seen that the voltage U^b has an essential influence and may lead to results on the non-conservative side outside the lower border of -5% , if the voltage U^b is higher than $U_n = 110$ kV.

Figure 21b gives the deviations after the introduction of K_T from equation (63). The deviations now are small, if the voltage at the high-voltage side of the transformer has a range of 380 kV ... 420 kV.

2.3.3 Statistical examination of 150 network transformers

A statistical examination is carried out for about 150 network transformers with on-load tap changer. Data are collected from public service companies and industrial plants [19]. The nominal system voltages at the high-voltage side of the transformers are between 20 kV and 380 kV, the rated apparent power between 10 MVA and 600 MVA. The following ratios and data are found:

$$U_{rTHV}/U_{nQ} = 0,91 \dots \underline{0,987} \dots 1,105$$

$$U_{rTLV}/U_n = 1,0 \dots \underline{1,075} \dots 1,125$$

$$u_{kr} = 7,2 \dots 19,0 \% \text{ [IEC 60909-2: } u_{kr} = 7 \% \dots \underline{12 \%} \dots 15 \%]$$

in case of $S_{rT} = 10 \text{ MVA} \dots 40 \text{ MVA}$

Only network transformers in the UK have extremely high values of u_{kr} up to 32 %; in these cases the second term of equation (63) is of special interest.

Table 1 gives the boundary conditions I and II for the statistical evaluation.

Table 1 – Voltages and currents before the short circuit at the low-voltage side of the network transformers

Boundary condition	$\frac{U^b}{U_n}$	$\frac{U_Q^b}{U_{nQ}}$	$\frac{I_T^b}{I_{rT}}$	$\cos \varphi^b$
I	0,95 ... 1,05	0,95 ... 1,05	0 ... 0,6 (1,0) ^a	1 ... 0,8
II	0,95 ... 1,1	1,0 ... 1,1		1 ... 0,8
^a In the case of one transformer: $I_T^b / I_{rT} = 0 \dots 1,0$ In the case of two parallel transformers: $I_T^b / I_{rT} = 0 \dots 0,6$				

The short-circuit power $S_{kQ}'' = \sqrt{3} U_{nQ} I_{kQ}''$ at Q is varied in the following ranges:

U_{nQ}	kV	20	110	220	380
S_{kQ}''	GVA	0,25 ... 1,25	1,0 ... 7,5	4,0 ... 30	7,0 ... 50

The results are given as cumulative frequency curves H of the deviations Δ_{NT} calculated with equation (64) and using the maximum values of $I_{kT(S)}''$ found with the superposition method and I_{kT}'' from equation (59) with $K_T=1$ or K_T from equation (63) with $I_T^b / I_{rT}=1$.

The difference between figure 21a and 21b is mainly related to the higher ratio U^b / U_{rTLV} on the low-voltage side of the transformer before the short circuit. Because the operation condition I in table 1 is true for most practical cases, the correction factor K_T from equation (63), equal to IEC 60909-0, equation (12b), should give sufficient results.

If the operating conditions U^b , I^b and $\cos \varphi^b$ are unknown, for instance in the planning stage, the following expression for K_T may give sufficient results [19] and [23]:

$$K_T = 0,95 \frac{c_{\max}}{1 + 0,6x_T} \quad (65)$$

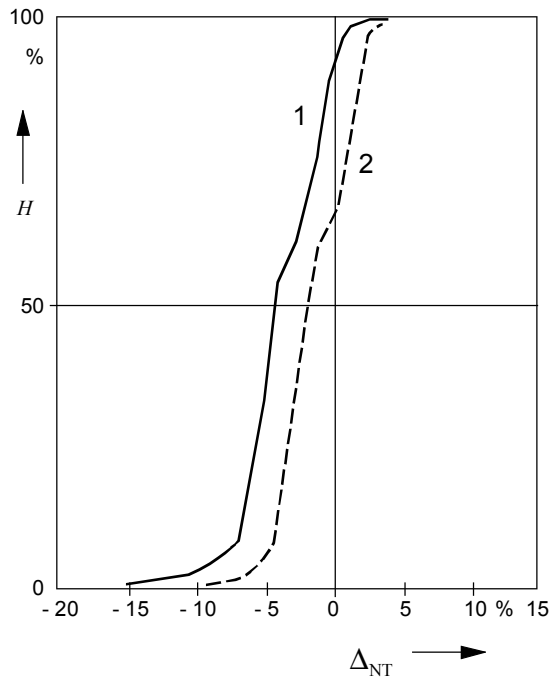


Figure 22a – Operating condition I (table 1)

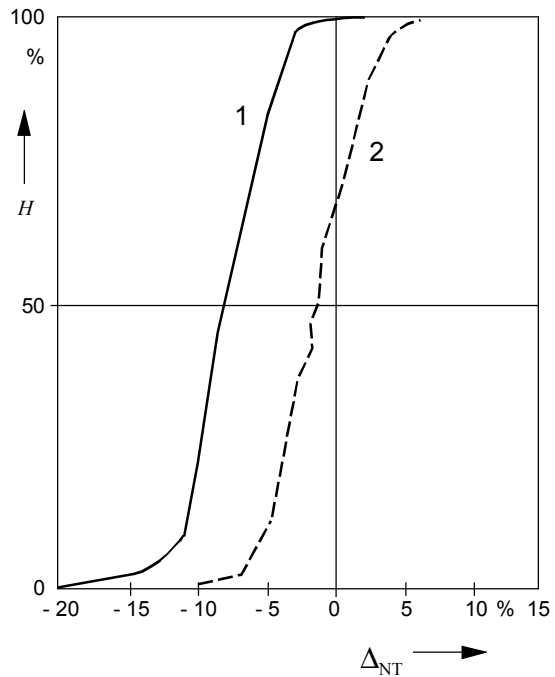


Figure 22b – Operating condition II (table 1)

Figure 22 – Cumulative frequency H of the deviations Δ_{NT} calculated with equation (64)
1: $K_T = 1,0$; 2: K_T according to equation (63) with $I_T^b / I_{rT} = 1$

The correction factor from equation (65), given in IEC 60909-0, equation (12a), shall be used for network transformers with on-load tap-changer as well as for network transformers with a constant transformation ratio, for instance in the case of low-voltage transformers $U_{rTHV} / U_{rTLV} = 20 \text{ kV} / 0,41 \text{ kV}$ or $U_{rTHV} / U_{rTLV} = 20 \text{ kV} / 0,42 \text{ kV}$ feeding low-voltage networks with $U_n = 400 \text{ V}$ [23]. The factor $c_{\max}$ is related in every case to the nominal voltage of the network connected to the low-voltage side of the network transformer.

2.3.4 Impedance correction factors for network transformers in meshed networks

An impedance correction factor for network transformers when calculating short-circuit currents with the equivalent voltage source at the short-circuit location shall be able to reduce the negative deviations calculated in comparison with I_{kMAX}'' without a general increase of the short-circuit currents in the whole network [19].

The statistical evaluation examines only network transformers [19] and [25]. Unit transformers are dealt together with power-station units (see 2.2). The statistical calculation dealing with the effect of K_T in meshed networks was carried out with those impedance correction factors K_T , which had already shown a sufficient effect during the calculations for single fed short-circuit currents at the low-voltage side of network transformers [19]. The results when using K_T according to equation (63) with $\cos \varphi^b = 0,8$ ($\sin \varphi^b = 0,6$), $I_T^b / I_{rT} = 1$ and $U_n / U^b = 0,95$ (operation condition I in table 1: $U^b / U_n = 0,95 \dots 1,05$) are incorporated in the results of figure 22a (see equation (65)).

2.4 Factor κ for the calculation of the peak short-circuit current

2.4.1 General

The factor κ is used for the calculation of the peak short-circuit current i_p . The basic equation is equation (54) of IEC 60909-0:

$$i_p = \kappa \sqrt{2} I_k'' \quad (67)$$

In the case of a single-fed far-from-generator short circuit the factor κ is given as a function of R/X or X/R in figure 15 of IEC 60909-0, or can be calculated with the following numerical approximation:

$$\kappa = \kappa_{IEC} = 1,02 + 0,98 e^{-3R/X} \quad (68)$$

$\kappa = \kappa_{IEC}$ is valid for both 50 Hz systems and 60 Hz systems. The factor κ shall lead to the highest possible instantaneous value of the short-circuit current; therefore, it is assumed that the short circuit starts at zero voltage and i_p is reached approximately 10 ms (in 50 Hz systems) or 8,33 ms (in 60 Hz systems) after the beginning of the short circuit (see 2.4.2 and 2.4.3).

In the case of a near-to-generator or near-to-motor short circuit, the symmetrical a.c. component of the short-circuit current is also decaying. To account for this effect during the first 10 ms (or 8,33 ms) after the beginning of the short circuit, special fictitious ratios R_{Gf}/X_d'' (IEC 60909-0, 3.6.1) and R_M/X_M (IEC 60909-0, 3.8.1) are introduced. These ratios are considerably higher than the natural ratios of R/X of the synchronous and asynchronous machines. These fictitious ratios are found from measurements and calculations.

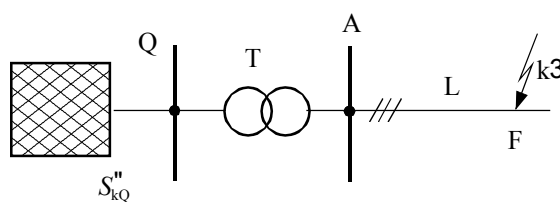
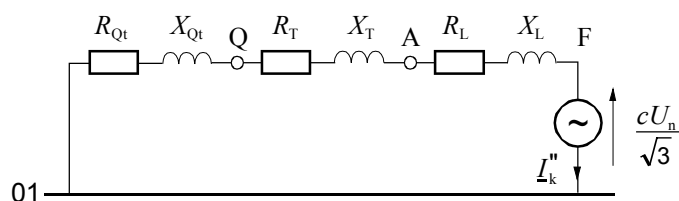
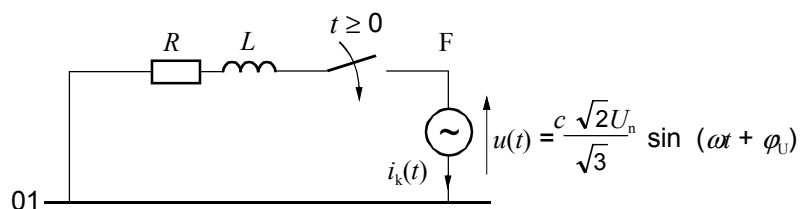
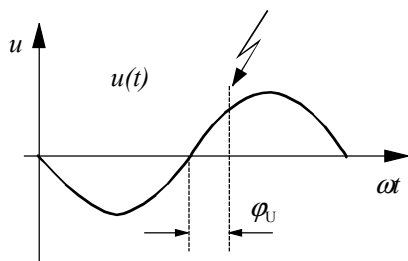
In the case of three-phase short circuits fed from non-meshed sources (see for instance IEC 60909-0, figure 12) the peak short-circuit current at the short-circuit location is found as the sum of the peak short-circuit currents of the different branches (IEC 60909-0, 4.3.1.1).

In meshed networks with parallel branches and combinations of parallel and series branches, (IEC 60909-0, figure 14) the calculation of the peak short-circuit current i_p is performed with one of the three methods given in IEC 60909-0, 4.3.1.2. Details of these three methods and their applications are given in 2.4.4.

The guiding principle for the methods a), b) and c) to calculate κ and i_p is the requirement that the methods be applicable both for hand calculation and for calculations with a suitable digital programme (see IEC 60909-4, clause 6). Furthermore, the results should not differ more than 5 % from the exact value on the non-conservative side. In this concept the viewpoint of safety is the main goal, but economic aspects also have to be taken into account.

2.4.2 Factor κ in series R-L-circuits

In the case of a single-fed three-phase short circuit (near-to-generator or far-from-generator short circuit) i.e. in a series R-L-circuit, i_p is calculated in accordance with IEC 60909-0, 4.3.1.1. To demonstrate this, the example in figure 23 is given with the assumption $Z_{Qt} \ll Z_T + Z_L$.

Figure 23a – Example with the assumption $Z_{Qt} \ll Z_T + Z_L$ Figure 23b – Equivalent circuit diagram of the positive-sequence system for the calculation of I_k'' Figure 23c – Equivalent circuit diagram of the positive-sequence system for the calculation of i_p with $R = R_{Qt} + R_T + R_L$ and $L = L_{Qt} + L_T + L_L$, $\omega = 2\pi f$, $f = 50$ Hz or 60 HzFigure 23d – Definition of the voltage angle φ_U Figure 23 – Calculation of the factor κ in the case of a single-fed three-phase short circuit (series R-L-circuit)

From figure 23c the following differential equation for the calculation of i_k is found ($t \geq 0$):

$$Ri_k + L \frac{di_k}{dt} = \frac{c\sqrt{2}U_n}{\sqrt{3}} \sin(\omega t + \varphi_U) \quad (69)$$

The solution of equation (69) is:

$$i_k = \sqrt{2} I_k'' [\sin(\omega t + \varphi_U - \gamma) + e^{-\frac{R}{X} \omega t} \sin(\gamma - \varphi_U)] = \sqrt{2} I_k'' \kappa \quad (70)$$

where

I_k'' is the initial three-phase short-circuit current (IEC 60909-0, equation (29)),

$$I_k'' = \frac{c U_n}{\sqrt{3} |Z_k|} \text{ with } Z = R + jX = Z_k;$$

Z is the short-circuit impedance, $Z = Z_k = \sqrt{R^2 + X^2}$;

γ is the impedance angle, $\gamma = \arctan \frac{X}{R}$;

φ_U is the voltage angle at the beginning of the short circuit (see figure 23d).

i_k in equation (70) is composed of a periodic term i_{kAC} and a decaying aperiodic term $i_{kDC} = i_{DC}$ (IEC 60909-0, figure 1):

$$i_k = i_{k,a.c.} + i_{k,d.c.} = \sqrt{2} I_k'' \sin(\omega t + \varphi_U - \gamma) + \sqrt{2} I_k'' e^{-\frac{R}{X} \omega t} \sin(\gamma - \varphi_U) \quad (71)$$

At the beginning of the short circuit ($t = 0$), the short-circuit current is equal to zero: $i_k(t = 0) = i_{kAC}(t=0) + i_{kDC}(t = 0) = 0$.

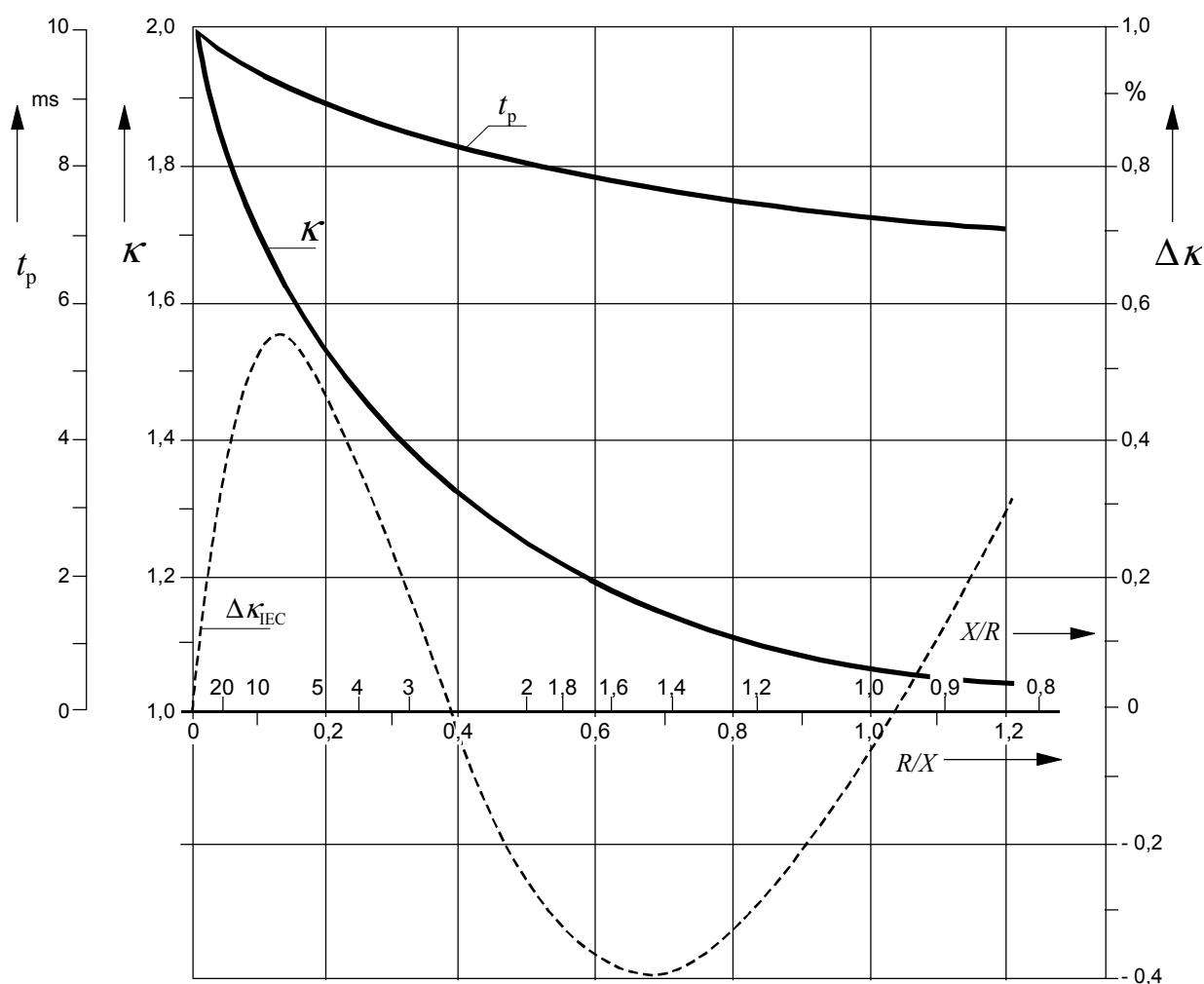
Only in the special case of $R/X = 0$, the peak value of the short-circuit current i_k is reached exactly 10 ms (in 50 Hz systems) or 8,33 ms (in 60 Hz systems) (see figure 24) after the beginning of the short circuit at zero voltage ($\varphi_U = 0$). With $R/X > 0$ the peak short-circuit current i_p occurs at a time $t_p < 10$ ms or $t_p < 8,33$ ms.

The maximum of i_p is found by partial differentiation of κ from equation (70) with respect to t and φ_U . From this procedure $\varphi_U = 0$ is found independent of the ratio R/X and therefore $\kappa = \kappa_{\max}$ from equation (70):

$$\kappa = \sin(\omega t - \gamma) + e^{-\frac{R}{X} \omega t} \sin \gamma \quad (72)$$

The factor κ and the time t_p (when the peak short-circuit current occurs) is found from equation (72) by an iteration process. Figure 24 gives the exact values of κ depending on R/X or X/R . Furthermore the time $t_p = t_{p50}$ is given in the case of nominal frequencies 50 Hz ($t_{p60} = t_{p50}$ 50 Hz/60 Hz).

The deviation $\Delta \kappa_{IEC}$ from the exact value κ is less than approximately 0,6 % as long as $0 \leq R/X \leq 1,2$ (figure 24).



$$\text{Deviation } \Delta\kappa_{\text{IEC}} = \frac{\kappa_{\text{IEC}} - \kappa}{\kappa} \cdot 100 \% ; \kappa_{\text{IEC}} \text{ according to equation (68)}$$

Figure 24 – Factor κ and t_p ($f = 50$ Hz) as a function of R/X or X/R

In the case of $\varphi_U \neq 0$ a factor $\kappa(\varphi_U \neq 0) < \kappa$ is found. Considering these cases, it should be remembered that the current in a three-phase short circuit is assumed to be made simultaneously in all three poles (IEC 60909-0), and therefore it is not necessary to take a range outside $0 \leq \varphi_U \leq \pm 30^\circ$ into account. The factor $\kappa(\varphi_U = \pm 30^\circ)$ has a maximum deviation only of about -7% against $\kappa(\varphi_U = 0)$.

2.4.3 Factor κ of parallel R-L-Z branches

The impedance of a meshed network at the short-circuit location, depending on frequency, can be transformed by partial divisions to a network with parallel R-L-Z branches where each of the parallel branches represents one of the eigenvalues of the meshed network.

The simplest configuration is therefore an equivalent of two parallel branches as shown in figure 25 with the impedances $Z_I = R_I + jX_I$ and $Z_{II} = R_{II} + jX_{II}$ and the variation of the ratios R_I/X_I and R_{II}/X_{II} at various relations Z_I/Z_{II} . The possible range of the ratio R/X in the positive-sequence system of low- and high-voltage lines lies between 0,05 and ≈ 10 . The ratios R/X of transformers depend essentially on the rated apparent power S_{rT} . A range of 0,007 up to 0,4 can be found. Therefore, it seems to be sufficient to consider the range $0,005 \leq R/X \leq 10$ for the two parallel branches.

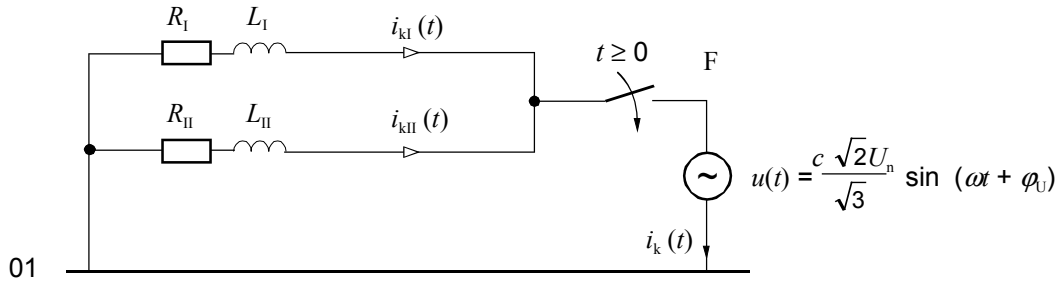


Figure 25 – Equivalent circuit diagram for the calculation of κ in the case of two parallel branches (positive-sequence system)

Taking figure 24 it is possible to derive the following equation for the calculation of the time dependent short-circuit current i_k at the short-circuit location F:

$$\begin{aligned}
 i_k &= i_{kI} + i_{kII} \\
 &= \sqrt{2} I_{kI}'' [\sin(\omega t + \varphi_U - \gamma_I) + e^{-\frac{R_I}{X_I} \omega t} \sin(\gamma_I - \varphi_U)] \\
 &\quad + \sqrt{2} I_{kII}'' [\sin(\omega t + \varphi_U - \gamma_{II}) + e^{-\frac{R_{II}}{X_{II}} \omega t} \sin(\gamma_{II} - \varphi_U)]
 \end{aligned} \tag{73}$$

The peak short-circuit current i_p is defined as

$$i_p = i_{kmax} = \kappa \sqrt{2} I_k'' \tag{74}$$

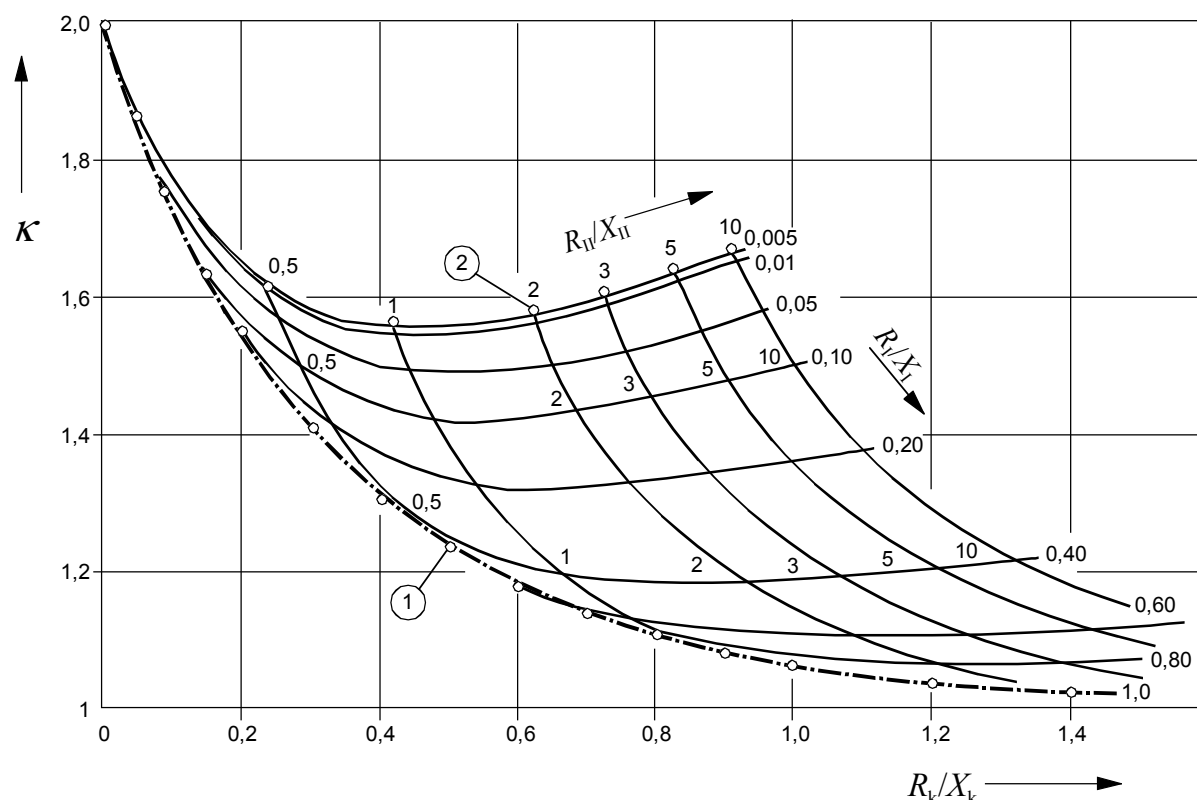
with $I_k'' = \frac{c U_n}{\sqrt{3} |Z_k|}$ and $Z_k = \frac{Z_I Z_{II}}{Z_I + Z_{II}} = R_k + jX_k$

By partial differentiation of equation (73) with respect to t and φ_U it is possible, as in 2.4.2, to demonstrate that also in this case the short-circuit current reaches its maximum at $\varphi_U = 0$.

The factor κ is:

$$\kappa = \frac{Z_k}{Z_I} \sin(\omega t - \gamma_I) + \frac{Z_k}{Z_{II}} \sin(\omega t - \gamma_{II}) + \frac{Z_k X_I}{Z_I^2} e^{-\frac{R_I}{X_I} \omega t} + \frac{Z_k X_{II}}{Z_{II}^2} e^{-\frac{R_{II}}{X_{II}} \omega t} \tag{75}$$

Figure 26 gives the factor κ for the configuration of figure 25 plotted against R_k/X_k with the parameters $0,005 \leq R_I/X_I \leq 1,0$ and $0,005 \leq R_{II}/X_{II} \leq 10,0$ in the special case $Z_I = Z_{II}$. The lower envelope is equivalent to the factor κ of equation (72). The upper envelope is found in the case of $R_I/X_I = 0$. The lower envelope for the time t_p is given by the values of t_p in figure 24.



① Lower envelope: κ according to equation (72)

② Upper envelope: $R_I/X_I = 0$

Figure 26 – Factor κ for the calculation of $i_p = \kappa \sqrt{2} I_k''$ in the case of two parallel branches as shown in figure 25, with $Z_I = Z_{II}$, $0,005 \leq R_I/X_I \leq 1,0$ and $0,005 \leq R_{II}/X_{II} \leq 10,0$

Figure 27 gives the deviations $\Delta\kappa$ of the approximate values κ_a , $1,15 \kappa_b$ and κ_c , calculated with the methods a, b and c of IEC 60909-0, 4.3.1.2, from the exact values found with equation (75).

$$\Delta\kappa_a = \frac{\kappa_a - \kappa}{\kappa} 100 \% ; \quad \Delta(1,15 \kappa_b) = \frac{1,15 \kappa_b - \kappa}{\kappa} 100 \% ; \quad \Delta\kappa_c = \frac{\kappa_c - \kappa}{\kappa} 100 \% \quad (76)$$

The conclusions from the results of figure 27 are as follows.

- The factor κ_a always gives results on the conservative side ($\kappa_a \geq \kappa$). In extreme cases the factor κ_a is about 100 % greater than κ .
- The factor $1,15 \kappa_b$ for the calculation of i_p , $i_{p(b)} = 1,15 \kappa_b \sqrt{2} I_k''$ (IEC 60909-0, 4.3.1.2, method b) gives results on the conservative and on the non-conservative side. It is only when both the ratios R_I/X_I and R_{II}/X_{II} are within the range $0,005 \leq R/X \leq 1,0$ that the results of $\Delta\kappa$ are not lower than –5 %, on the non-conservative side. Therefore, method B (figure 27) is limited to cases where $0,005 \leq R/X \leq 1,0$ is valid for all branches of a network. This range is normally found in high-voltage networks with $U_n \geq 110$ kV.
- The factor κ_c leads to deviations $\Delta\kappa_c$ in the range between +3,7 % and –6,5 % against the exact values of κ . The limits ± 5 % (see 2.4.1) are met in the case of $0,005 \leq R_I/X_I \leq 1,0$ and $0,005 \leq R_{II}/X_{II} \leq 5,0$.

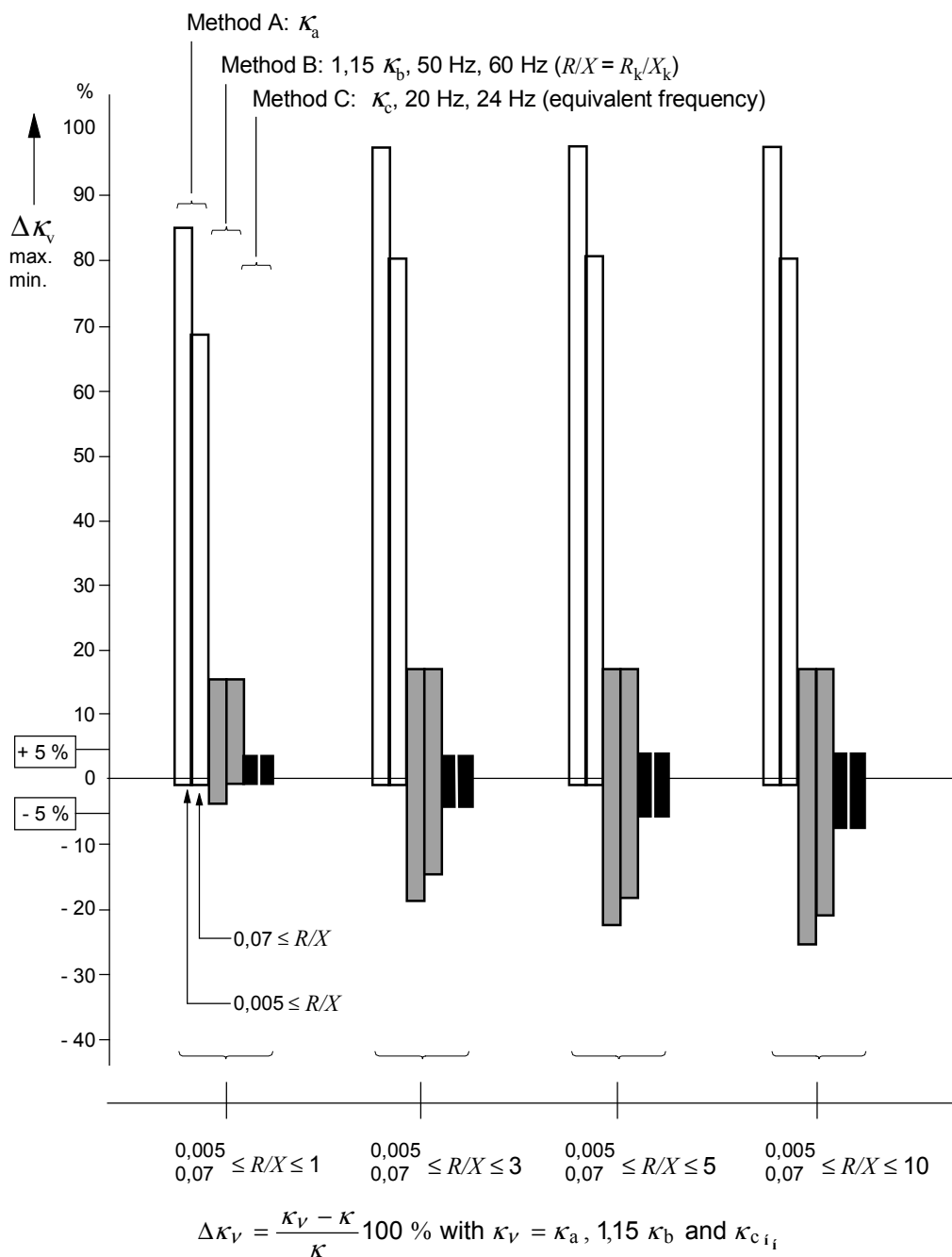


Figure 27 – Deviations $\Delta\kappa_a, \Delta(1,15 \kappa_b)$ and $\Delta\kappa_c$ from the exact value κ with $0,005 \leq Z_I/Z_{II} \leq 1,0$ for the configuration of figure 25

2.4.4 Calculation of the peak short-circuit current i_p in meshed networks

The methods a), b) and c) are given in IEC 60909-0, 4.3.1.2 for both far-from-generator and near-to-generator short circuits. The basic equation (67) (IEC 60909-0, equation (54)) is applied independently of the method. Some additional information may be useful.

Method a) Uniform ratio R/X or $X/R \Rightarrow$ use $\kappa = \kappa_a$

The factor κ_a is determined from IEC 60909-0, figure 15 or equation (55), taking the smallest ratio of R/X or the largest ratio of X/R of those branches of the network which carry partial short-circuit currents and which belong to a network part with the same nominal voltage. Branches with transformers adjacent to the short-circuit location also have to be taken into account.

In accordance with figure 27, method a) leads to results on the conservative side. Furthermore, figure 27 demonstrates that the calculation with κ_a may lead to results with high safety margins if the ratios R/X are different. Therefore, method a) should be used as a first estimation only, limited to those cases of short-circuit current calculations with reactances only and $R_k < 0,3 X_k$ (IEC 60909-0, 4.2.1.1).

Method b) Ratio R/X or X/R at the short-circuit location $\Rightarrow$ use $\kappa = 1,15 \kappa_b$

The factor κ_b is determined from IEC 60909-0, figure 15 or equation (55), using the ratio R_k/X_k ($Z_k = R_k + jX_k$) at the short-circuit location F calculated with the frequency $f = 50$ Hz or $f = 60$ Hz. The factor 1,15 is a safety factor to cover inaccuracies caused by using the ratio R_k/X_k , of a network reduction with complex impedances, for the calculation of κ_b (see figure 26). The calculations in 2.4.3, however, demonstrate that it is also necessary to have the ratio $R/X \leq 1$ in all branches to avoid deviations higher than -5 % on the non-conservative side (see figure 27: $\Delta(1,15 \kappa_b)$).

As long as the ratio R/X remains smaller than 0,3 in all branches, it is not necessary to use the factor 1,15.

The factor $1,15 \kappa_b$ is limited to 1,8 in low-voltage networks. To avoid unrealistic results in high-voltage networks with ratios $R_k/X_k < 0,1$, the factor $1,15 \kappa_b$ is limited to 2,0 (IEC 60909-0, 4.3.1.2, method b).

It is recommended in IEC 60909-0, 4.3.1.2, to use method c) instead of method b) for the calculation of the peak short-circuit current in meshed networks. This is possible when using an adequate digital programme (see IEC 60909-4).

Method c) Equivalent frequency $f_c = 20$ Hz or $f_c = 24$ Hz [6] $\Rightarrow$ use $\kappa = \kappa_c$

The factor κ_c is determined from IEC 60909-0, figure 15 or equation (55) using the ratio

$$\frac{R}{X} = \frac{R_c}{X_c} \cdot \frac{f_c}{f} \quad \text{or} \quad \frac{X}{R} = \frac{X_c}{R_c} \cdot \frac{f}{f_c}$$

The impedance $Z_c = R_c + jX_c$ is the impedance as seen from the short-circuit location F if an equivalent voltage source as the only active voltage is applied with the frequency $f_c = 20$ Hz (in systems with 50 Hz) or $f_c = 24$ Hz (in systems with 60 Hz).

Figure 27 demonstrates that the deviations $\Delta\kappa_c$ are between the boundaries of ± 5 % as long as the ratios R/X of all branches are inside the range $0,005 \leq R/X \leq 5,0$ (see also [6]).

Method c) is therefore applied in low-voltage and high-voltage networks if it is necessary to find a good approximation for the peak short-circuit current, considering the viewpoints of both safety and economy. Using suitable digital programmes for the calculation of the complex short-circuit impedance Z_k it is possible, with small additions in programming, to find Z_c and from this κ_c for the equivalent frequency 20 Hz or 24 Hz.

2.4.5 Example for the calculation of κ and i_p in meshed networks

Figure 28 gives an example for the calculation of κ in a meshed network, which can be reduced to two parallel branches in the positive-sequence system having different ratios R/X .

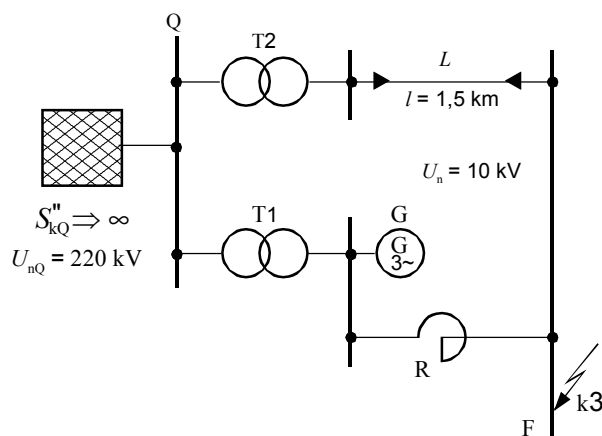


Figure 28a – Network, $f = 50$ Hz

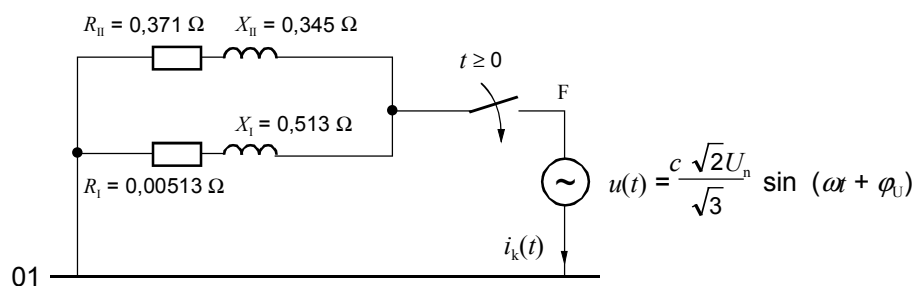


Figure 28b – Equivalent circuit diagram (positive-sequence system)

Generator G	$S_{rG} = 100$ MVA	$U_{rG} = 10$ kV	$x_d'' = 11$ %	$r_G = 0,11$ %	
Transformer T1	$S_{rT1} = 100$ MVA	$U_{rT1HV} = 220$ kV	$U_{rT1LV} = 10$ kV	$u_{XrT1} = 11$ %	$u_{RrT1} = 0,11$ %
Transformer T2	$S_{rT2} = 31,5$ MVA	$U_{rT2HV} = 220$ kV	$U_{rT2LV} = 10$ kV	$u_{XrT2} = 6$ %	$u_{RrT2} = 0,75$ %
Cable L	$R'_L = 0,231 \Omega / \text{km}$		$X'_L = 0,104 \Omega / \text{km}$		
Reactor R	$R_R = 0,00458 \Omega$		$X_R = 0,485 \Omega$		

Figure 28 – Example for the calculation of κ and i_p with the methods a), b) and c)
(IEC 60909-0, 4.3.1.2)

The impedances of the parallel branches (figure 28b) are:

$$\underline{Z}_I = (0,00513 + j0,513) \Omega; \quad R_I/X_I = 0,01; \quad Z_I = 0,51303 \Omega$$

$$\underline{Z}_{II} = (0,371 + j0,345) \Omega; \quad R_{II}/X_{II} = 1,0754; \quad Z_{II} = 0,5066 \Omega$$

$$Z_I/Z_{II} = 1,0127$$

The short-circuit impedance $\underline{Z}_k$ at the short-circuit location F is:

$$\underline{Z}_k = (0,11276 + j0,25349) \Omega; \quad Z_k = 0,27744 \Omega; \quad \gamma = 1,15225$$

The factor κ is found by an iteration process from equation (72):

$$\kappa = \kappa_e = 1,54655; \quad t_p = 8,58 \text{ ms } (f = 50 \text{ Hz})$$

Using the ratio $R_k/X_k \approx 0,445$ it is possible to find $\kappa \approx 1,55$ from figure 26 (the ratio Z_I/Z_{II} is only slightly different from 1).

Table 3 gives the results for the calculations of κ with the methods a), b) and c) (IEC 60909-0, 4.3.1.2) for the example in figure 28.

Table 3 – Values of κ for the example in figure 28

Method	R/X	κ equation (68)	$\Delta\kappa_V$ ^a %	Remarks
a	$\frac{R}{X} = \left[\frac{R}{X} \right]_{\min} = \frac{R_I}{X_I} = 0,01$	$\kappa = \kappa_a = 1,971$	+27,45	The application de la method a) is not adequate in this example because $R_k > 0,3 X_k$
b	$\frac{R}{X} = \frac{R_k}{X_k} = 0,4448$	$\kappa_b = 1,278$ $1,15 \kappa_b = 1,4697$	–4,967	$\Delta\kappa_b \leq -5 \%$ as long as $R/X \leq 1$ for all branches
c	$\frac{R}{X} = \frac{R_c}{X_c} \cdot \frac{20 \text{ Hz}}{50 \text{ Hz}} = 0,1732$ ^b $f = 50 \text{ Hz}; f_c = 20 \text{ Hz}$	$\kappa = \kappa_c = 1,6029$	+3,644	$ \Delta\kappa_c \leq 5 \%$ as long as $R/X \leq 5$ for all branches
^a $\Delta\kappa_V = \frac{\kappa_V - \kappa}{\kappa} \cdot 100 \%$ with $\kappa_V = \kappa_a, 1,15 \kappa_b$ or κ_c and $\kappa = \kappa_{\text{exact}} = 1,54655$				
^b $Z_c = (0,0634 + j0,14644) \Omega$; $R_c/X_c = 0,43294$				

2.5 Factor μ for the calculation of the symmetrical short-circuit breaking current

2.5.1 General

In the case of a near-to-generator short circuit the symmetrical short-circuit current decays appreciably within the first tenth of a second after the beginning of a short circuit as described in IEC 60909-0, figure 2. This phenomenon is due to the change of the flux in the rotor of the generator during the short circuit.

Even if digital programmes or analogue computers are used, it is not simple to simulate the actual conditions. The appropriate system of differential equations simulating the transient behaviour of the generator cannot be integrated in a simple manner due to the eddy currents in the forged rotor of turbine generators, the non-linear characteristic of the rotor iron, the rotor anisotropy (different conductivity in the direct and quadrature axes) and due to the tooth saturation in the stator. A considerable amount of data must be taken into account for the calculation of the transient decay of the symmetrical short-circuit current.

This includes primarily

- the reactances and time constants of the generator;
- the location of the balanced or unbalanced short circuit either within the power station or outside in the network;
- the operating conditions of the generator before the short circuit between no-load and rated load either over- or underexcited;
- the type and response of excitation device and voltage regulation;
- the kinetic behaviour of the generator-turbine set during the short circuit;
- the rated data and the operating values (for instance tap-changer position and saturation) of the transformers through which the short-circuit current flows.

Therefore, it is not surprising that even results obtained from detailed calculations are often less accurate than expected. Detailed calculations are used in special cases, for example, if particular safety requirements are necessary.

The most severe case under consideration may be described as follows.

- The balanced three-phase short circuit at the terminals of the generator leads to the highest initial short-circuit current. It is assumed that the neutral of the generator is isolated (if the neutral of the generator is directly earthed, for instance in low-voltage networks, the line-to-earth short circuit leads to the highest initial short-circuit current, because the zero-sequence reactance of the generator is smaller than the subtransient reactance of the positive-sequence system).
- Unbalanced short circuits (line-to-line short circuits) result in smaller initial short-circuit currents than balanced short circuits; however, the symmetrical short-circuit currents decay to a lower value (IEC 60909-0, figure 10).
- The symmetrical short-circuit breaking currents I_b are largest for low values of the minimum time delay $t_{\min}$.
- The short-circuit currents of large generators decay in general more slowly than those of smaller generators. The factor μ (IEC 60909-0, figure 16 and equation (70)) is chosen above the highest values found from measurements and calculations (see figure 30 for generators of different type and rated power).

2.5.2 Basic concept

The following equation for the calculation of the symmetrical short-circuit breaking current I_b is used (IEC 60909-0, 4.5.2.1):

$$I_b = \mu I_k'' \quad (77)$$

The factor μ is used for the most severe case as described in 2.5.1. From the parameters listed above only the following are taken into account:

- The minimum time delay $t_{\min}$ in accordance with IEC 60909-0, 1.3.22. Four different μ -curves are given for different values of $t_{\min}$ (IEC 60909-0, figure 16).
- The initial symmetrical short-circuit current of the generator related to the rated current of the generator, I_{kG}'' / I_{rG} . This value is entered on the abscissa of figure 16 of IEC 60909-0.

Equation (77) shall be used only in the case of single-fed short circuits (short circuits fed from one generator, according to IEC 60909-0, 4.5.2.1) or in the case of short circuits fed from non-meshed sources (IEC 60909-0, 4.5.2.2). The calculation of short-circuit breaking currents in meshed networks needs special considerations (IEC 60909-0, 4.5.2.3) [12].

In order to demonstrate that in the case of a single-fed short circuit, the abscissa of figure 16 of IEC 60909-0 may be used both for short circuits of generators with varying reactances and for power-system short circuits with a reactance X_N between the generator terminals and the short-circuit location, the decay of the symmetrical a. c. component of the short-circuit current is considered in a simplified manner.

For this purpose the following equation for the calculation of the a.c. part $I_{kAC}(t)$ of the symmetrical short-circuit current $I_k(t) = I_{kAC}(t) + I_{kDC}(t)$ is introduced:

$$I_{ka.c.}(t) = (I_k'' - I_k') e^{-t/T_N''} + (I_k' - I_k) e^{-t/T_N'} + I_k \quad (78)$$

where

I_k'' is the initial symmetrical short-circuit current (IEC 60909-0, 1.3.5), or the subtransient short-circuit current (equation (79a));

I_k' is the transient short-circuit current (equation (79b));

I_k is the steady-state short-circuit current (IEC 60909-0, 1.3.10), (equation (79c));

T_N'' is the subtransient time constant as a medium value between T_{dN}'' and T_{qN}'' (equation (80a));

T_N' is the transient time constant, $T_N' = T_{dN}'$ (equation (80b)).

The currents I_k'' , I_k' and I_k depend on the internal voltages E'' , E' and E of the generator, and the sum of reactances $X_d'' + X_N$, $X_d' + X_N$ and $X_d + X_N$, as shown in figure 29.

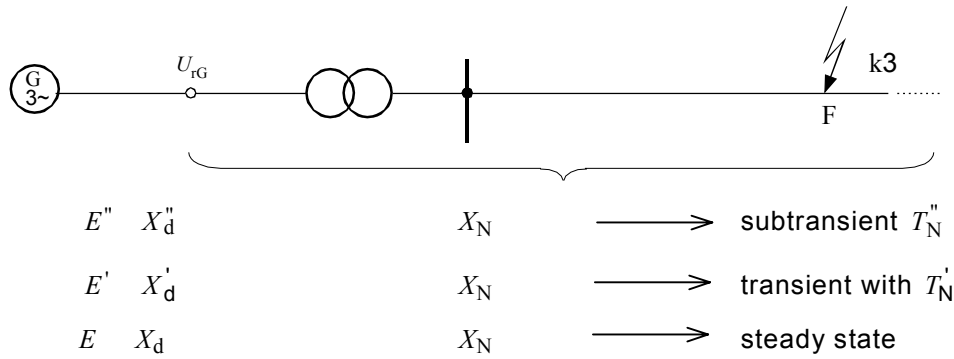


Figure 29 – Network configuration (single-fed short circuit) and relevant data to demonstrate the decay of the symmetrical a.c. component of a near-to-generator short circuit

In the case of a power-system short circuit under no-load conditions of the generator and therefore $E'' = E' = E$ the following currents are to be introduced:

$$I_k'' = \frac{E''}{X_d'' + X_N} = \frac{E}{X_d'' + X_N} \quad (79a)$$

$$I_k' = \frac{E'}{X_d' + X_N} = \frac{E}{X_d' + X_N} \quad (79b)$$

$$I_k = \frac{E}{X_d + X_N} \quad (79c)$$

and with $T_N'' \approx T_{dN}''$ for simplification, the time constants:

$$T_N'' = T_0'' \frac{X_d'' + X_N}{X_d' + X_N} = T'' \frac{X_d'}{X_d''} \cdot \frac{X_d'' + X_N}{X_d' + X_N} \quad \text{with } T_0'' = T'' \frac{X_d'}{X_d''} \quad (80a)$$

$$T_N' = T_0' \frac{X_d' + X_N}{X_d' + X_N} = T' \frac{X_d}{X_d'} \cdot \frac{X_d' + X_N}{X_d' + X_N} \quad \text{with } T_0' = T' \frac{X_d}{X_d'} \quad (80b)$$

Typical time constants are $T'' \approx 0,03$ s and $T' \approx 1$ s and therefore from equation (80a) $T_N'' \approx (1,0 \dots 1,3) T''$ and from equation (80b) $T_N' \approx (1,0 \dots 2,5) T'$. With the approximation $e^{-t_{\min}/T_N'} \approx 1$ at $t = t_{\min} = (0,02 \dots 0,25)$ s, the following equation for the factor μ may be found:

$$\mu = \frac{I_{kAC}(t)}{I_k''} \approx \left[1 - \frac{X_d'' + X_N}{X_d' + X_N} \right] e^{-t_{\min}/T_N''} + \frac{X_d'' + X_N}{X_d' + X_N} \quad (81)$$

Because $t_{\min}/T_N''$ may be smaller or greater than 1, it is necessary to introduce the infinite progression ($t = t_{\min}$)

$$e^{-t/T_N''} = 1 - \frac{t}{T_N''} + \frac{1}{2} \left[\frac{t}{T_N''} \right]^2 - \frac{1}{6} \left[\frac{t}{T_N''} \right]^3 + \frac{1}{24} \left[\frac{t}{T_N''} \right]^4 - \dots + \dots$$

to find the following equation for μ at no-load conditions:

$$\mu = 1 - 0,5 x_d'' \frac{t}{T_0''} \cdot \frac{I_k''}{I_{rG}} t_{\min} + \frac{1}{3 + 2x_N/x_d''} \left[\frac{t_{\min}}{T_N''} + e^{-t_{\min}/T_N''} - 1 \right] \quad (82)$$

It is obvious from this equation (82) that the factor μ can be introduced depending on I_k''/I_{rG} and $t_{\min}$ (IEC 60909-0, figure 16 and equation (70)). In the case of a far-from-generator short circuit with $x_N \gg x_d''$ the factor μ approaches 1. Equation (82) should not be used for calculations.

2.5.3 Calculation of the symmetrical short-circuit breaking current I_b with the factor μ

The simplifications described in 2.5.2 were checked on the basis of numerous experiments in testing stations and by detailed calculations using the data of existing generators of different rated power and different types. Examples are depicted in figure 30 both for the case of no-load conditions and for the case of rated load conditions of the generator before the short circuit.

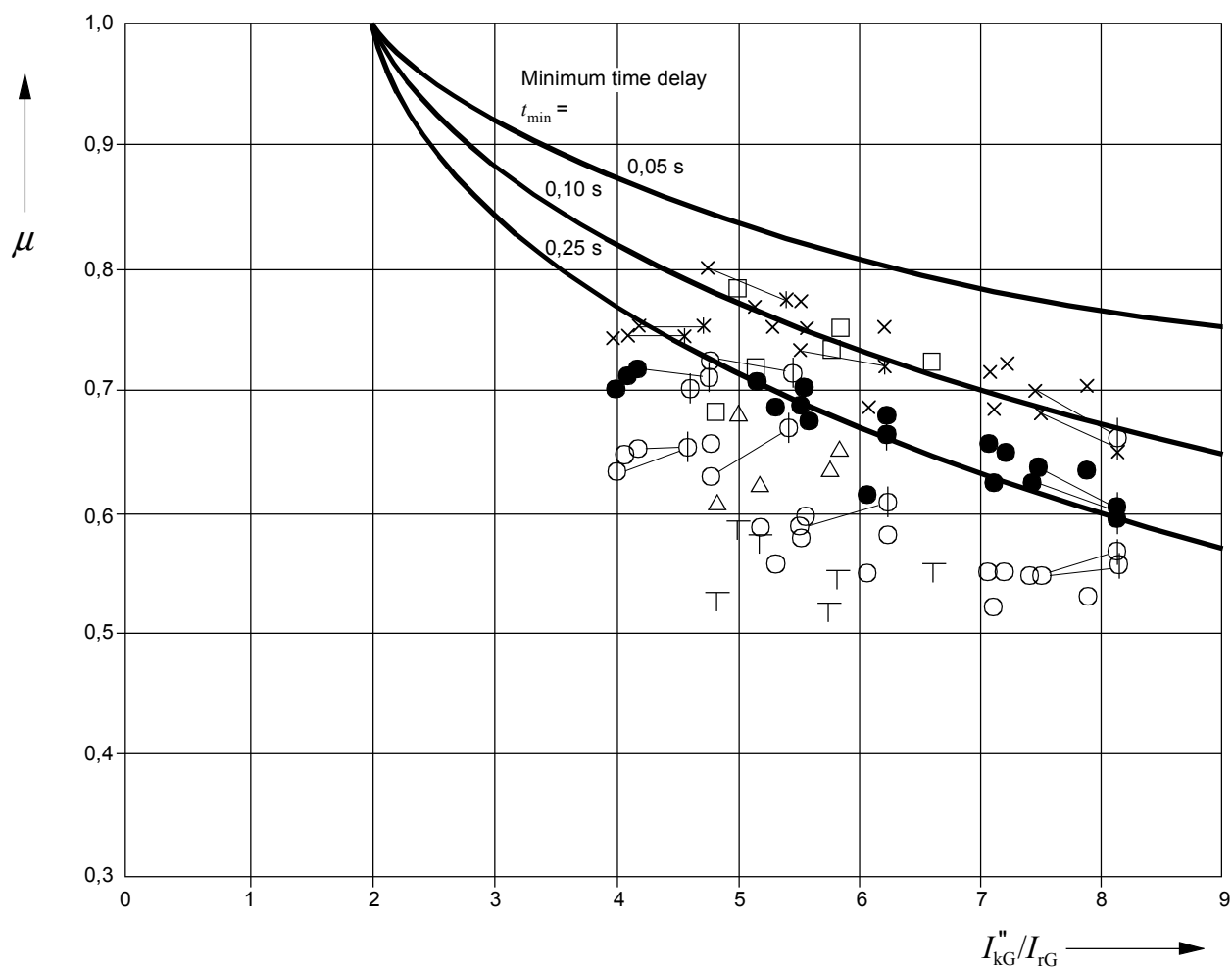
In order to simplify the short-circuit calculations, $\mu = 1$ is assumed for all values $I_{kG}''/I_{rG} \leq 2$ (IEC 60909-0, figure 16) and furthermore the family of curves for different values $t_{\min}$ are numerically described (IEC 60909-0, equation (70)). The values from both are identical.

In the case of static converter excitation and in the case of compound excited low-voltage generators the remarks of IEC 60909-0, 4.5.2.1 should be taken into account. They are based on measurements in testing stations.

The values of μ for $t_{\min} = 0,02$ s are introduced especially for the calculation of the breaking currents of low-voltage motors and groups of low-voltage motors (IEC 60909-0, 3.8). In high-voltage networks the minimum time delay $t_{\min}$ is normally higher than 0,02 s.

Equation (77) (IEC 60909-0, equation (69)) is limited to single-fed short circuits from non-meshed sources (IEC 60909-0, figures 11 and 12). If, however the short-circuit current is generated both by generators and motors and in addition by network feeders, and if the short-circuit current flows through a meshed power system (IEC 60909-0, figure 14), then equation (77) (IEC 60909-0, equation (69)) cannot be used directly for the calculation of the partial short-circuit breaking currents fed from generators and motors. As a result of the coupling of the partial short-circuit breaking currents caused by the meshed network, the partial short-circuit current of the network feeder increases as the currents fed from the generators and motors decrease [12].

If the approximation $I_b = I_k''$ (IEC 60909-0, equation (74)) is insufficient in the case of near-to-generator or near-to-motor short circuits in meshed networks, a more accurate result on the conservative side can be found from IEC 60909-0, equation (75). The terms $(1 - \mu_i) I_{kGi}''$ and $(1 - \mu_j q_j) I_{kMj}''$ are evaluated by the initial voltage differences $\Delta \underline{U}_{Gi}''$ and $\Delta \underline{U}_{Mi}''$ (IEC 60909-0, equations (76) and (77)). The factors μ_i and μ_j are found from IEC 60909-0, figure 16 or equation (70), with I_{kGi}'' / I_{rGi} or I_{kMj}'' / I_{rMj} [12].



Turbine generators:

$\times$ $t_{\min} = 0,05 \text{ s}$
 $\bullet$ $t_{\min} = 0,10 \text{ s}$
 $\circ$ $t_{\min} = 0,25 \text{ s}$

no-load condition

$\times$ $t_{\min} = 0,05 \text{ s}$
 $\bullet$ $t_{\min} = 0,10 \text{ s}$
 ϕ $t_{\min} = 0,25 \text{ s}$

rated-load condition
(calculated)

Salient pole generators:

$\square$ $t_{\min} = 0,05 \text{ s}$
 Δ $t_{\min} = 0,10 \text{ s}$
 T $t_{\min} = 0,25 \text{ s}$

no-load condition

$\times - \times$; $\bullet - \bullet$; $\circ - \phi$

Connection lines between measurements at no-load conditions and calculations at rated-load conditions

Figure 30 – Decay of the symmetrical short-circuit current (factor μ) based on test measurements and calculations [5]

2.6 Factor λ ($\lambda_{\max}$, $\lambda_{\min}$) for the calculation of the steady-state short-circuit current

2.6.1 General

The typical time behaviour of a symmetrical short-circuit current in the case of a near-to-generator short circuit is described in IEC 60909-0, figure 2. The initial symmetrical short-circuit current I_k'' changes in a few seconds to the steady-state short-circuit current I_k with $I_k < I_k''$. The decay depends on the method of excitation and voltage regulation and to a high degree on the saturation and the highest possible excitation voltage $U_{f\max}$ (IEC 60909-0, 4.6.1.1). The decaying d.c. component of the short-circuit current is zero before the symmetrical steady-state short-circuit current is reached.

In most cases the maximum steady-state short-circuit current $I_{k\max}$ is of interest (IEC 60909-0, equation (78)):

$$I_{k\max} = \lambda_{\max} I_{rG} \quad (83)$$

The current $I_{k\max}$ of one generator is referred to I_{rG} by reason of its thermal effect. The factor $\lambda_{\max}$ depends on the relation I_k'' / I_{rG} and the saturated value $x_{dsat} = X_{dsat} / Z_{rG}$ of the direct-axis synchronous reactance (IEC 60909-0, figures 18 and 19). The relation $I_{k\max} / I_k''$ of the relevant generator is used to find the thermal stresses of electrical equipment in accordance with IEC 60909-0, 4.8, using the factors m and n (see 2.8).

The amount of data to be taken into account for the calculation of $I_{k\max}$ is considerably greater than in the case of the calculation of the symmetrical a.c. breaking current I_b (see 2.5.1). The following additional parameters are of influence:

- the synchronous reactance particularly in the direct axis (X_d);
- the iron saturation particularly of the rotor;
- the effectiveness of voltage regulation;
- the maximum possible excitation voltage $U_{f\max}$ produced by the excitation equipment and usually referred to the excitation voltage U_{fr} at rated operation with $u_{f\max} = U_{f\max} / U_{fr}$.

Since cylindrical rotor generators and salient-pole machines differ widely with respect to their synchronous reactances X_d and are fitted with different types of excitation equipment, separate factors $\lambda_{\max}$ are required for cylindrical rotor generators (IEC 60909-0, figure 18) and salient-pole machines (IEC 60909-0, figure 19).

Special considerations are necessary for synchronous machines with terminal-fed static exciters (IEC 60909-0, 4.5.1). The $\lambda_{\max}$ -curves (IEC 60909-0, figures 18 and 19) may be applied also in the case of terminal-fed static exciters, if the short circuit occurs at the high-voltage side of the unit transformer of a power station unit or in the network, and if the maximum excitation voltage is chosen with respect to the reduction of the terminal voltage during the short circuit.

The minimum short-circuit current $I_{k\min}$ is of interest in particular for the selection and setting of protection devices in power systems (IEC 60909-0, equation (79)):

$$I_{k\min} = \lambda_{\min} I_{rG} \quad (84)$$

The factor $\lambda_{\min}$ is given in IEC 60909-0, figures 18 and 19, for constant no-load excitation under the assumption that no voltage regulator is provided or active.

In the case of generators with compound excitation special considerations and information from the manufacturer are necessary (IEC 60909-0, 4.6.1.2).

The above-mentioned assumptions made to find the factor λ are not always fulfilled in practice. When calculating $I_{k\max}$ larger deviations may occur between the real and the calculated values than in the case of I_k'' and I_b .

If the short circuit is fed from more than one generator, the generators may fall out of synchronism before the steady-state short-circuit current is reached, especially in meshed networks. In these cases the steady-state short-circuit current can be estimated only roughly as described in IEC 60909-0, 4.6.3.

In extreme cases of asynchronism the maximum steady-state short-circuit current may even be higher than the initial symmetrical short-circuit current.

The value of $I_{k\min}$ may also deviate considerably from real values obtained in practice.

2.6.2 Influence of iron saturation

2.6.2.1 Characteristic curve method (Potier reactance)

The Potier reactance is found as the difference between two saturation curves, one taken at no load, and the other taken at rated current, zero power factor. The no-load characteristic in figure 31 describes the relationship between the voltage $E_0(I_f)$ and the field current I_f for an unloaded synchronous machine ($I_G = 0$). This characteristic is clearly non-linear. The load characteristic $U_L(I_f)$ in figure 31 for the rated current I_{rG} at $\cos \varphi_G = 0$ has a similar non-linear form. In the case of $U_L = 0$ a terminal short circuit exists. In this case complete linearity can be assumed.

On subtraction of the field current I_{fkr} , required in the case of a terminal short circuit with I_{rG} , from the field current $I_f(U_L)$ at point A of figure 31, the remainder is due to the saturation of the magnetic circuit. If it is further assumed that all leakage flux in the generator, external to the iron, is directly proportional to I and is independent of U_L , the equivalent circuit of figure 32 is obtained.

From the circuit of figure 32 it can be seen that $E_0 - U_L = X_p I$ is the difference value between the source and the terminal voltage. The effective reactance X_p is called the Potier reactance [2] and [4]. The value of X_p is found on drawing a tangent to the no-load characteristic of figure 31 passing through the origin and by drawing a parallel BC thereto passing through point B. The hatched triangles ADC and A'D'C' are congruent. The height of the triangles in figure 31 is:

$$X_p / Z_{rG} = X_p I_{rG} / (U_{rG} / \sqrt{3})$$

The no-load characteristic $E_0(I_f)$ can easily be measured in the testing station. Standard characteristic curves are sometimes used in the literature, but specific curves are dependent on the design of the magnetic circuits of the synchronous machine.

The value of the Potier reactance X_p lies between the stator leakage reactance X_σ (no saturation in the poles, all saturation being in the teeth) and the transient reactance X_d' (all saturation in the poles, and the teeth are unsaturated): $X_\sigma < X_p < X_d'$ [4]. Experience has shown that either the relationship $X_p \approx 0,82 X_d'$ or the relationship $X_p \approx X_d'' + 0,63 (X_d' - X_d'')$ is sufficiently accurate.

On the basis of the no-load characteristic curve $E_0(I_f)$ and the Potier reactance X_p , the steady-state short-circuit current in the case of a single-fed short circuit for any value of the field current I_f and the network reactance X_N (see figure 29) may be calculated. Figure 32 leads to the following equation:

$$I_k = \frac{E_0(I_f)}{X_p + X_N} \quad (85)$$

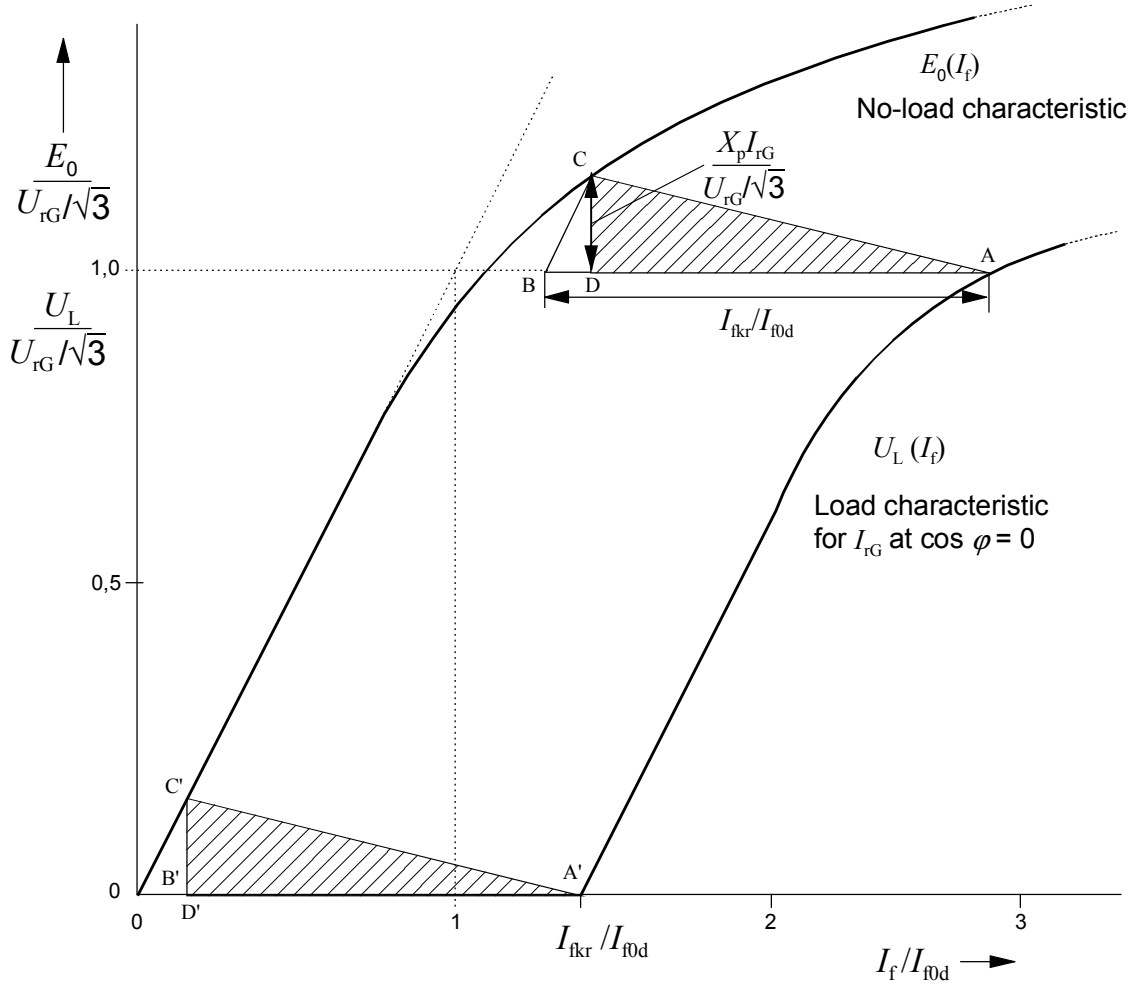


Figure 31 – Characteristic saturation curve method to find the Potier reactance X_p in accordance with [4]

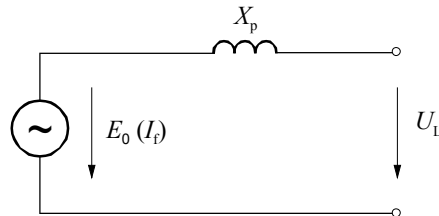


Figure 32 – Equivalent circuit with the source voltage $E_0(I_f)$ and the Potier reactance X_p

On the basis of the relationship $I_{kG}''(X_d'' + X_N) = E''$ the following result is obtained for the factor λ :

$$\lambda = \frac{I_k}{I_{rG}} = \frac{E_0(I_f)}{(X_p - X_d'')I_{rG} + E'' \cdot I_{rG} / I_{kG}''} \quad (86)$$

$E_0(I_f)$ is dependent on the synchronous reactance X_d only in the case of $\lambda_{\max}$ for the calculation of $I_{k\max}$ (IEC 60909-0, figures 18 and 19).

2.6.2.2 Simplified method for the calculation of λ

In view of the large number of assumptions required for the characteristic curve method in 2.6.2.1 and the limitation of the calculation to the steady-state short-circuit currents which are fed by one generator only, it is useful to simplify the method of calculation. The following method was used to find the λ -curves (IEC 60909-0, figures 18 and 19).

On the basis of figure 31 both the source voltage E_0 for a given field current I_f and the Potier reactance X_p are reduced as a result of saturation. Both influences partially compensate for each other as shown in figure 32. It would thus appear possible to develop an equation without these effects [8]. As a result of this assumption equation (85) is replaced by the following equation:

$$I_k = \frac{u_{f\max} E_r}{X_{dsat} + X_N} \quad (87)$$

and instead of equation (86) the following relationship is found:

$$\begin{aligned} \lambda = \lambda_{\max} &= \frac{I_k}{I_{rG}} = \frac{u_{f\max} E_r}{(X_{dsat} - X_d'') I_{rG} + E'' I_{rG} / I_{kG}} \\ &= \frac{u_{f\max} \sqrt{1 + 2 x_{dsat} \sin \varphi_{rG} + x_{dsat}^2}}{x_{dsat} - x_d'' + (1 + x_d'' \sin \varphi_{rG}) I_{rG} / I_{kG}} \end{aligned} \quad (88)$$

where for the synchronous machine

- E'' is the subtransient internal voltage (IEC 60909-0, 1.3.16);
- E_r is the steady-state internal voltage at rated operation U_{rG} , I_{rG} and $\cos \varphi_{rG}$;
- $u_{f\max} = U_{f\max}/U_{fr}$, the referred highest possible excitation voltage;
- U_{fr} is the excitation voltage at rated operation;
- X_d'' is the subtransient reactance (IEC 60909-0, 1.3.21);
- X_{dsat} is the saturated synchronous reactance (IEC 60909-0, 4.6.1.1);
- I_{kG}'' / I_{rG} is the initial symmetrical short-circuit current referred to the rated current;
- x_{dsat}, x_d'' are per unit quantities (see equations (89) and (90)).

The internal subtransient voltage is obtained to a good approximation:

$$E'' = \frac{U_{rG}}{\sqrt{3}} + I_{rG} X_d'' \sin \varphi_{rG} = \frac{U_{rG}}{\sqrt{3}} (1 + x_d'' \sin \varphi_{rG}) \quad (89)$$

The internal voltage for rated condition is found to be:

$$E_r = \sqrt{\left(\frac{U_{rG}}{\sqrt{3}}\right)^2 + (I_{rG} X_{dsat})^2 + 2 \frac{U_{rG}}{\sqrt{3}} I_{rG} X_{dsat} \sin \varphi_{rG}} \quad (90)$$

$$\frac{U_{rG}}{\sqrt{3}} \sqrt{1 + x_{dsat}^2 + 2x_{dsat} \sin \varphi_{rG}}$$

where

$$x_d'' = X_d'' / Z_{rG}, \quad x_{dsat} = X_{dsat} / Z_{rG} \text{ with } Z_{rG} = U_{rG} / (\sqrt{3} I_{rG})$$

Since for the individual generator under consideration a near-to-generator short circuit is applicable for $I_{kG}'' / I_{rG} > 2$ only (IEC 60909-0, 1.3.18), the characteristic curves in IEC 60909-0, figures 18 and 19 and the above equation (88), apply to this region.

The factors λ calculated in accordance with the characteristic curve method (see 2.6.2.1) and in accordance with the simplified method [8] are in good agreement taking the values $X_d'' = 0,2 Z_{rG}$ and $\cos \varphi_{rG} = 0,85$. The following usual values for $u_{fmax} = U_{fmax} / U_{fr}$ are assumed for the calculation of the λ_{max} -curves (IEC 60909-0, figures 18 and 19):

Series	1	2
Cylindrical rotor generators	$u_{fmax} = 1,3$	$u_{fmax} = 1,6$
Salient-pole machines	$u_{fmax} = 1,6$	$u_{fmax} = 2,0$

Since the factor λ_{max} is clearly defined by equation (88) through (90) in the range $I_{kG}'' / I_{rG} > 2$, it need not be approximated by simplified numerical expressions as in the case of the factors μ (IEC 60909-0, equation (70)) and q (IEC 60909-0, equation (73)).

2.7 Factor q for the calculation of the short-circuit breaking current of asynchronous motors

2.7.1 General

The factor q is used together with the factor μ (see 2.5) to determine the symmetrical short-circuit breaking current of asynchronous motors or groups of asynchronous motors in medium- and low-voltage networks (IEC 60909-0, 4.5.2 and 4.7).

The symmetrical short-circuit current of asynchronous motors in the case of a three-phase short circuit at the terminals decays considerably faster than the short-circuit current of synchronous generators. The effective time constant T_{AC} increases with the square root of the active power per pair of poles (see figure 35).

The factor q is given in IEC 60909-0, figure 17 or equation (73), depending on P_{rM}/p and t_{min} . The factor μ of the product μq (IEC 60909-0, equations (75) and (96)) takes into consideration the distance between the short-circuit location and the motor terminals. The factor μ is taken from IEC 60909-0, figure 16 or equation (70).

If the symmetrical short-circuit breaking current I_b of asynchronous motors is known from short-circuit measurements, it is possible to determine the factor q from equation (91).

The initial symmetrical short-circuit current of asynchronous motors is to be calculated with the equivalent voltage source $cU_n / \sqrt{3}$ (IEC 60909-0, table 2) although the internal voltage of the motor before the short circuit is smaller than the terminal voltage. The reasons for this are:

- unification of the calculation procedure in all cases (IEC 60909-0, 2.3);
- the voltage at the terminals of the asynchronous motor before the short circuit may be higher than the rated voltage U_{rM} of the motor;

- the ratio I_{LR}/I_{rM} is normally allowed to be 20 % higher than the rated value given by the manufacturer. (The ratio I_{LR}/I_{rM} takes care of the saturation of the reactances of asynchronous motors. This effect should be considered, especially if the results of transient calculations are compared the results found with IEC 60909-0).

When calculating the breaking current in meshed networks with asynchronous motors, IEC 60909-0, equation (75) shall be used, even in the case of a far-from-generator short circuit with $\mu_j = 0$. If the short circuit is a far-from-motor short circuit with $\mu_j = 1$, then, independent of the value q_j , take $1 - \mu_j q_j = 0$.

2.7.2 Derivation of factor q

To find the factor q (IEC 60909-0, figure 17 or equation (73)) several short-circuit measurements [5] and transient calculations [14] for short-circuit breaking currents I_{bM} at different times t_{min} (IEC 60909-0, 1.3.22) are used:

$$q = \frac{I_{bM}}{\mu I_{kM}''} \quad (91)$$

In figure 33 the factor q from equation (91) is given for 8 low-voltage motors with $P_{rM} = (0,011 \dots 0,16)$ MW and 20 medium-voltage motors with $P_{rM} = (0,16 \dots 10,0)$ MW at the minimum time delay $t_{min} = 0,02$ s, $0,05$ s, $0,10$ s and $\geq 0,25$ s. The main data of these asynchronous motors are given in table 4.

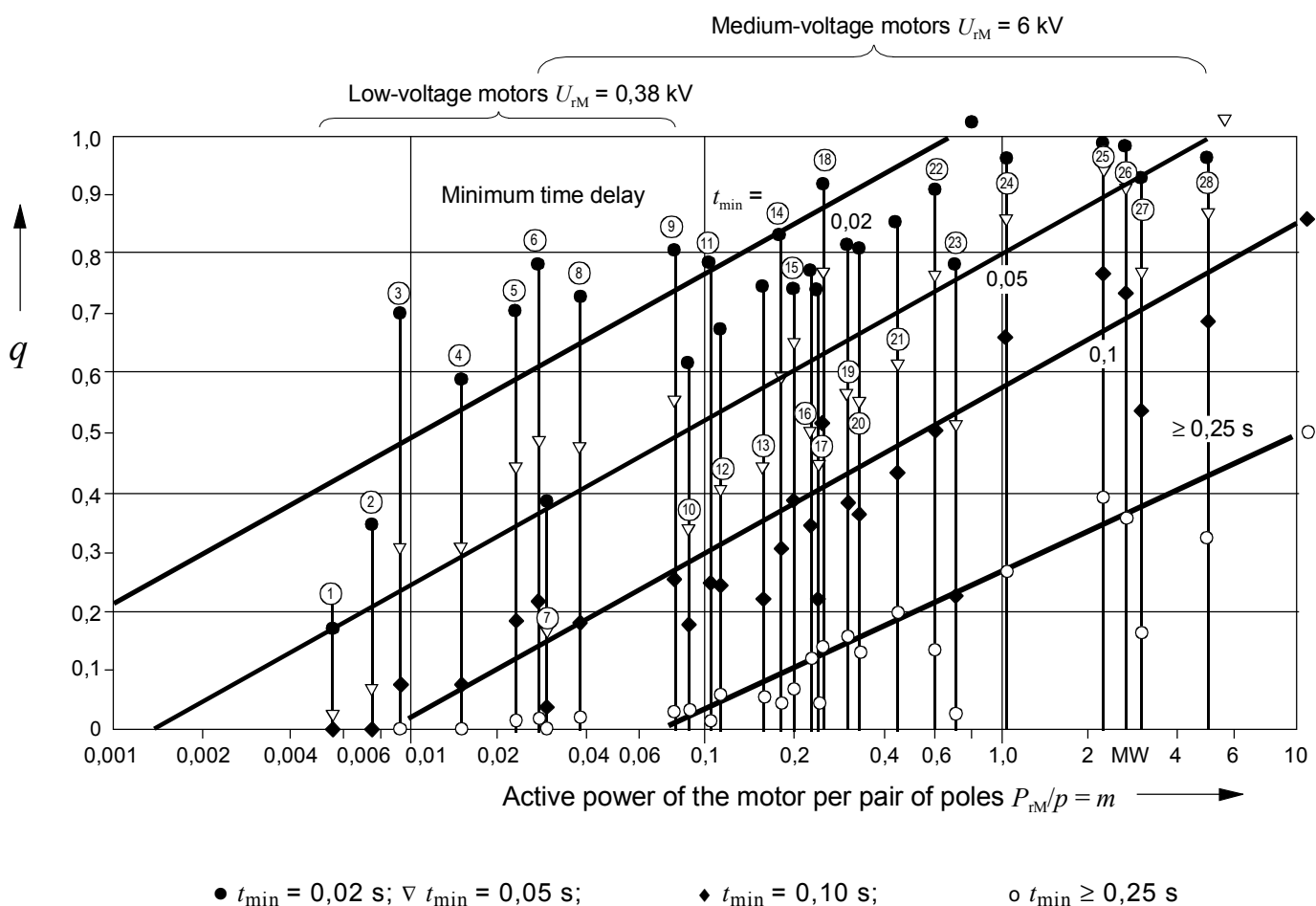
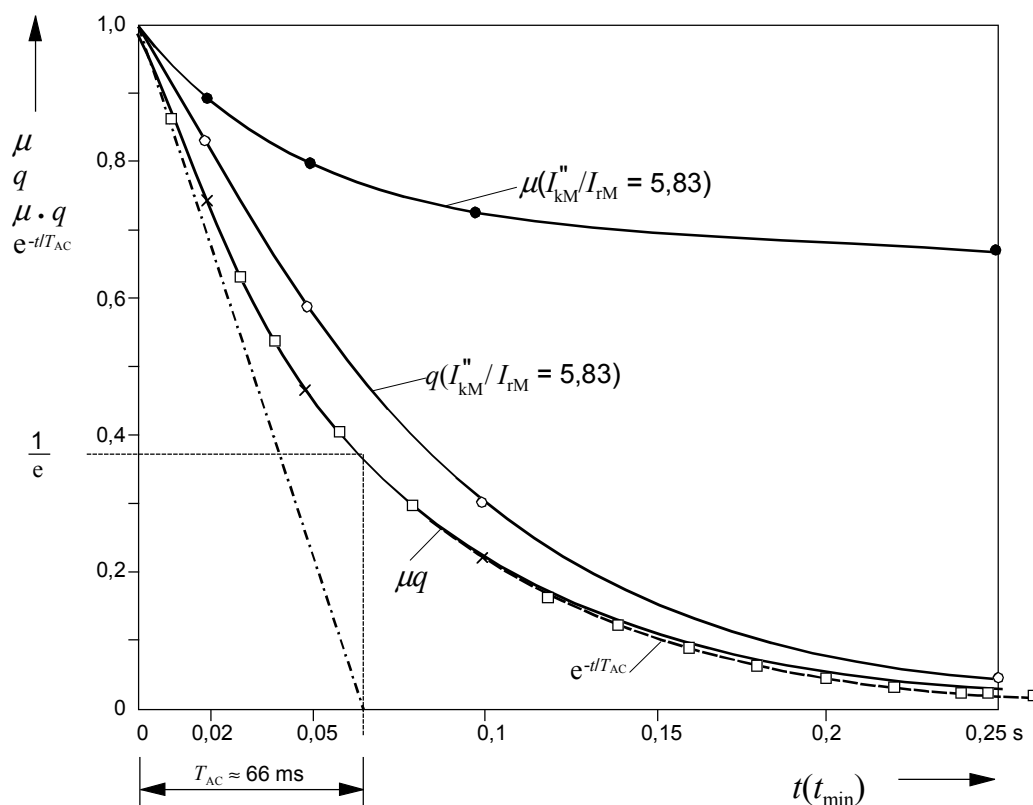


Figure 33 – Factor q from measured and calculated values of $I_{bM} = \mu q I_{kM}''$, equation (91), at different values t_{min} in comparison to $q = q_{IEC}$ (IEC 60909-0, figure 17)

The values of q_{IEC} (IEC 60909-0, figure 17 or equation (73)) are medium values as can be seen from figure 33. The 50 % frequency of the deviations between the single value q and q_{IEC} is found in the range of $\Delta q_{\text{IEC}} \approx 0 \% \dots +5 \%$ for the four different curves q_{IEC} ($t_{\text{min}} = 0,02 \text{ s}, 0,05 \text{ s}, 0,10 \text{ s}$ and $\geq 0,25 \text{ s}$).

In figure 34 the time functions of μ , q and μq are given for one example (motor No. 14 in table 4). In this case, and as well as in the others, it is possible with sufficient accuracy to substitute the product $\mu q = I_{\text{bM}} / I_{\text{kM}}''$ by the function $e^{-t/T_{\text{AC}}}$.



Example: $P_{\text{rM}} = 550 \text{ kW}$; $U_{\text{rM}} = 6 \text{ kV}$; $I_{\text{rM}} = 66 \text{ A}$; $I_{\text{LR}}/I_{\text{rM}} = 5,3$; $n_{\text{r}} = 990 \text{ min}^{-1}$; $f = 50 \text{ Hz}$; $T_{\text{AC}} = 66 \text{ ms}$.

Figure 34 – Time functions μ , q , μq and $e^{-t/T_{\text{AC}}}$ for the calculation of the symmetrical short-circuit breaking current $I_{\text{bM}} = \mu q I_{\text{kM}}''$ in the case of a short circuit at the terminals of an asynchronous motor

Table 4 – Data of low-voltage and medium-voltage asynchronous motors (50 Hz) and calculated values

No.	U_{rM}	I_{rM}	S_{rM}^a	P_{rM}^b	n_r^c	p^d	P_{rM} / p^e	I_{LR} / I_{rM}	Z_M^f
	kV	A	MVA	MW	min ⁻¹	–	MW	–	Ω
1	0,38	22,5	0,0149	0,011	1425	2	0,0055	8,5	1,147
2	0,38	45,0	0,0295	0,022	960	3	0,0073	6,0	0,813
3	0,38	36,6	0,024	0,0185	1460	2	0,0093	6,0	1,002
4	0,38	61,0	0,040	0,030	1465	2	0,015	6,5	0,533
5	0,38	88,0	0,0578	0,045	1475	2	0,0225	6,7	0,372
6	0,38	106,0	0,0696	0,055	1475	2	0,0275	7,3	0,284
7	6,0	25,0	0,259	0,175	485	6	0,0292	4,3	32,22
8	0,38	137,0	0,090	0,075	1475	2	0,0375	6,8	0,236
9	0,38	291,0	0,192	0,160	1480	2	0,080	6,3	0,120
10	6,0	21,0	0,218	0,180	1480	2	0,090	5,7	28,94
11	6,0	80,5	0,837	0,630	492	6	0,105	5,2	8,275
12	6,0	26,0	0,270	0,225	1485	2	0,113	5,8	22,97
13	6,0	19,0	0,197	0,160	2970	1	0,160	6,0	30,39
14	6,0	66,0	0,685	0,550	990	3	0,183	5,3	9,903
15	6,0	45,0	0,467	0,400	1487	2	0,200	5,33	14,44
16	6,0	27,0	0,281	0,230	2980	1	0,230	6,0	21,38
17	6,0	81,0	0,842	0,700	990	3	0,233	6,0	7,128
18	6,0	29,0	0,299	0,250	2973	1	0,250	5,3	22,54
19	6,0	34,0	0,353	0,300	2980	1	0,300	6,2	16,43
20	6,0	36,0	0,374	0,320	2980	1	0,32	6,0	16,04
21	6,0	52,0	0,540	0,460	2985	1	0,460	7,0	9,517
22	6,0	201,0	2,090	1,800	992	3	0,600	5,2	3,314
23	6,0	163,0	1,697	1,400	1478	2	0,700	5,0	4,250
24	6,0	231,0	2,400	2,100	1492	2	1,050	5,1	2,940
25	6,0	504,0	5,245	4,500	1492	2	2,250	4,7	1,462
26	6,0	296,0	3,070	2,650	2985	1	2,650	5,0	2,341
27	6,0	659,0	6,850	6,000	1489	2	3,000	5,5	0,956
28	6,0	1120,0	11,640	10,000	1490	2	5,000	4,0	0,773

^a $S_{rM} = \sqrt{3} U_{rM} I_{rM}$
^b $P_{rM} = S_{rM} \cos \varphi_{rM} \cdot \eta_{rM}$
^c $f = 50$ Hz
^d pair of poles
^e active power per pair of poles
^f Z_M according to equation (26) of IEC 60909-0

The time constants $T_{a,c}$ for the 28 asynchronous motors of table 4 are given in figure 35 together with the time constant $T_{\mu q} = -t_{\min} \ln(\mu q)_{IEC\mu}$ with μ in accordance with IEC 60909-0, equation (70) and $q = q_{IEC}$ in accordance with IEC 60909-0, equation (73). The relatively high variations in figures 33 and 34 are mainly due to the different design of the asynchronous motors even in the case of equal rated active power and number of poles.

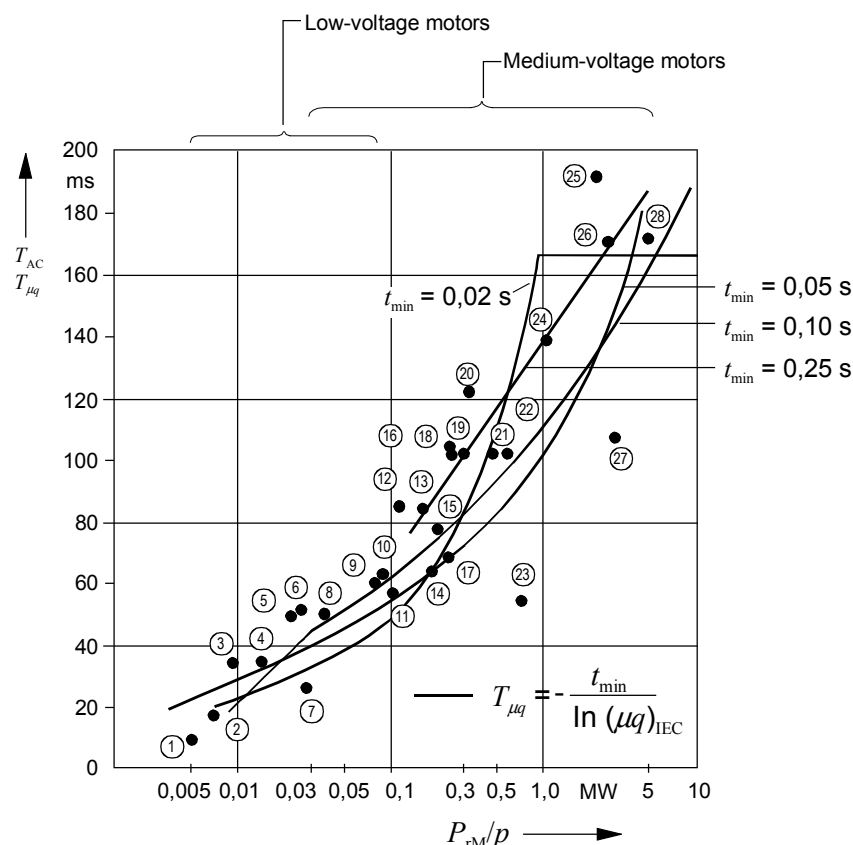


Figure 35 – Effective time constants T_{AC} for the determination of the symmetrical short-circuit breaking current I_{bM} and in comparison $T_{\mu q} = -t_{min}/\ln(\mu q)_{IEC}$ (for the number of motors see table 4)

2.7.3 Short-circuit breaking currents in the case of unbalanced short circuits

In the case of unbalanced short circuits, asynchronous motors or groups of asynchronous motors contribute to the initial asymmetrical short-circuit current, to the peak short-circuit current, to the short-circuit breaking current and to the steady-state short-circuit current (IEC 60909-0, table 3). The influence of motors on the line-to-earth short-circuit current has to be taken into account only in the case of solidly earthed networks.

Figure 36 gives an example for the time functions I_{bM} / I_{kM}'' in the case of a balanced three-phase short circuit and a line-to-line short circuit at the terminals of an asynchronous motor.

The current of the third line in the case of a line-to-line short circuit can reach the magnitude of about I_{k3M}'' , if the time delay is large enough. The steady-state short-circuit current in both the short-circuited lines reaches a maximum of $(\sqrt{3}/2) I_{k3M}''$ (see figure 36 and IEC 60909-0, equation (100)). The short-circuit current in the case of a line-to-earth short circuit, if the network is solidly earthed, will decay more slowly than the line-to-line short-circuit current shown.

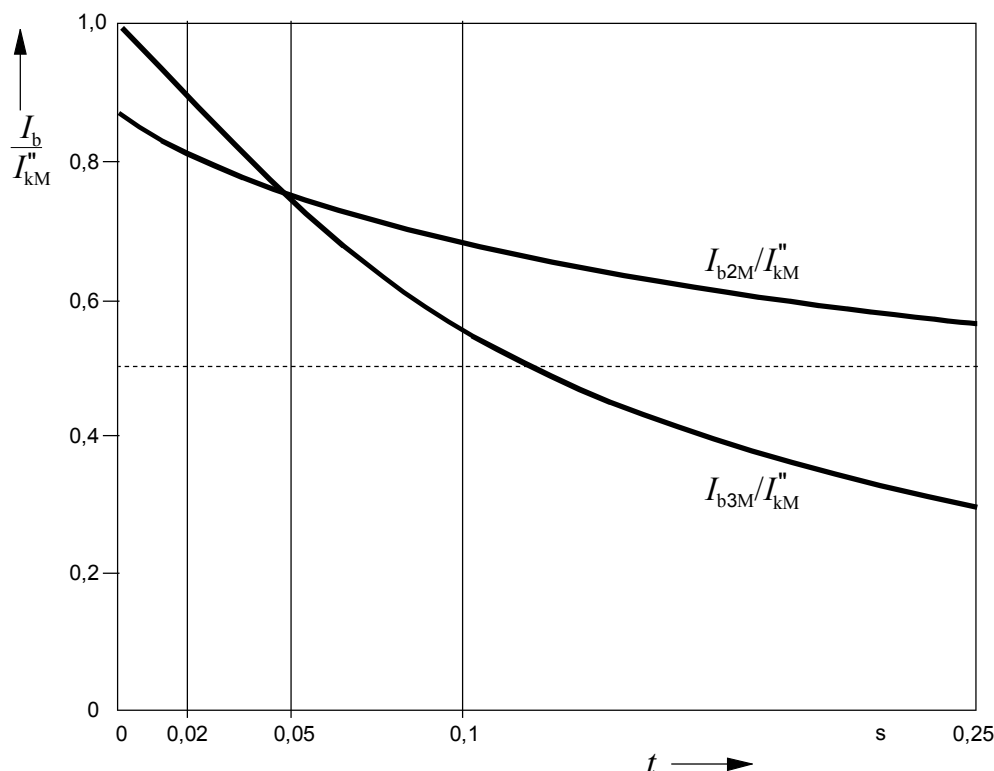


Figure 36 – Time function I_{bM} / I''_{kM} in the case of a balanced short circuit (I_{b3M} / I''_{kM}) and a line-to-line short circuit (I_{b2M} / I''_{kM}) at the terminals of an asynchronous motor (example: Motor No. 28 of table 4)

2.8 Factors m and n for the calculation of the Joule integral or the thermal equivalent short-circuit current

2.8.1 General

The factors m and n are used for the calculation of the Joule integral $\int i^2 dt$ or the thermal equivalent short-circuit current I_{th} (IEC 60909-0, 4.8 and annex A). The factor m is used to express the time-dependent heat effect of the d.c. component of the short-circuit current and the factor n to express the time-dependent heat effect of the a.c. component.

For one individual three-phase short-circuit current $i_k(t)$ the following equation is given for the calculation of the Joule integral (IEC 60909-0, equation (102)):

$$\int_0^{T_k} i^2 dt = I_k''^2 (m + n) T_k = I_{th}^2 T_k \quad (92)$$

$$\text{with} \quad I_{th} = I_k'' \sqrt{m + n} \quad (93)$$

where

I_k'' is the initial symmetrical short-circuit current (IEC 60909-0, equation (29));

T_k is the duration of the short-circuit current;

m is the factor for the heat effect of the d.c. component of the short-circuit current;

n is the factor for the heat effect of the a.c. component of the short-circuit current.

The calculation procedure for the heat effect of a series of three-phase successive individual short-circuit currents is given in IEC 60909-0, equation (104).

2.8.2 Time-dependent three-phase short-circuit current

For the short-circuit current $i_k(t)$ at the terminals of a synchronous machine it is possible to write the following equation with sufficient accuracy [3] and [15]:

$$i_k(t) = \sqrt{2} \left[(I_k'' - I_k') e^{-t/T_d''} + (I_k' - I_k) e^{-t/T_d'} + I_k \right] \cos(\omega t + \varphi_U) - \sqrt{2} I_k'' e^{-t/T_a} \cos \varphi_U \quad (94)$$

where

I_k' is the transient short-circuit current;

T_d'' is the subtransient time constant of a synchronous machine in the direct axis;

T_d' is the transient time constant of a synchronous machine in the direct axis;

T_a is the d.c. time constant of a synchronous machine;

φ_U is the voltage angle at the beginning of the short circuit (see 2.4.2).

The decaying transient current within the quadrature axis of the synchronous machine is neglected in equation (94). The maximum d.c. component of the short-circuit current (equation (94)) is reached in the case $\varphi_U = 0$ (see 2.4.2).

If the short circuit does not occur at the terminals of the synchronous machine but at a location in the network with a reactance X_N between the terminals and the short-circuit location, the following currents and time constants are to be introduced [3] and [15]:

$$I_k'' = \frac{E''}{X_d'' + X_N} \quad (95a)$$

$$I_k' = \frac{E'}{X_d' + X_N} \quad (95b)$$

$$I_k = \frac{E}{X_d + X_N} \quad (95c)$$

$$T_{dN}'' \approx 0,1 \cdot T_{dN}' \quad (96a)$$

$$T_{dN}' \approx T_{d0}' \frac{X_d' + X_N}{X_d + X_N} \quad (96b)$$

$$T_{aN} \approx -\frac{1}{2f \ln(\kappa - 1)} \quad (96c)$$

T_{aN} is an approximation, using $\kappa = 1 + e^{-3R/X}$ instead of $\kappa_{IEC} = 1,02 + 0,98 e^{-3R/X}$ (IEC 60909-0, equation (55)) and $R/X = 1/(\omega T_{aN})$ with $\omega = 2\pi f$ and $\pi \approx 3$.

2.8.3 Factor m

For the determination of the factor m , used to describe the heat effect of the d.c. component of the short-circuit current, a far-from-generator short circuit ($I_k = I_k''$ and $n = 1$) can be presumed [15].

In this case with $I_k'' \approx I_k' \approx I_k$ (far-from-generator short circuit) and $\varphi_U = 0$ equation (94) leads to:

$$i_k(t) = \sqrt{2}I_k'' \cos \omega t - \sqrt{2}I_k'' e^{-t/T_{aN}} = \sqrt{2}I_k'' (\cos \omega t - e^{-t/T_{aN}}) \quad (97)$$

and to the Joule integral (equation (92)) with $T_k = T_{k \max}$:

$$\int_0^{T_k} i_k^2(t) dt = I_k''^2 \int_0^{T_k} \left[1 + \cos 2\omega t - 4 \cos \omega t \cdot e^{-t/T_{aN}} + 2e^{-2t/T_{aN}} \right] dt \quad (98)$$

The terms in equation (97) with ω ($\cos 2\omega t$ and $\cos \omega t$) can be neglected because their contribution to the thermal effect is very small [3].

From equation (98) the following result is found:

$$\int_0^{T_k} i_k^2(t) dt = I_k''^2 T_k \left[1 + \frac{T_{aN}}{T_k} \left(1 - e^{-2T_k/T_{aN}} \right) \right] \quad (99)$$

Together with equation (92) and T_{aN} from equation (96c) the factor m depending on $f T_k$ and κ is found as follows:

$$m = \frac{1}{2f T_k \ln(\kappa - 1)} (e^{4f T_k \ln(\kappa - 1)} - 1) \quad (100)$$

The factor m calculated with equation (100) is given in IEC 60909-0, figure 21 (see also IEC 60909-0, annex A).

2.8.4 Factor n

The factor n for the thermal effect of the decaying a.c. component of the short-circuit current can be found from equation (101) if m is known.

$$n = \frac{1}{I_k''^2 T_k} \int_0^{T_k} i_k^2(t) dt - m \quad (101)$$

If $i_k(t)$ from equation (94) is introduced with $\cos \varphi_U = 1$, then the following result can be found, if the oscillatory components 50 Hz (60 Hz) or 100 Hz (120 Hz) are neglected, because their contribution to the thermal effect is very small [3].

$$n = \frac{1}{(I_k''/I_k)^2} \left[1 + \frac{T_{dN}''}{2T_k} (1 - e^{-2T_k/T_{dN}''}) \left(\frac{I_k''}{I_k} - \frac{I_k'}{I_k} \right)^2 + \frac{T_{dN}'}{2T_k} (1 - e^{-2T_k/T_{dN}'}) \left(\frac{I_k'}{I_k} - 1 \right)^2 \right. \\ \left. + \frac{2T_{dN}''}{T_k} (1 - e^{-T_k/T_{dN}''}) \left(\frac{I_k''}{I_k} - \frac{I_k'}{I_k} \right) + \frac{2T_{dN}'}{T_k} (1 - e^{-T_k/T_{dN}'}) \left(\frac{I_k'}{I_k} - 1 \right) \right. \\ \left. + \frac{2T_{SN}}{T_k} (1 - e^{-T_k/T_{SN}}) \left(\frac{I_k''}{I_k} - \frac{I_k'}{I_k} \right) \left(\frac{I_k'}{I_k} - 1 \right) \right] \quad (102a)$$

$$\text{with } \frac{1}{T_{SN}} = \frac{1}{T_{dN}''} + \frac{1}{T_{dN}'}$$

If the relation $T_{dN}'' \approx 0,1 \cdot T_{dN}'$ [3] is used, then $T_{SN} \approx (1/11) \cdot T_{dN}'$ is found and the following equation for the calculation of the factor n :

$$n = \frac{1}{(I_k'' / I_k)^2} \left[1 + \frac{T_{dN}'}{20T_k} \left(1 - e^{-20T_k / T_{dN}'} \right) \left(\frac{I_k''}{I_k} - \frac{I_k'}{I_k} \right)^2 + \frac{T_{dN}'}{2T_k} \left(1 - e^{-2T_k / T_{dN}'} \right) \left(\frac{I_k'}{I_k} - 1 \right)^2 \right. \\ \left. + \frac{T_{dN}'}{5T_k} \left(1 - e^{-10T_k / T_{dN}'} \right) \left(\frac{I_k''}{I_k} - \frac{I_k'}{I_k} \right) + \frac{2T_{dN}'}{T_k} \left(1 - e^{-T_k / T_{dN}'} \right) \left(\frac{I_k'}{I_k} - 1 \right) \right. \\ \left. + \frac{T_{dN}'}{5,5T_k} \left(1 - e^{-11T_k / T_{dN}'} \right) \left(\frac{I_k''}{I_k} - \frac{I_k'}{I_k} \right) \left(\frac{I_k'}{I_k} - 1 \right) \right] \quad (102b)$$

To find a unique decaying characteristic of the a.c. part of the short-circuit current, it is necessary to find a relation between I_k' / I_k and I_k'' / I_k , if I_k'' / I_k is known from the short-circuit current calculation (IEC 60909-0), and to choose a transient time constant T_{dN}' . This is only possible for one characteristic model generator.

2.8.5 Factor n in IEC 60909-0, figure 22

The data for the model generator leading to the factor n in IEC 60909-0, figure 22, are given in the following table.

Table 5 – Data for the model generator [15]

Cylindrical rotor generator:	
Subtransient reactance	$x_d'' = 0,225$
Transient reactance	$x_d' = 0,34 \approx 1,5 \cdot x_d''$
Synchronous reactance	$x_d = 2,35$
No-load transient time constant	$T_{d0}' = 10 \text{ s}$
Rated power factor	$\cos \varphi_{rG} = 0,8$
Highest excitation voltage	$u_{fmax} = U_{fmax} / U_{fr} = 1,3$

The network reactance X_N between the terminals of the generator and the short-circuit location can be found from the quotient of I_k'' and I_k taking the equations (95a) and (95c):

$$\frac{I_k''}{I_k} = \frac{E''}{X_d'' + X_N} \cdot \frac{X_d + X_N}{E} \quad (103a)$$

and therefore

$$X_N = \frac{I_k E'' X_d - I_k' E X_d''}{I_k' E - I_k E''} \quad (103b)$$

The equations (95b) and (95c) lead to the quotient I_k' / I_k and, if X_N from equation (103b) is introduced, to:

$$\frac{I_k'}{I_k} = \frac{E'}{X_d' + X_N} \cdot \frac{X_d + X_N}{E} = \frac{I_k'' / I_k}{\frac{E''}{E'} \cdot \frac{X_d - X_d'}{X_d - X_d''} + \frac{E}{E'} \cdot \frac{X_d' - X_d''}{X_d - X_d''} \cdot (I_k'' / I_k)} \quad (104)$$

with

$$E'' = (U_{rG} / \sqrt{3}) \sqrt{1 + 2x_d'' \sin \varphi_{rG} + x_d''^2} \quad (105a)$$

$$E' = (U_{rG} / \sqrt{3}) \sqrt{1 + 2x_d' \sin \varphi_{rG} + x_d'^2} \quad (105b)$$

$$E = u_{fmax} (U_{rG} / \sqrt{3}) \sqrt{1 + 2x_d \sin \varphi_{rG} + x_d^2} \quad (105c)$$

if $U_G = U_{rG}$, $I_G = I_{rG}$ and $\cos \varphi_G = \cos \varphi_{rG}$.

With the data given in table 5 for the model generator the following results are found:

$$E'' = (U_{rG} / \sqrt{3}) \cdot 1,149$$

$$E' = (U_{rG} / \sqrt{3}) \cdot 1,234$$

$$E = u_{fmax} (U_{rG} / \sqrt{3}) \cdot 3,057$$

and from equation (103):

$$\frac{I_k'}{I_k} = \frac{I_k'' / I_k}{0,88 + u_{fmax} \cdot 0,134 (I_k'' / I_k)} \quad (106a)$$

If $u_{fmax} = 1,3$ is introduced, equation (106a) leads to equation (106b) equal to the equation given in IEC 60909-0, annex A:

$$\frac{I_k'}{I_k} = \frac{I_k'' / I_k}{0,88 + 0,17 (I_k'' / I_k)} \quad (106b)$$

If the equations (95b) and (95c) are introduced to equation (96b), the following equation is found for the transient time constant:

$$T_{dN}' = T_{d0}' \frac{X_d' + X_N}{X_d + X_N} = T_{d0}' \frac{E' / E}{I_k' / I_k} \quad (107a)$$

If E' and E , found with the data of table 5, are introduced, the following transient time constant is found:

$$T_{dN}' = \frac{4,04 \text{ s}}{u_{fmax} (I_k' / I_k)} \quad (107b)$$

With $u_{fmax} = 1,3$ this equation becomes equal to the equation given in IEC 60909-0, annex A:

$$T_{dN}' = \frac{3,1 \text{ s}}{I_k' / I_k} \quad (107c)$$

2.9 Statement of the contribution of asynchronous motors or groups of asynchronous motors (equivalent motors) to the initial symmetrical short-circuit current

2.9.1 General

Asynchronous motors or groups of asynchronous motors (equivalent motors) contribute to the initial symmetrical short-circuit current I_k'' , especially in the case of near-to-motor short circuits, and furthermore to the peak short-circuit current i_p , to the symmetrical short-circuit breaking current I_b and in the case of unbalanced short circuits, to the steady-state short-circuit current I_k (IEC 60909-0, 3.8.1). If the contribution to the initial symmetrical short-circuit current remains smaller than 5 % of the total short-circuit current, this contribution may be neglected. IEC 60909-0 gives two equations to estimate whether the contribution is less than 5 % either to a short circuit at the terminal of motors (IEC 60909-0, equation (25)) or to a short circuit fed from motors or motor groups through transformers (IEC 60909-0, equation (28)) without an exact calculation.

Large numbers of low-voltage synchronous motors for instance in industrial installations or in the case of power station auxiliaries, and in some cases the lack of sufficient data for each of the motors leads to the assumption of equivalent motors including their connection cables to a common busbar (IEC 60909-0, 3.8.2). The assumptions for equivalent motors are:

$$\sum P_{rM} \Rightarrow P_{rM}, I_{LR} / I_{rM} = 5, \cos \varphi_{rM} \eta_{rM} = 0,8 \text{ and } P_{rM} / p = 0,05 \text{ MW}$$

in the case of low-voltage motor groups (IEC 60909-0, 3.8.2).

2.9.2 Short circuit at the terminals of asynchronous motors

The contribution of one asynchronous motor or a group of asynchronous motors (equivalent motor), as shown in figure 37, to the total initial symmetrical short-circuit current I_k'' may be neglected if

$$I_{kM}'' \leq 0,05 I_{kQ}'' \quad (108)$$

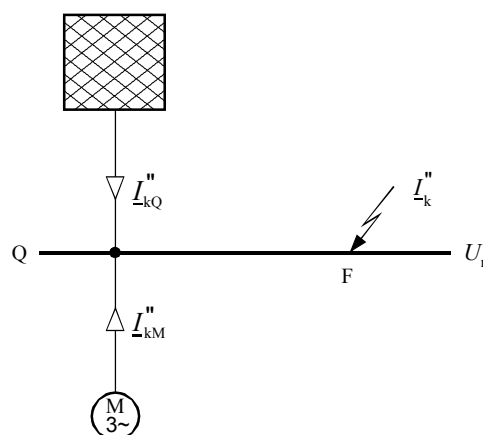


Figure 37 – Contribution of one asynchronous motor or a group of asynchronous motors to the initial symmetrical short-circuit current $I_k'' = I_{kQ}'' + I_{kM}''$

Taking Z_M from equation (26) of IEC 60909-0, neglecting the impedance of the feeding cables, the partial short-circuit current I_{kM}'' becomes:

$$I_{kM}'' = \frac{cU_n}{\sqrt{3}Z_M} = \frac{I_{LR}}{I_{rM}} \cdot \frac{cU_n}{U_{rM}} \cdot I_{rM} \quad (109)$$

In the case of a motor group the sum of the rated currents $\sum I_{rM}$ of the motors is introduced instead of I_{rM} . Estimating $I_{LR}/I_{rM} = 5$ and $cU_n/U_{rM} = 1$, equation (108) takes the following form (see IEC 60909-0, equation (25)):

$$\sum I_{rM} \leq 0,01 I_{kQ}'' \quad (110)$$

2.9.3 Partial short-circuit currents of asynchronous motors fed through transformers

For the derivation of IEC 60909-0, equation (28), the simple case in figure 38 is used.

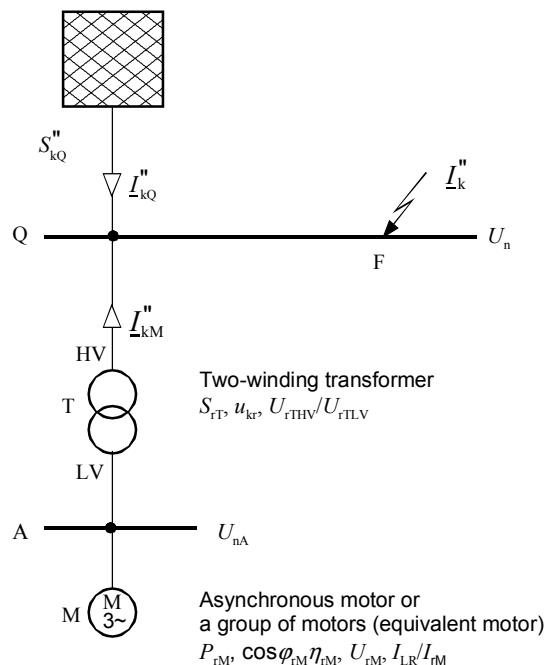


Figure 38 – Example for the estimation of the partial short-circuit current I_{kM}'' supplied by a single asynchronous motor or an equivalent motor

The total short-circuit current I_k'' at the short-circuit location F is given as the sum of the two partial short-circuit currents I_{kQ}'' and I_{kM}'' at the high-voltage side of the transformer in figure 38.

If the contribution I_{kM}'' of the asynchronous motor or the equivalent motor is smaller than, or equal to, 5 % of the short-circuit current I_{kQ}'' , then this can be stated as

$$I_{kM}'' \leq 0,05 I_{kQ}'' \quad (111)$$

The short-circuit current I_{kM}'' is found from the short-circuit impedance Z_{Mt} transferred to the high-voltage side of the transformer and the short-circuit impedance Z_{THV} of the transformer at the high-voltage side:

$$Z_{\text{Mt}} = \frac{1}{I_{\text{LR}} / I_{\text{rM}}} \cdot \frac{U_{\text{rM}}^2}{S_{\text{rM}}} \cdot \frac{U_{\text{rTHV}}^2}{U_{\text{rTLV}}^2} \quad (112)$$

$$Z_{\text{THV}} = \frac{u_{\text{kr}}}{100\%} \cdot \frac{U_{\text{rTHV}}^2}{S_{\text{rT}}} \quad (113)$$

Neglecting the impedance correction factor for network transformers and supposing $U_{\text{rM}} \approx U_{\text{rTLV}}$, $U_{\text{nQ}} \approx U_{\text{rTHV}}$, both the impedances given in equations (112) and (113) can be introduced in the following equation:

$$I_{\text{kM}}'' \approx \frac{c U_{\text{nQ}}}{\sqrt{3}(Z_{\text{THV}} + Z_{\text{Mt}})} \approx \frac{c U_{\text{nQ}}}{\sqrt{3} U_{\text{rTHV}}^2} \cdot \frac{1}{\frac{u_{\text{kr}}}{100\%} \cdot \frac{1}{S_{\text{rT}}} + \frac{1}{I_{\text{LR}} / I_{\text{rM}}} \cdot \frac{1}{S_{\text{rM}}}} \quad (114)$$

If I_{kM}'' from equation (114) is introduced to equation (111), the following equation is found:

$$\frac{u_{\text{kr}}}{100\%} \cdot \frac{1}{S_{\text{rT}}} + \frac{1}{I_{\text{LR}} / I_{\text{rM}}} \cdot \frac{1}{S_{\text{rM}}} \geq \frac{c}{0,05 \sqrt{3} U_{\text{nQ}} I_{\text{kQ}}''} \quad (115)$$

Introducing $P_{\text{rM}} = S_{\text{rM}} \cos \varphi_{\text{rM}} \eta_{\text{rM}}$ equation (115) leads to

$$\frac{P_{\text{rM}}}{S_{\text{rT}}} \leq \frac{\cos \varphi_{\text{rM}} \eta_{\text{rM}}}{\left| \frac{I_{\text{LR}}}{I_{\text{rM}}} \cdot \frac{c}{0,05} \cdot \frac{S_{\text{rT}}}{\sqrt{3} U_{\text{nQ}} I_{\text{kQ}}''} - \frac{I_{\text{LR}}}{I_{\text{rM}}} \cdot \frac{u_{\text{kr}}}{100\%} \right|} \quad (116a)$$

Introducing $\cos \varphi_{\text{rM}} \eta_{\text{rM}} = 0,8$, $I_{\text{LR}} / I_{\text{rM}} = 5$ and $u_{\text{kr}} = 6\%$ for low-voltage transformers, equation (116a) leads to:

$$\frac{P_{\text{rM}}}{S_{\text{rT}}} \leq \frac{0,8}{\left| \frac{c \cdot 100 \cdot S_{\text{rT}}}{\sqrt{3} U_{\text{nQ}} I_{\text{kQ}}''} - 0,3 \right|} \quad (116b)$$

It is necessary to take the absolute value of the denominator in equation (116), if the first term becomes smaller than the second one.

Asynchronous motors or equivalent motors contribute less than 5 % to the short-circuit current without motors if equation (116) holds (IEC 60909-0, 3.8.2).

2.9.4 Sum of partial short-circuit currents of several groups of asynchronous motors fed through several transformers

For configurations as shown in figure 39 where several groups of asynchronous motors (low-voltage or medium-voltage motors) fed through several different transformers, equations (116) can be used only as a first estimation under certain restrictive conditions, to find out if the motors contribute more or less than 5 % to the short-circuit current without motors.

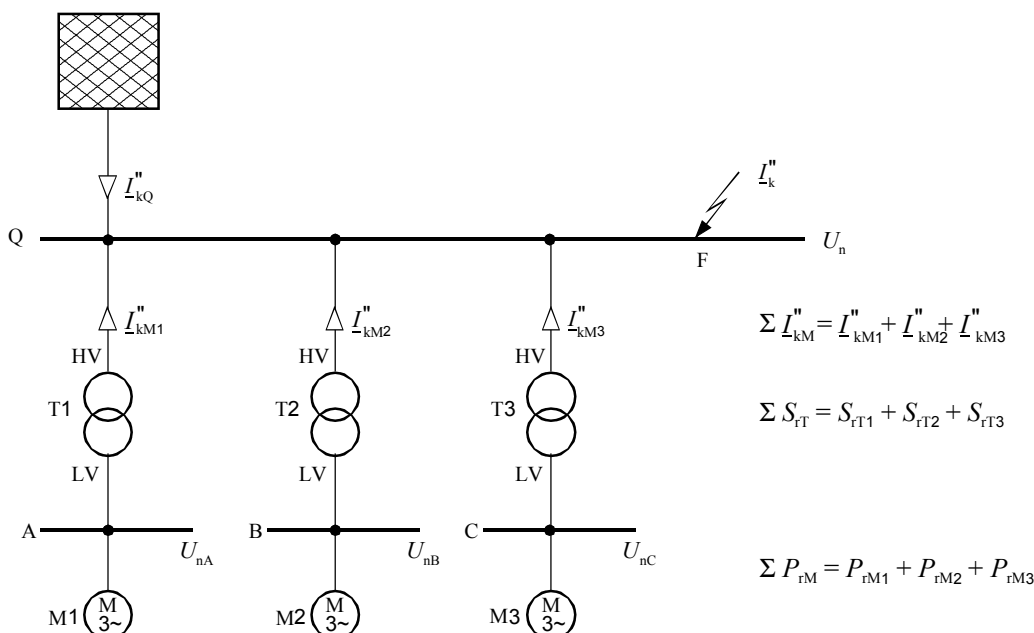


Figure 39 – Partial short-circuit currents from several groups of asynchronous motors fed through several transformers (see text for restrictive conditions)

The condition $\Sigma I_{kM}'' \leq 0,05 I_{kQ}''$ with $I_{kM}'' = I_{kM1}'' + I_{kM2}'' + I_{kM3}''$ (figure 39) shall be considered.

Introducing $Z_{M1t}, Z_{M2t}, Z_{M3t}$ in accordance with equation (112) and $Z_{T1HV}, Z_{T2HV}, Z_{T3HV}$ in accordance with equation (113) and assuming $u_{kr} = u_{krT1} = u_{krT2} = u_{krT3}$ as well as equal ratios I_{LR} / I_{rM} for the motor groups and furthermore, as already used in 2.9.3, $U_{rTHV} = U_{nQ}$ and $U_{rM} = U_{rTLV}$, the following can be found:

$$\begin{aligned}
 & \Sigma \sqrt{3} U_{nQ} I_{kM}'' = \sqrt{3} U_{nQ} (I_{kM1}'' + I_{kM2}'' + I_{kM3}'') \\
 & = \frac{c}{\frac{u_{kr} / 100 \%}{S_{rT1}} + \frac{1}{I_{LR} / I_{rM1}} \frac{1}{S_{rM1}}} + \frac{c}{\frac{u_{kr} / 100 \%}{S_{rT2}} + \frac{1}{I_{LR} / I_{rM2}} \frac{1}{S_{rM2}}} + \frac{c}{\frac{u_{kr} / 100 \%}{S_{rT3}} + \frac{1}{I_{LR} / I_{rM3}} \frac{1}{S_{rM3}}} \\
 & 0,05 \sqrt{3} U_{nQ} I_{kQ}''
 \end{aligned} \tag{117}$$

Equation (117) can be simplified using the following approximation:

$$\begin{aligned} & \frac{\frac{u_{kr}/100\%}{S_{rT1}} + \frac{1}{I_{LR}/I_{rM1}} \cdot \frac{1}{S_{rM1}}}{\frac{u_{kr}/100\%}{S_{rT2}} + \frac{1}{I_{LR}/I_{rM2}} \cdot \frac{1}{S_{rM2}}} + \frac{\frac{u_{kr}/100\%}{S_{rT3}} + \frac{1}{I_{LR}/I_{rM3}} \cdot \frac{1}{S_{rM3}}}{\frac{u_{kr}/100\%}{S_{rT1} + S_{rT2} + S_{rT3}} + \frac{1}{I_{LR}/I_{rM}} \cdot \frac{1}{S_{rM1} + S_{rM2} + S_{rM3}}} \\ & \approx \frac{\frac{u_{kr}/100\%}{S_{rT1} + S_{rT2} + S_{rT3}} + \frac{1}{I_{LR}/I_{rM}} \cdot \frac{1}{S_{rM1} + S_{rM2} + S_{rM3}}}{\frac{u_{kr}/100\%}{\sum S_{rT}} + \frac{1}{I_{LR}/I_{rM}} \cdot \frac{1}{\sum S_{rM}}} \quad (118) \end{aligned}$$

With the approximation (right side of equation (118)) an equation similar to equation (116a) can be established assuming that $\cos \varphi_{rM} \eta_{rM}$ is similar for all motor groups:

$$\frac{\sum P_{rM}}{\sum S_{rT}} \leq \frac{\cos \varphi_{rM} \eta_{rM}}{\left| \frac{I_{LR}}{I_{rM}} \cdot \frac{c}{0,05} \cdot \frac{\sum S_{rT}}{\sqrt{3} U_{nQ} I_{kQ}''} - \frac{I_{LR}}{I_{rM}} \cdot \frac{u_{kr}}{100\%} \right|} \quad (119)$$

The right side of equation (118) is different from the left side if $S_{rT1} \neq S_{rM1}$ or $S_{rT2} \neq S_{rM2}$ or $S_{rT3} \neq S_{rM3}$. If in each case $S_{rTi} = S_{rMi}$ holds, both sides of equation (118) are identical.

To find out to what extent the right side (RS) of equation (118) may differ from the left side (LS), which gives the exact value, the following deviation

$$\Delta = \frac{RS}{LS} - 1 \quad (120)$$

is given in figure (40) for two different groups of motors feeding through two different transformers, depending on S_{rM2}/S_{rT2} , and the parameters S_{rM1}/S_{rT1} and S_{rT1}/S_{rT2} . As long as the relations S_{rM1}/S_{rT1} and S_{rM2}/S_{rT2} remain between 0,5 and 1,5 the deviation Δ is smaller than 0,05. Therefore, the approximation of the right side of equation (118) may be used to find equation (119) for the assessment of the contribution of different motor groups to the short-circuit current.

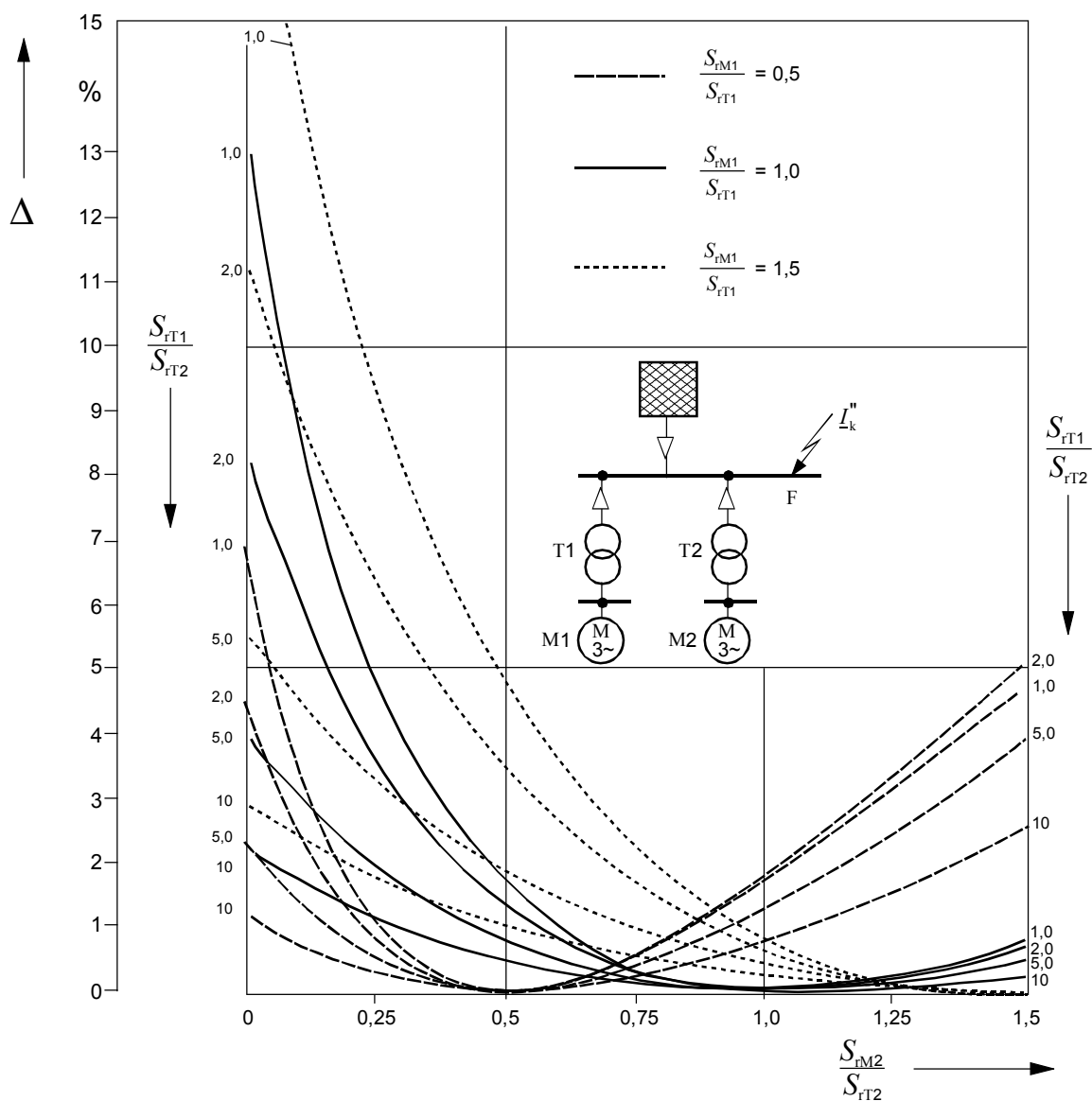


Figure 40 – Investigation of the left and right side of equation (118) to determine the deviation Δ according to equation (120): $u_{kr} = 0,06 \Rightarrow 6\%$, $I_{LR} / I_{rM} = 5$ for both the transformers and motor groups

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